

Time Series Analysis

Spectral analysis II

Dr. Ludovic Amruthalingam
ludovic.amruthalingam@hslu.ch

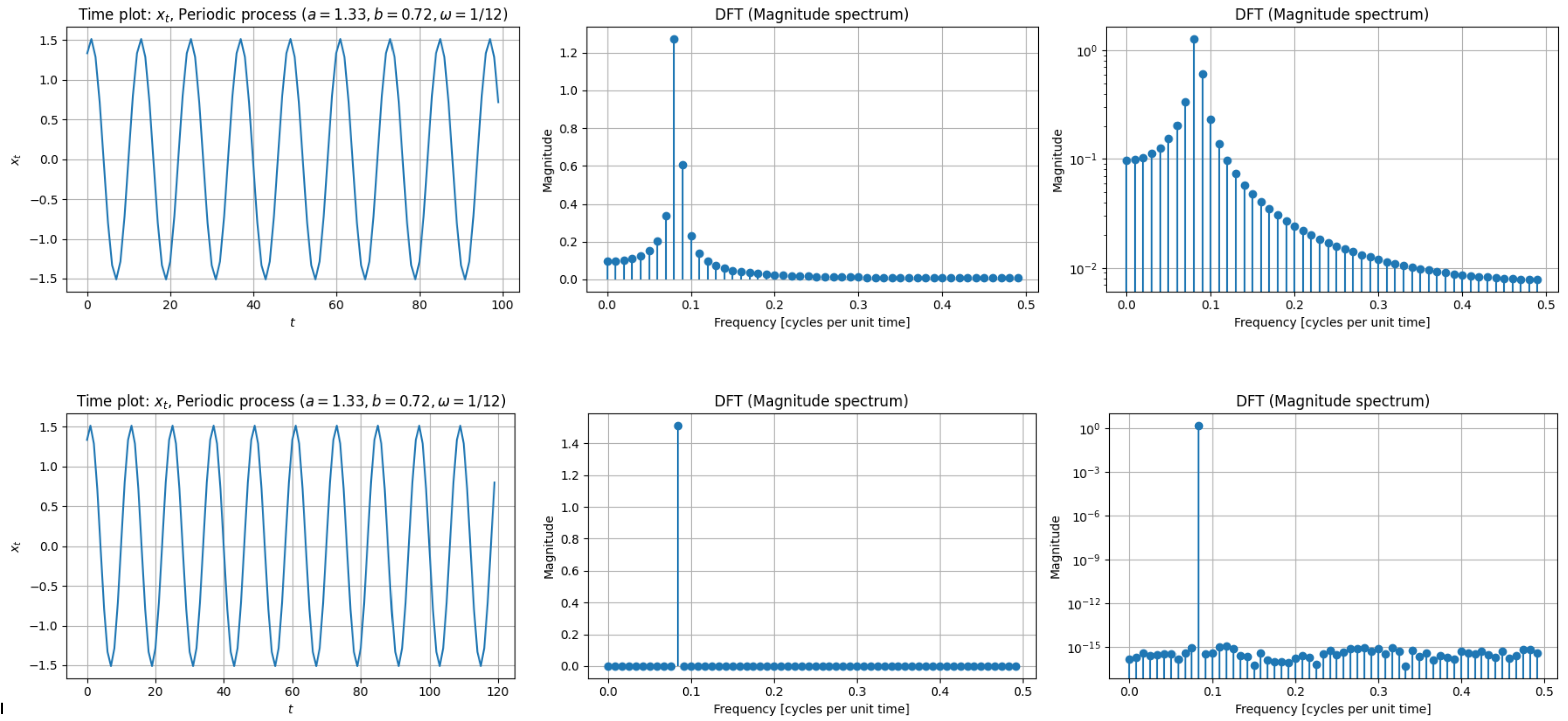
Informatik



Outline

- Spectral leakage
- Periodogram bias
- Tapering
- Periodogram smoothing
- Linear filtering

Spectral leakage



Spectral leakage

Consider a **zero-mean periodic process** $X_t = \cos(2\pi\omega't)$,

If ω' is a Fourier frequency, then $d(\omega_j) = \begin{cases} 0 & \omega_j \neq \omega' \\ 1 & \omega_j = \omega' \end{cases}$ i.e., there are **no leaks** into adjacent frequency bins.

Otherwise, the DFT converges to

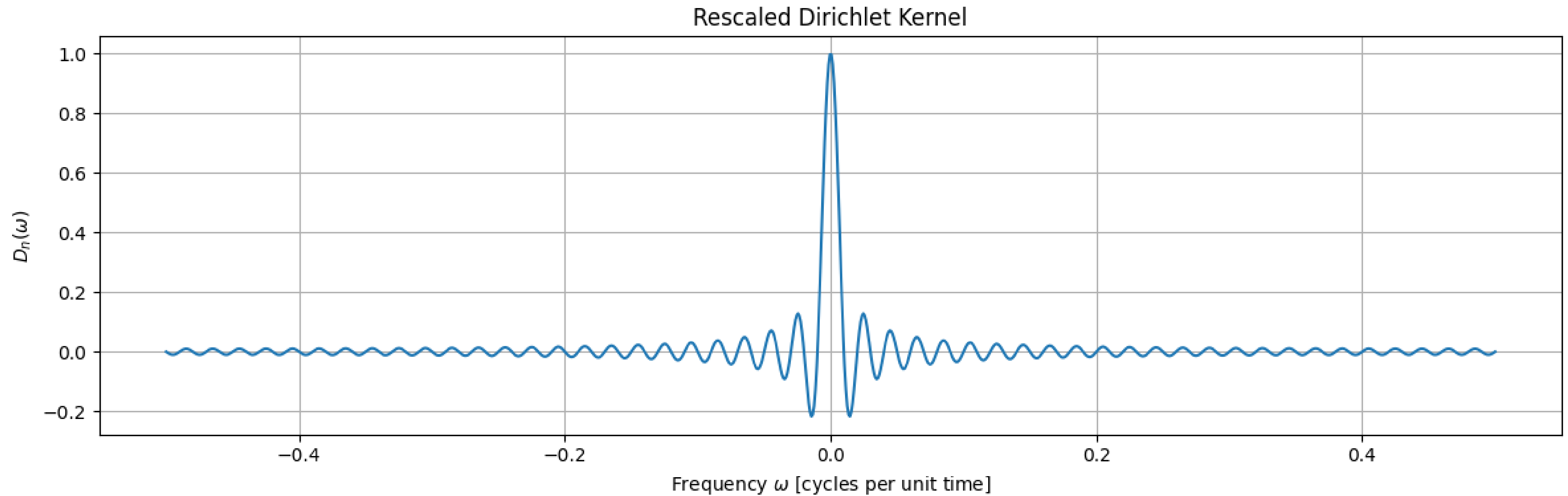
$$d(\omega_j) = e^{\pi i(\omega' - \omega_j)(n-1)} n D_n(\omega_j - \omega')$$

where $D_n(\omega)$ is the **rescaled Dirichlet kernel**

$$D_n(\omega) = \frac{\sin(\pi\omega n)}{n \sin(\pi\omega)}$$

The contribution of frequency ω' **leaks** into adjacent frequency bins, causing **spectral leakage**.

Rescaled Dirichlet kernel



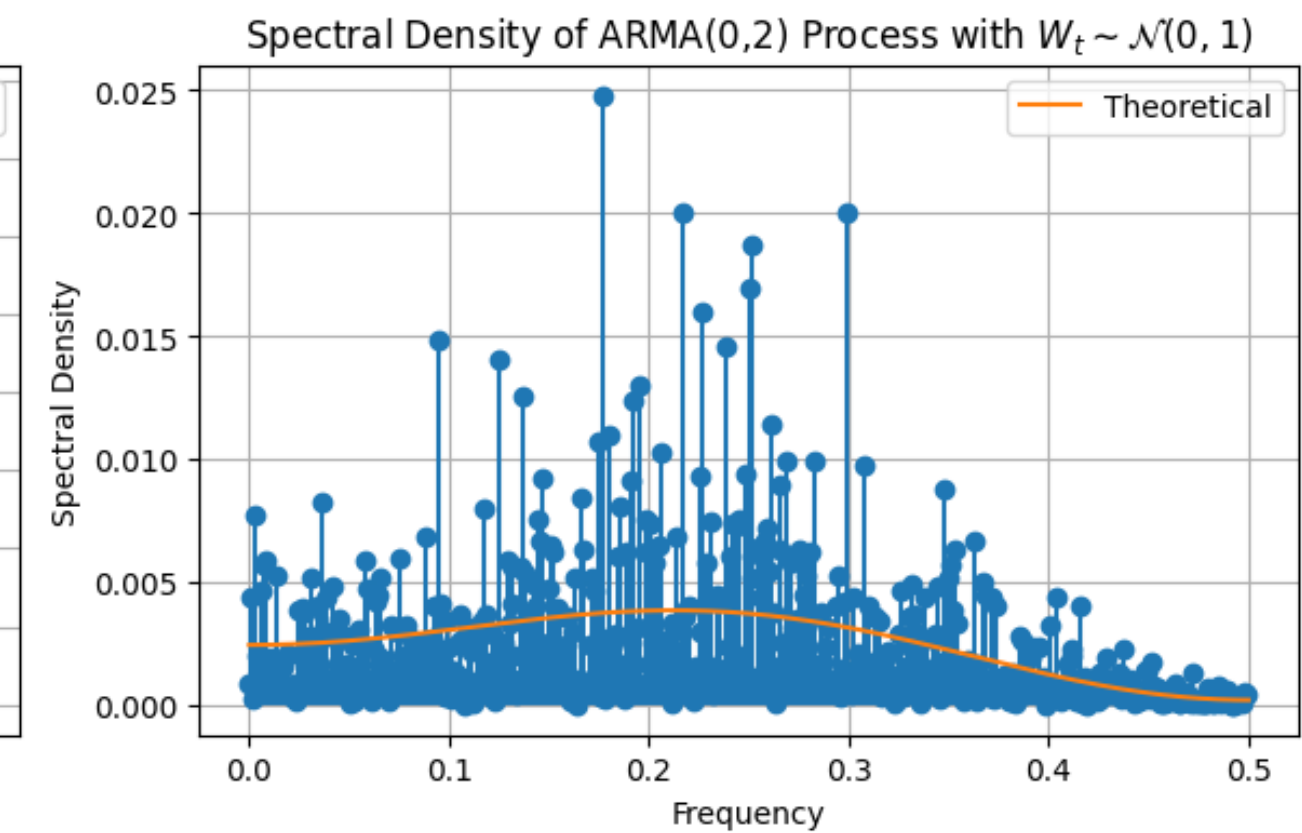
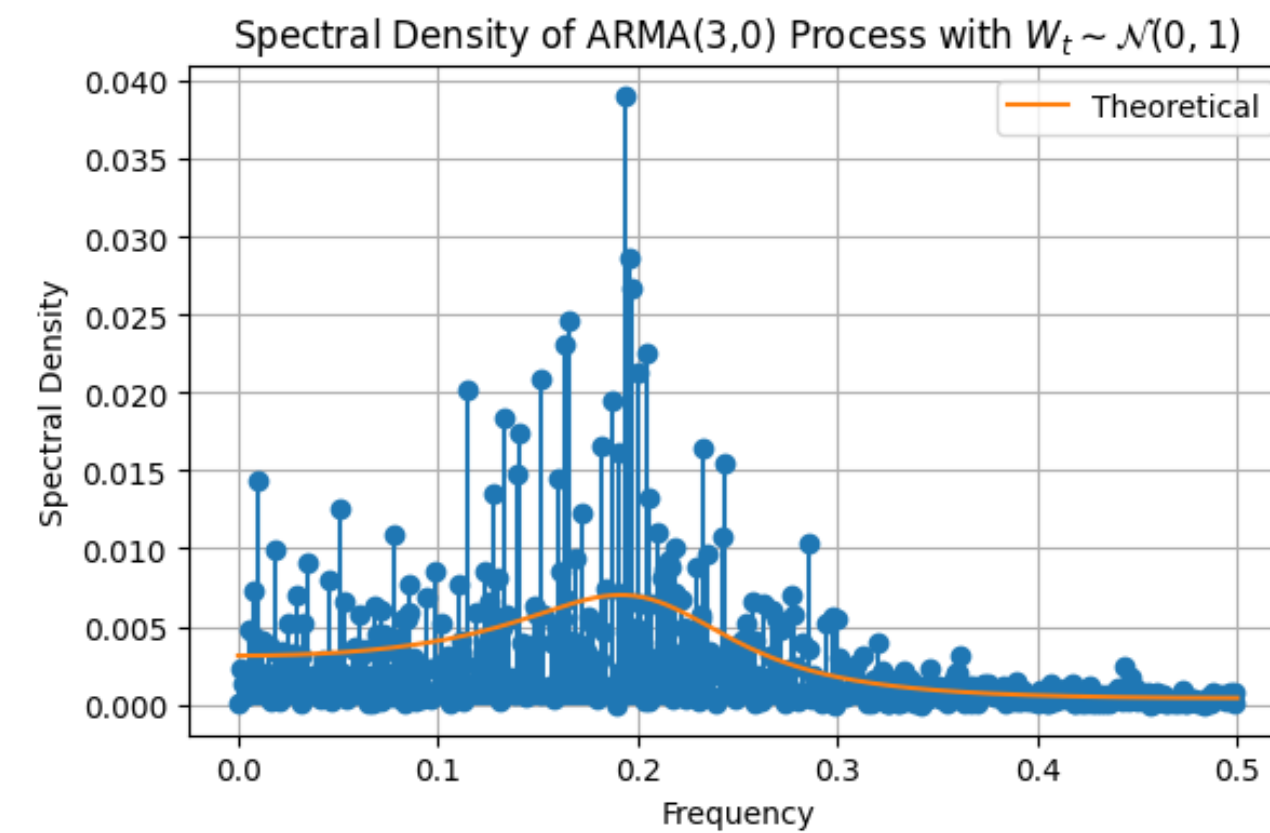
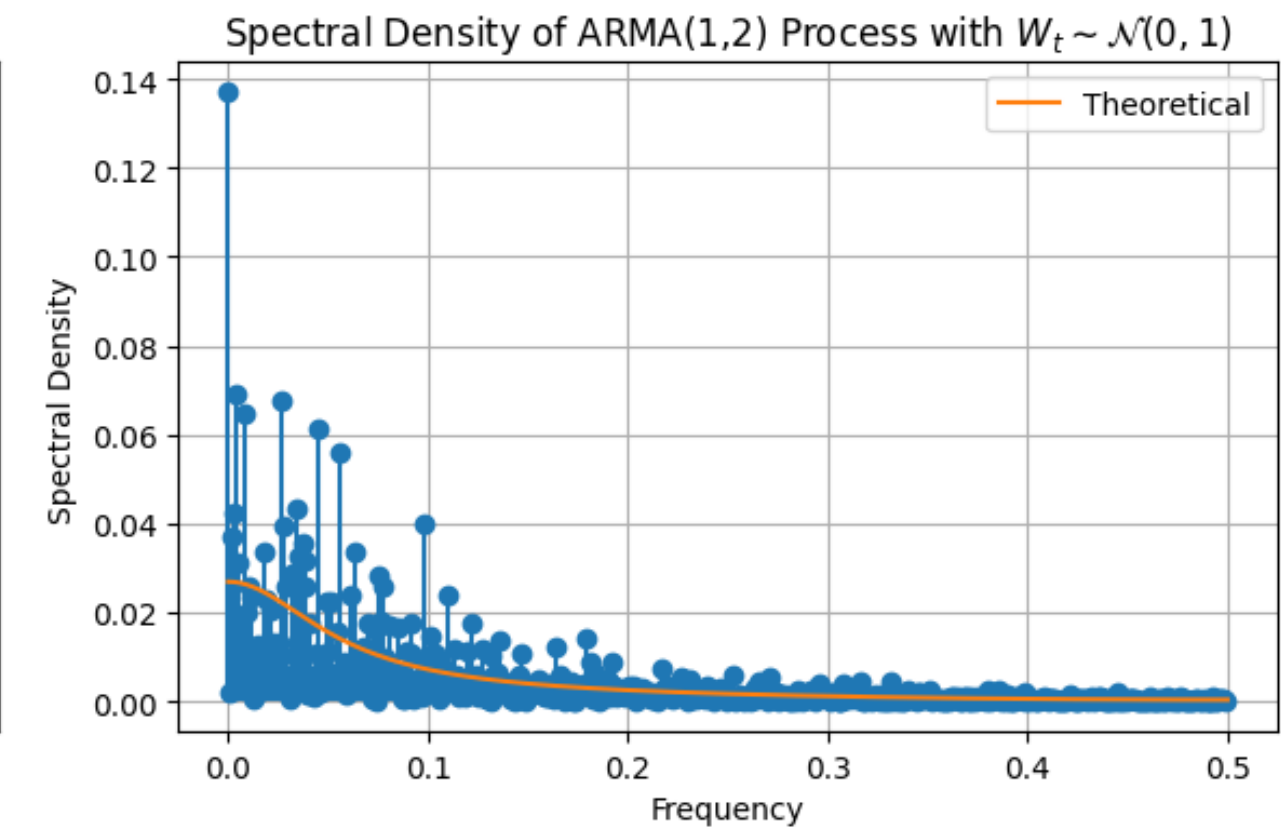
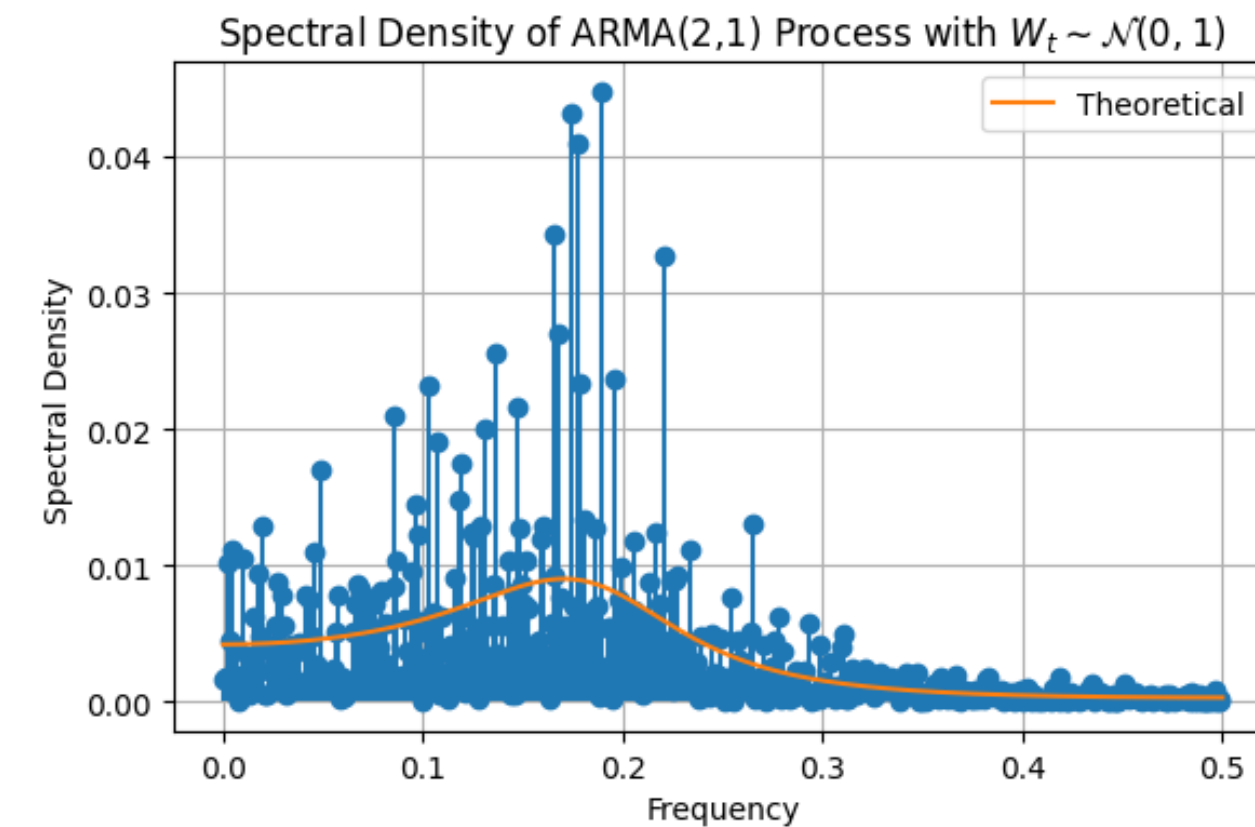
Periodogram bias

$$P(\omega_j) = \frac{1}{n} \sum_{h=-(n-1)}^{n-1} \hat{\gamma}(h) e^{-2\pi i \omega_j h} = \frac{1}{n} \hat{f}(\omega_j)$$

Small n leads to **high variance** (lack of **smoothness**) and **low frequency resolutions**.

However, for any n ,

- $\hat{\gamma}(n-1)$ is estimated from (x_0, x_{n-1}) only.
- $\hat{\gamma}(n-2)$ is estimated from (x_0, x_{n-2}) and (x_1, x_{n-1}) .
- and so on...



Periodogram bias

The **sample auto-covariance** $\hat{\gamma}(h)$ is defined as

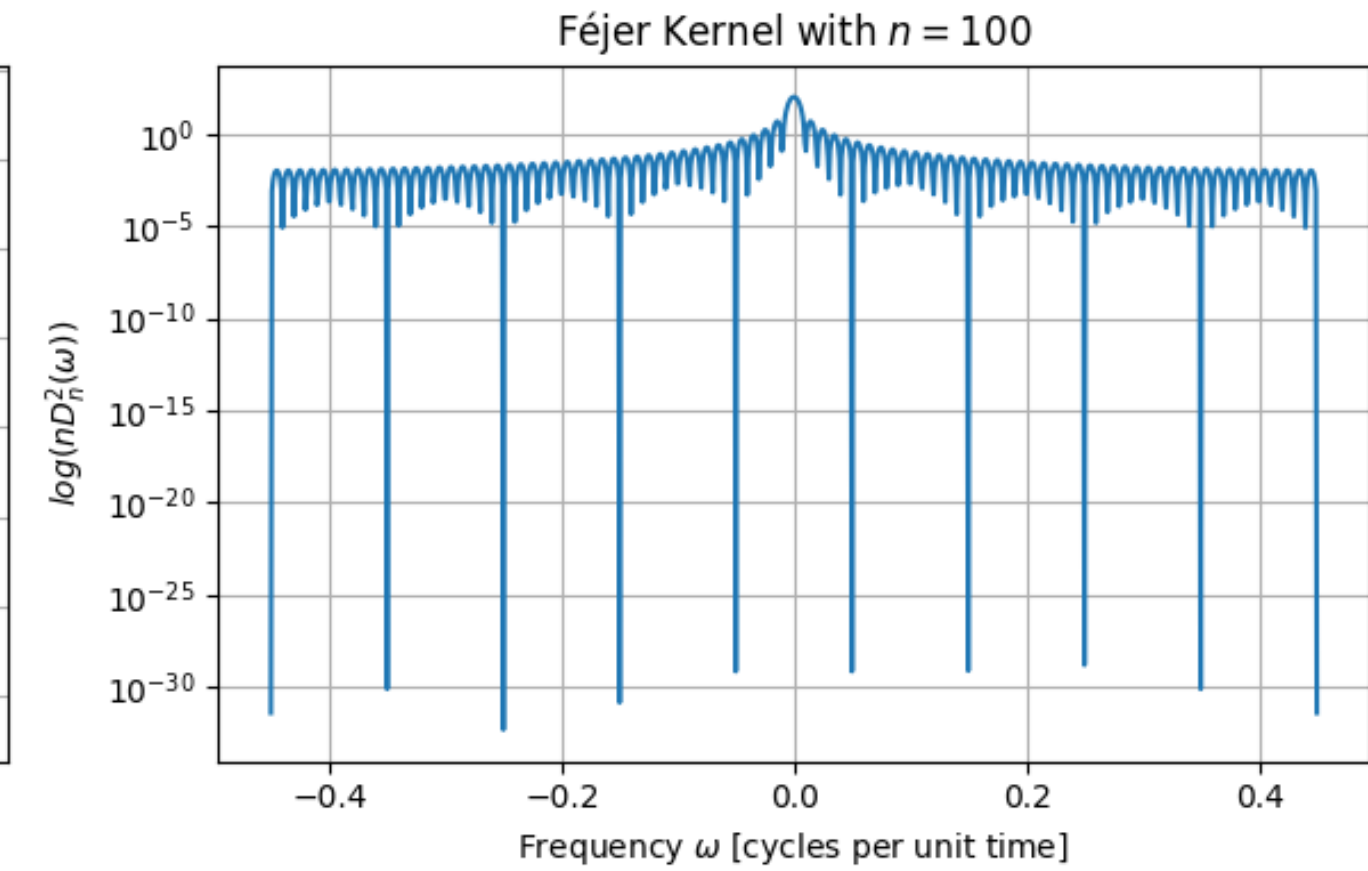
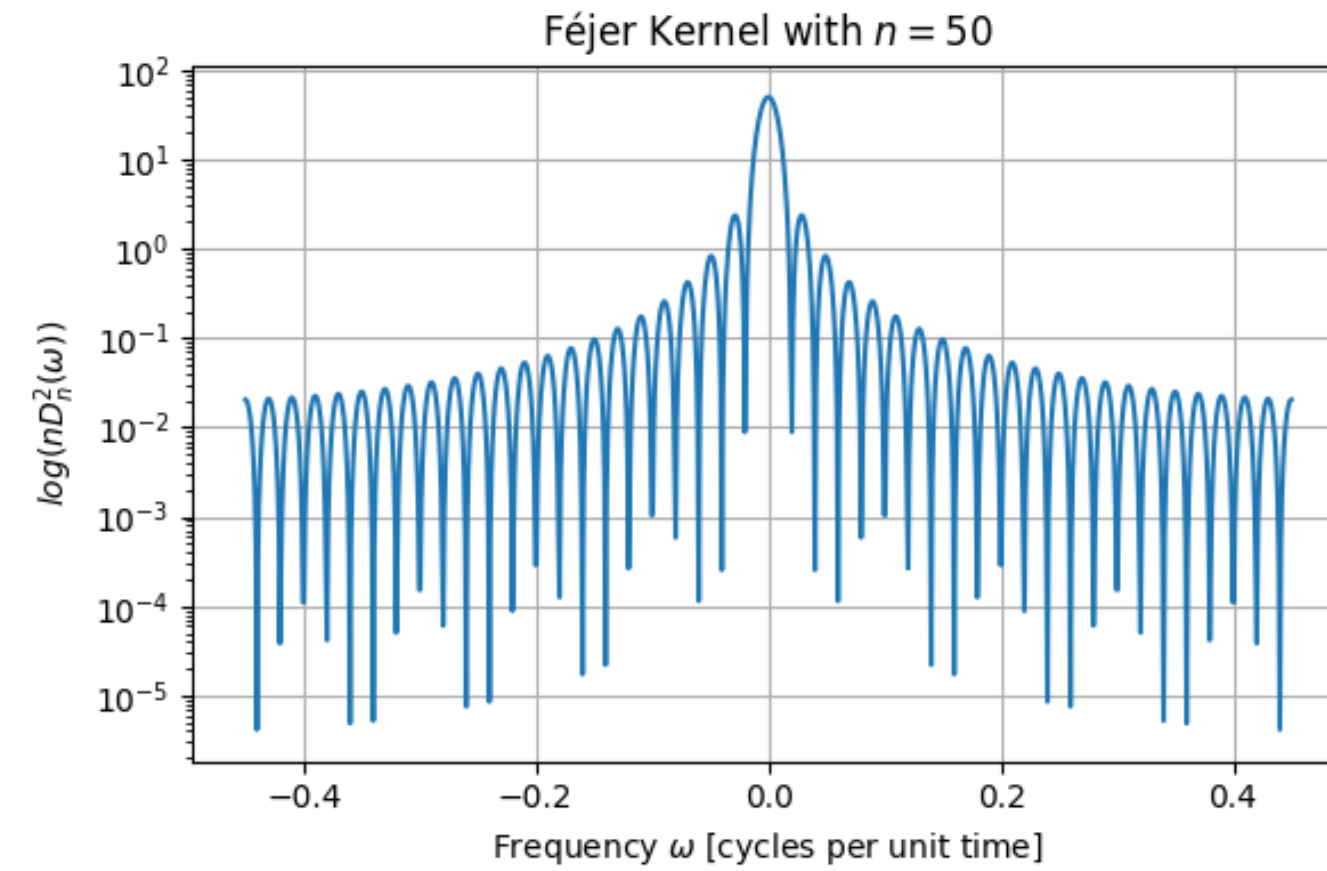
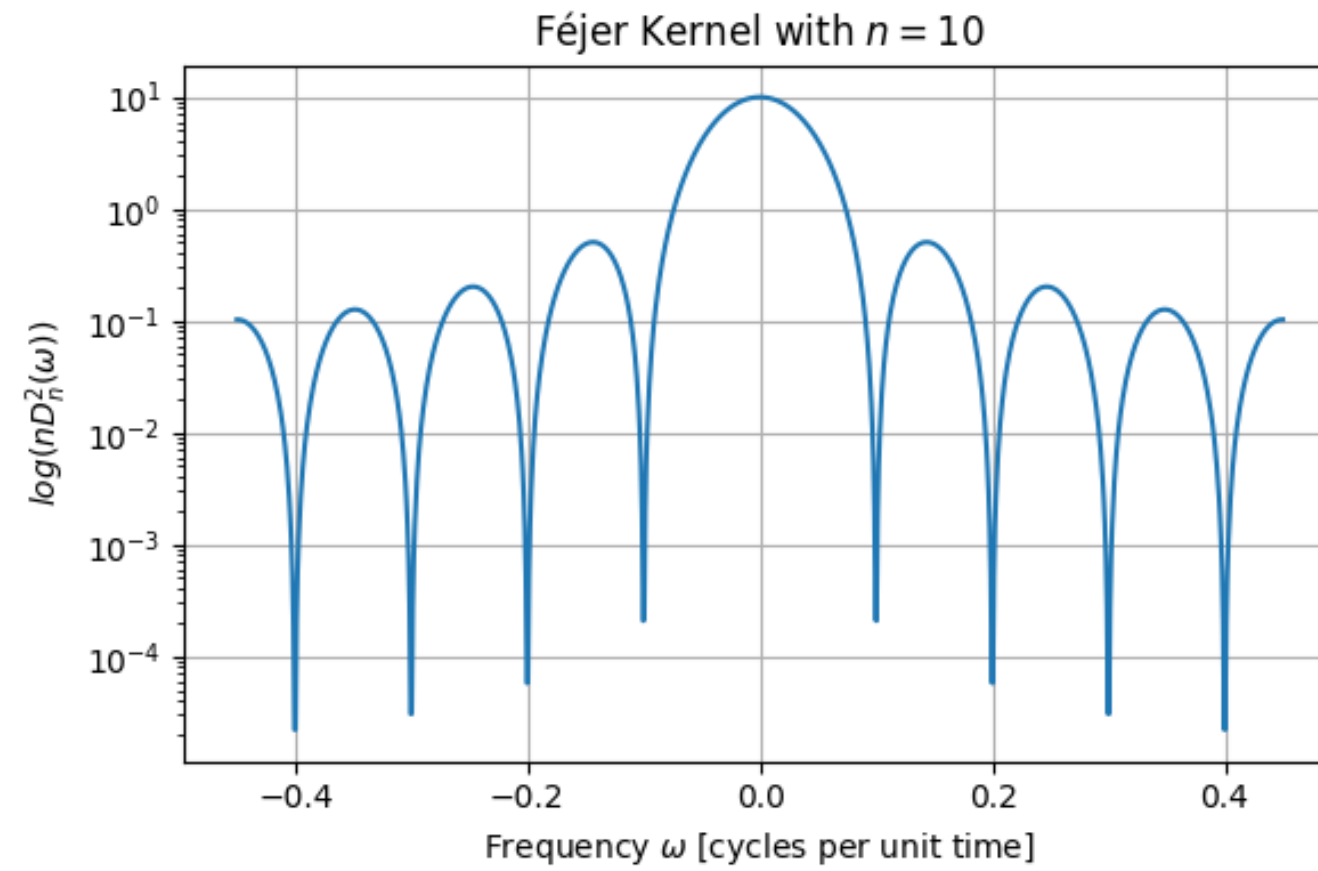
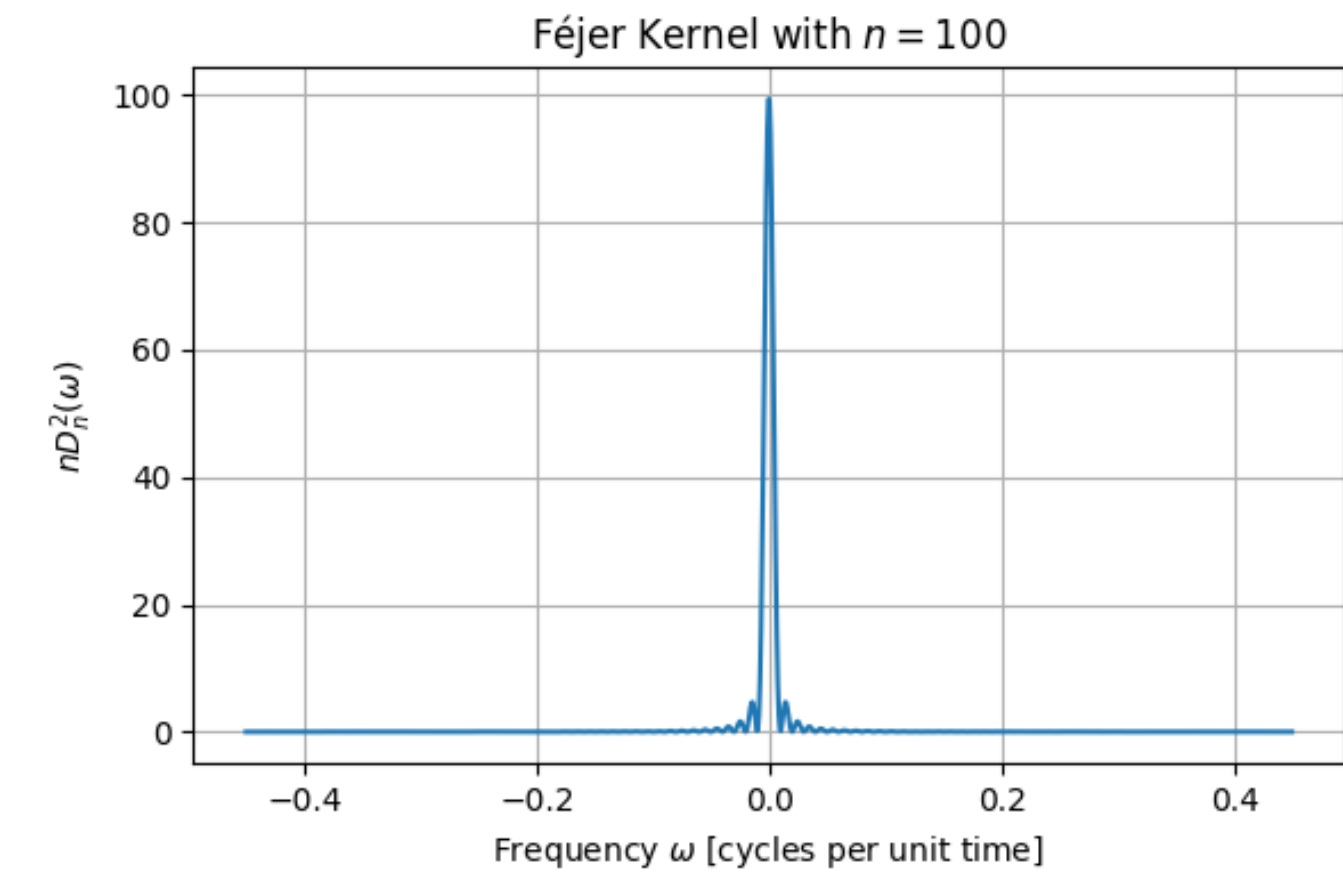
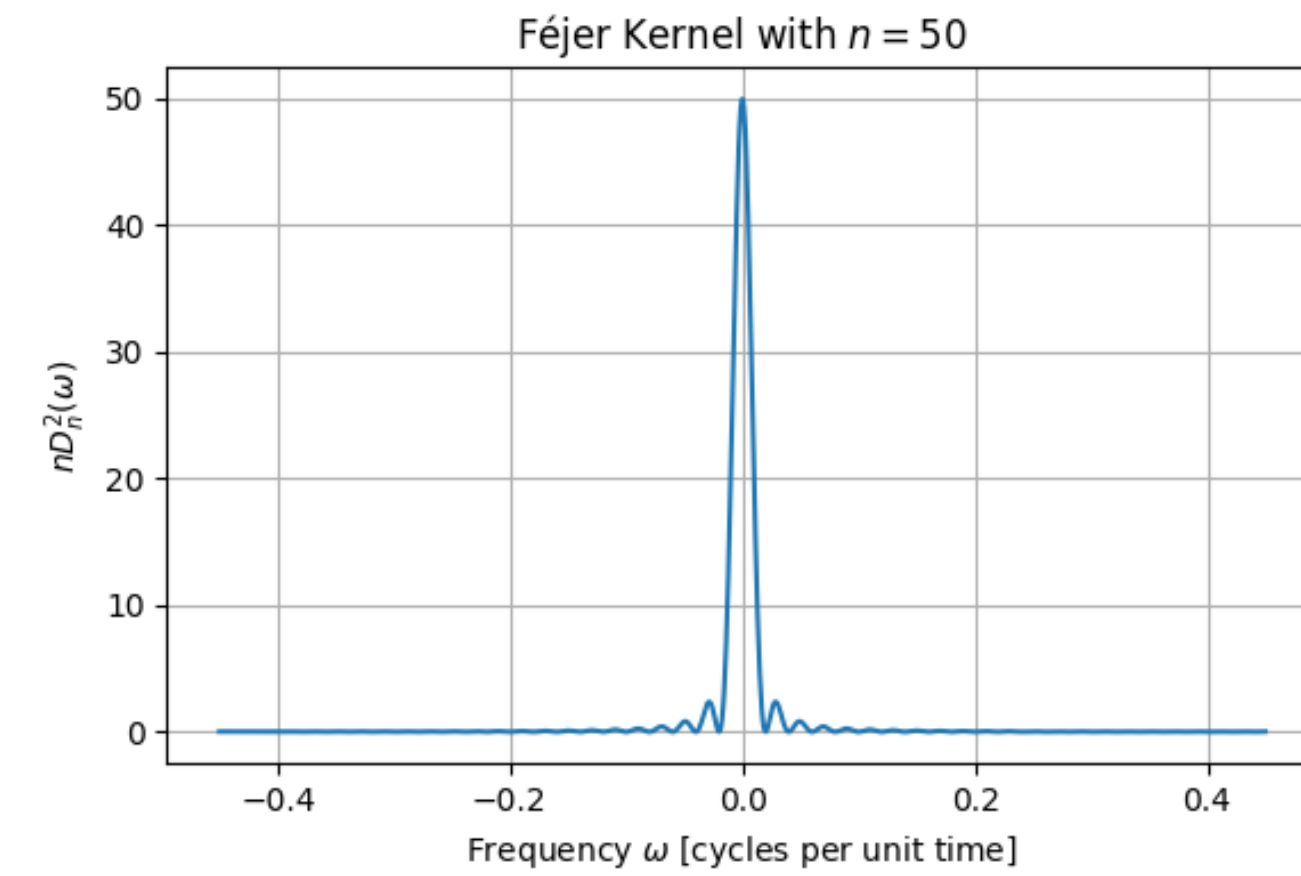
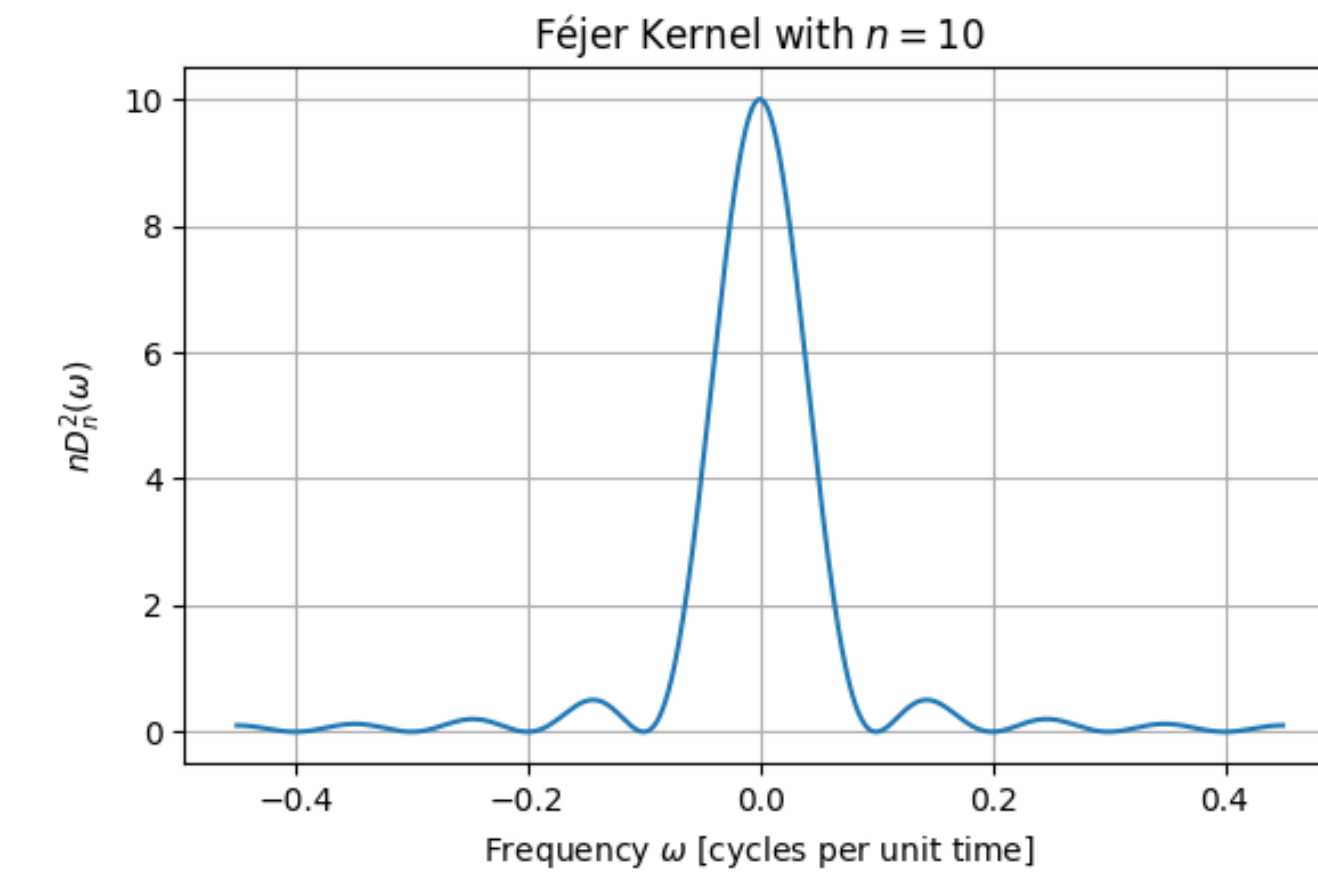
$$\hat{\gamma}(h) = \frac{1}{n} \sum_{i=0}^{n-|h|} (x_{i+|h|} - \hat{\mu})(x_i - \hat{\mu})$$

An **unbiased estimate** of $\gamma(h)$ is $\frac{n}{n-|h|} \hat{\gamma}(h)$. Thus, $E[\hat{\gamma}(h)] = \frac{n-|h|}{n} \gamma(h)$.

The **expected value** of the periodogram is

$$\begin{aligned} E[P(\omega_j)] &= \frac{1}{n} \sum_{h=-(n-1)}^{n-1} \frac{n-|h|}{n} \gamma(h) e^{-2\pi i \omega_j h} \\ &= \frac{1}{n} \int_{-1/2}^{1/2} nD_n^2(\omega_j - \omega) f(\omega) d\omega \end{aligned}$$

where $nD_n^2(\omega) = \frac{\sin^2(n\pi\omega)}{n \sin^2(\pi\omega)}$ is known as the **Fejér's kernel**.

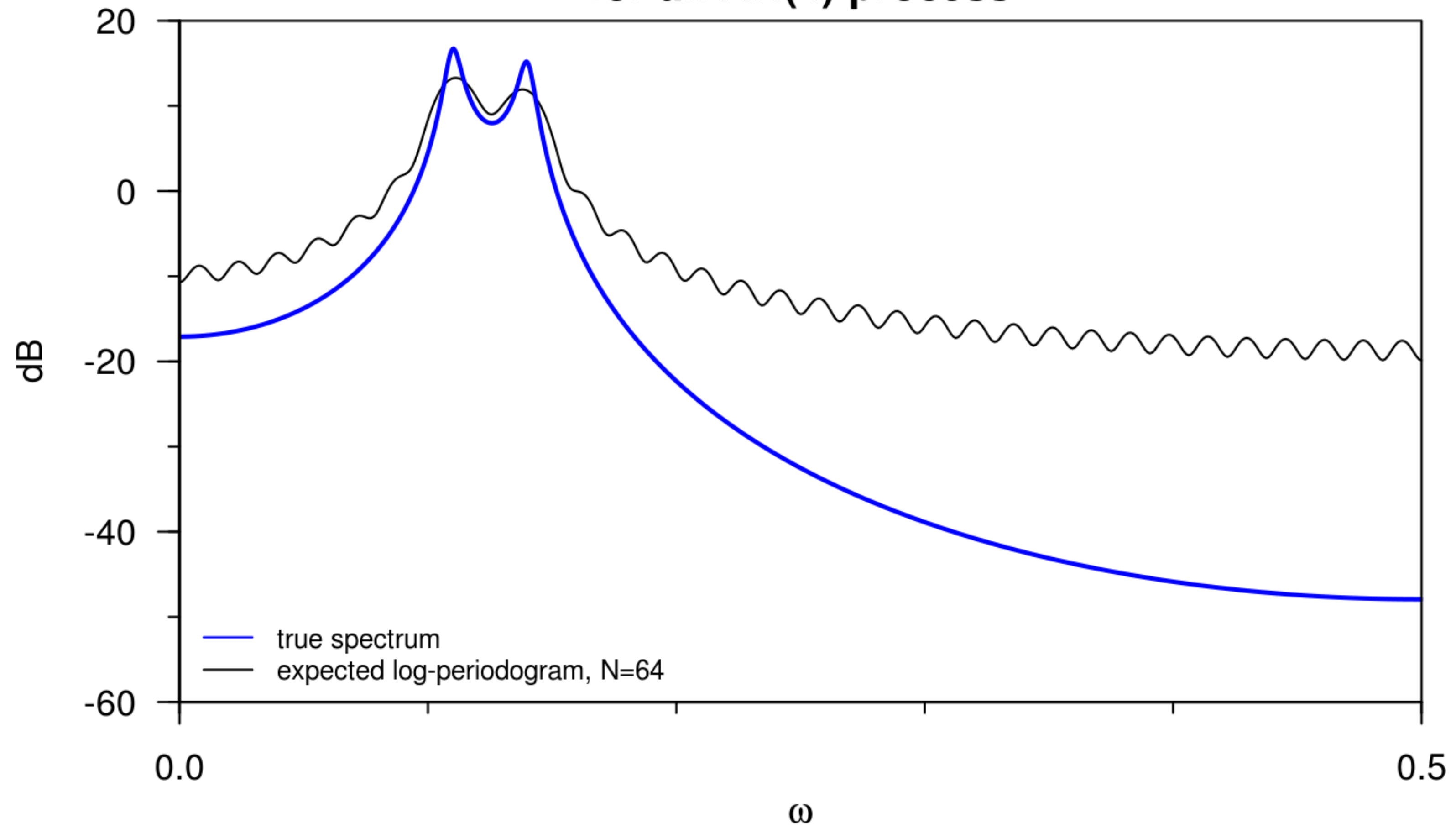


The Fejér kernel behaves asymptotically like the Dirac delta function (unit impulse) $\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$.

Thus, the bias depends mostly on the **dynamic range DR** of the spectral density

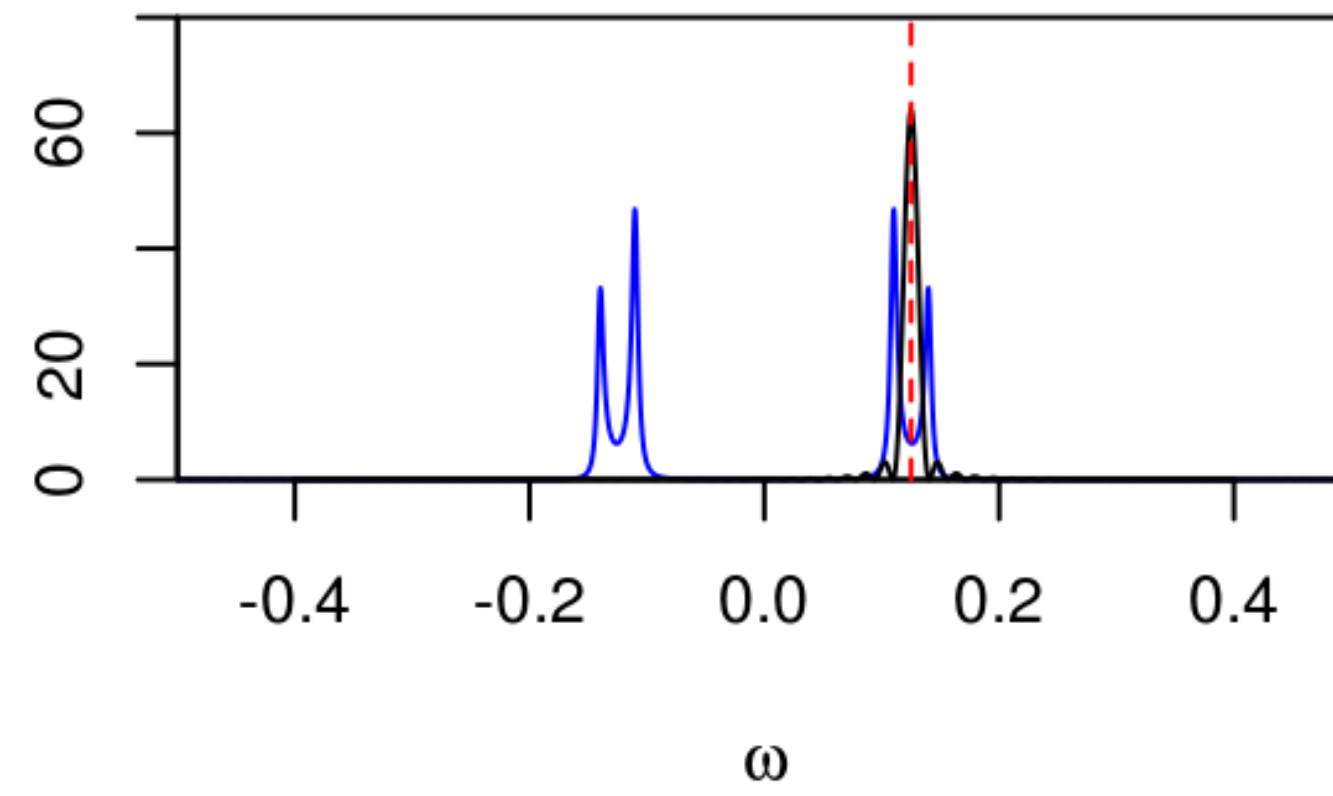
$$DR = \frac{\max f(\omega)}{\min f(\omega)}$$

Theoretical spectrum and expected log-periodogram for an AR(4) process

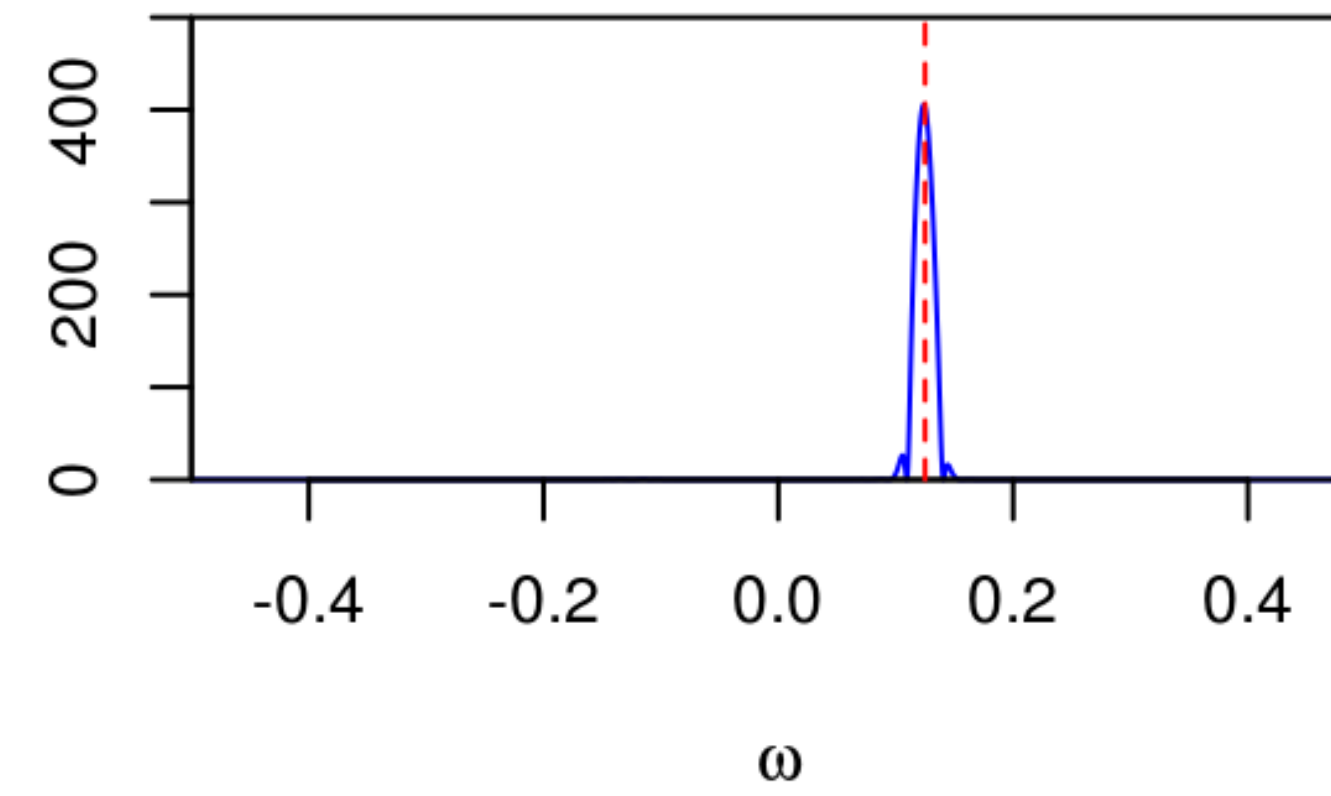


Léo Belzile

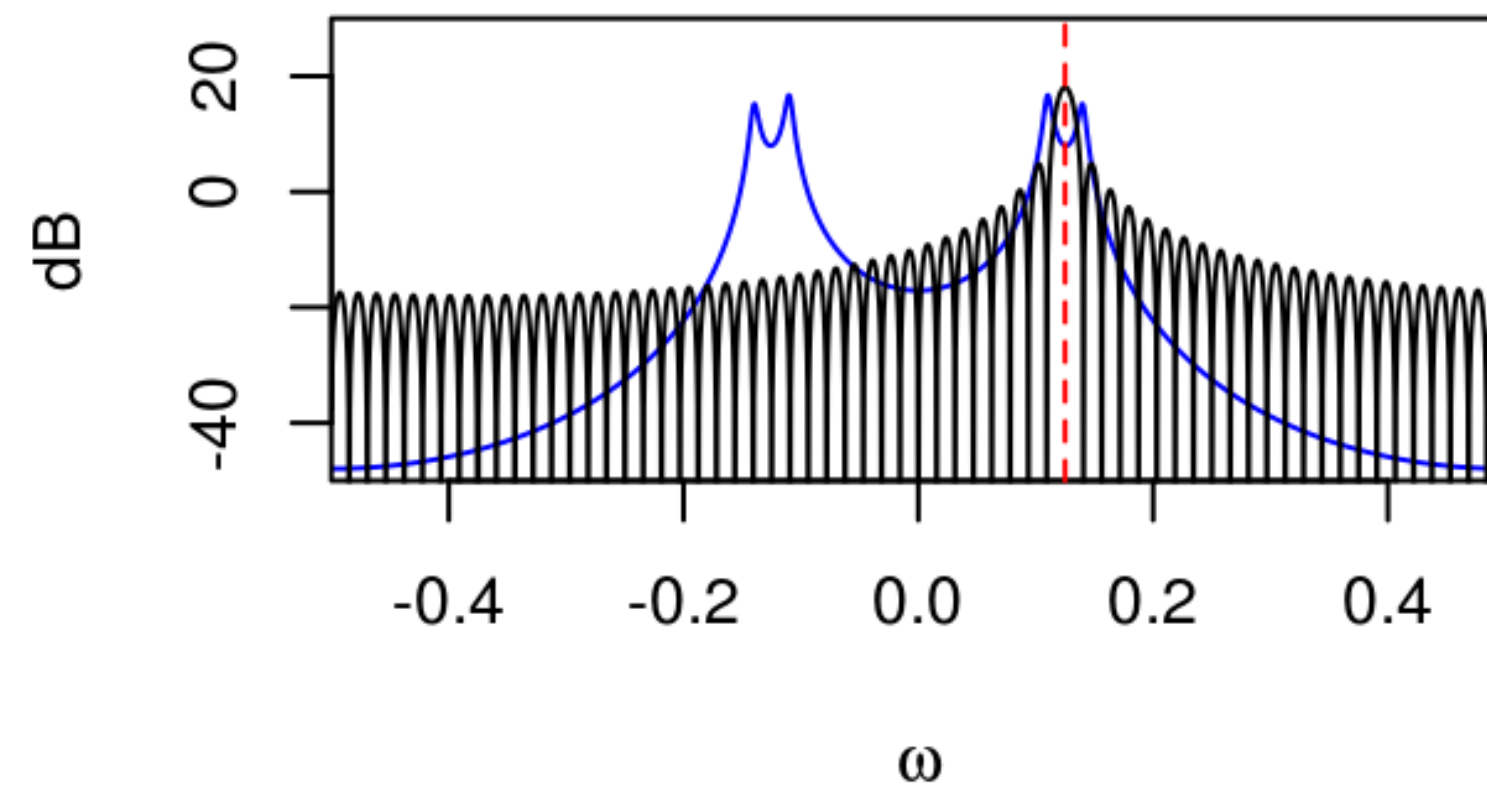
Fejer's kernel and AR(4) spectrum



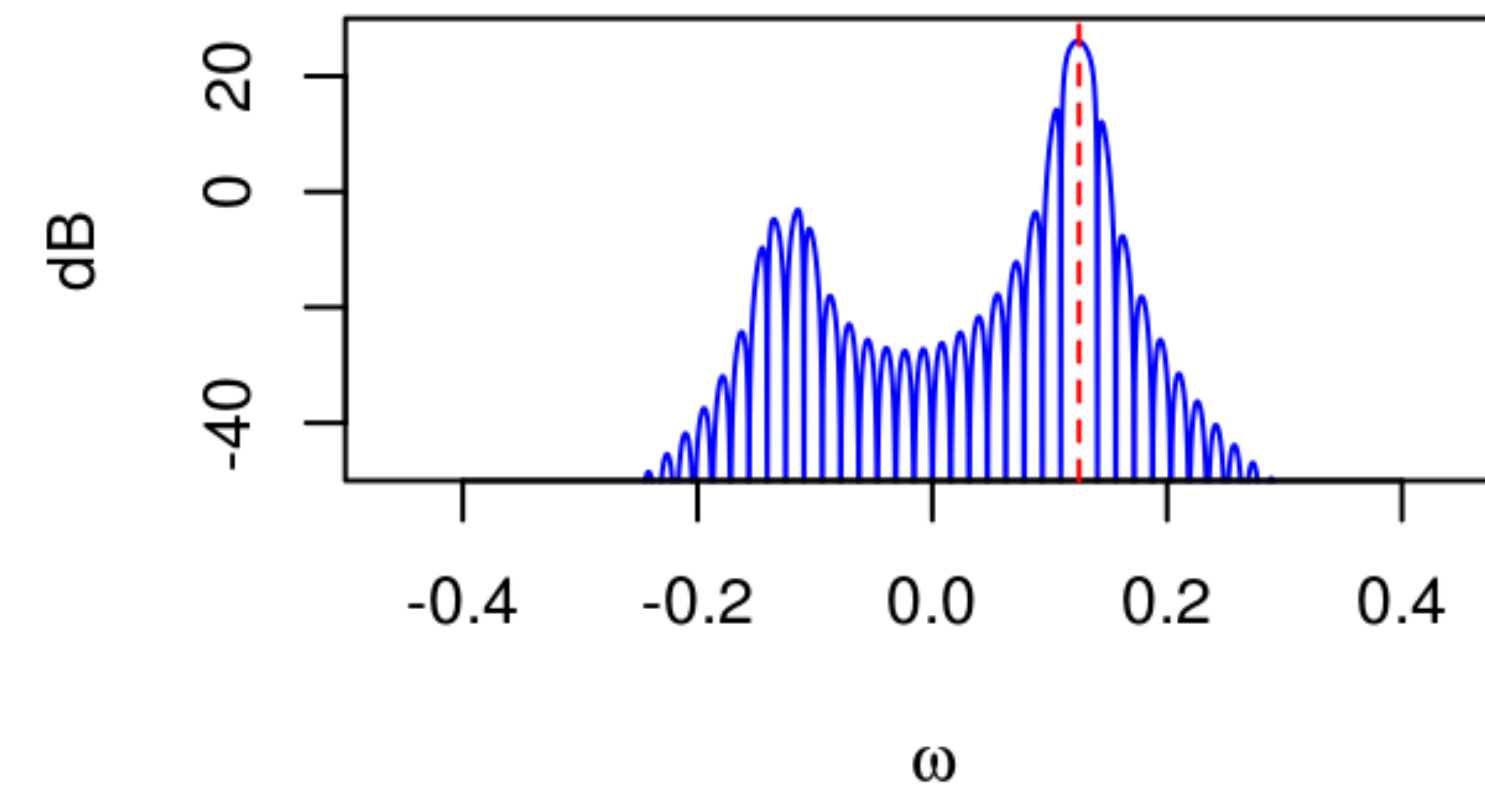
Fejer's kernel times AR(4) spectrum



Fejer's kernel and AR(4) spectrum

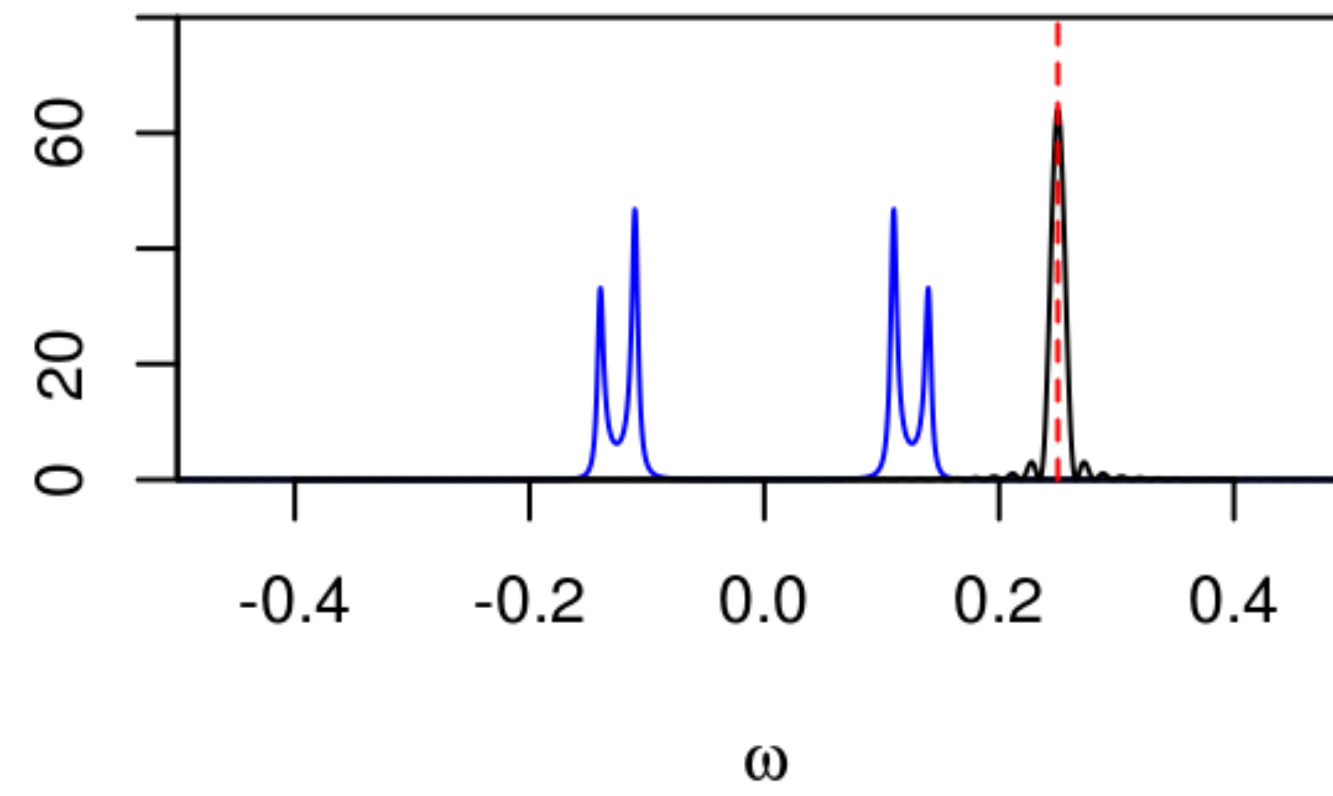


Fejer's kernel times AR(4) spectrum

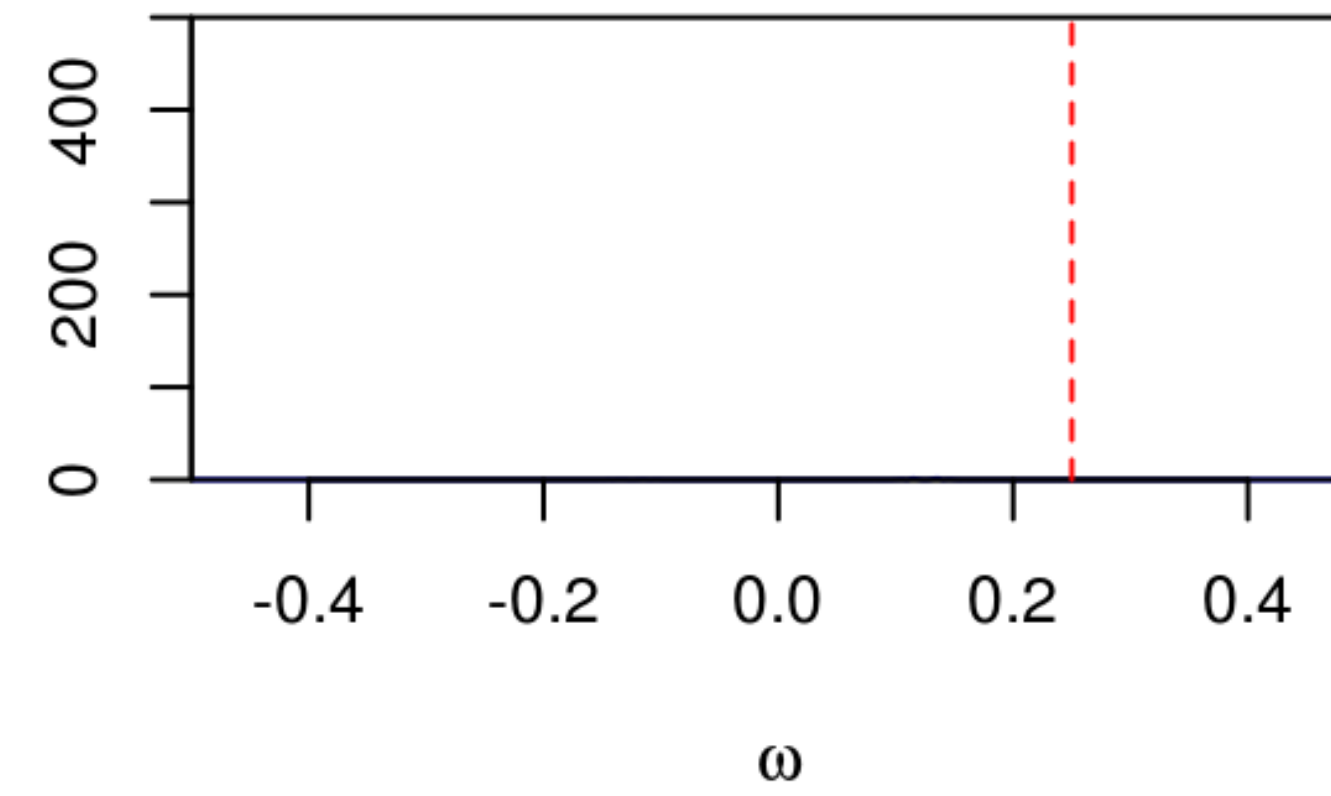


Léo Belzile

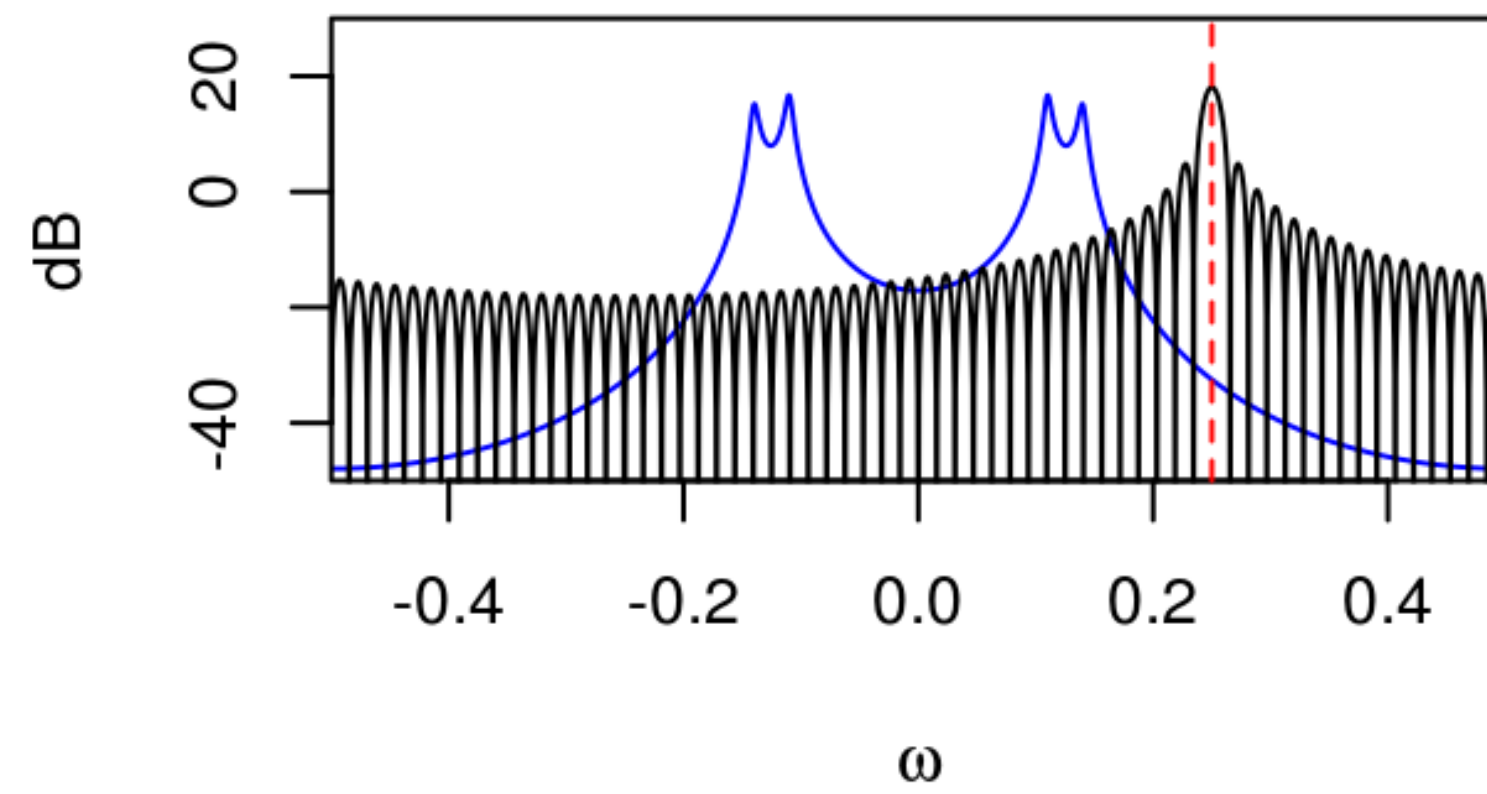
Fejer's kernel and AR(4) spectrum



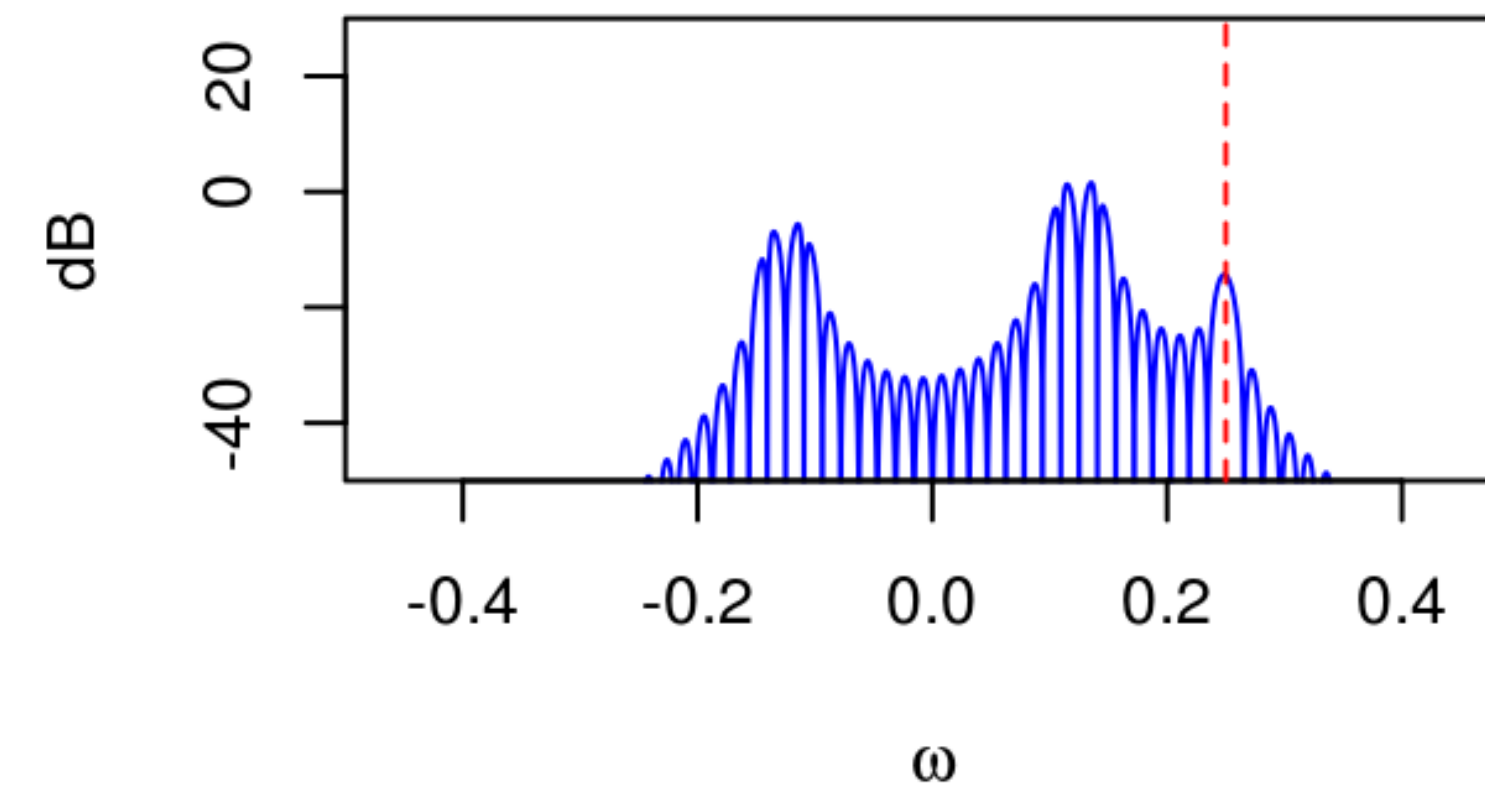
Fejer's kernel times AR(4) spectrum



Fejer's kernel and AR(4) spectrum



Fejer's kernel times AR(4) spectrum



Léo Belzile

Tapering (1)

Tapering aims to **reduce bias** by reducing the magnitude of the Fejér kernel's sidelobes.

Given a process X_t with spectral density $f_X(\omega)$, the **tapered series** with taper h_t is defined as $Y_t = h_t X_t$.

Provided X_t is **zero-mean and stationary**, the expected periodogram of Y_t is

$$E[P_Y(\omega_j)] = \frac{1}{n} \int_{-1/2}^{1/2} W_n(\omega_j - \omega) f_X(\omega) d\omega$$

where $W_n(\omega) = |H_n(\omega)|^2$ is called the **spectral window** and $H_n(\omega)$ is the DFT of h_t .

The **boxcar** taper is defined as $h_t = \begin{cases} 1 & 1 \leq t \leq n \\ 0 & \text{otherwise} \end{cases}$, it is equivalent to **no tapering**.

- In this case, $W_n(\omega)$ is the Fejér kernel.

Tapering (2)

To reduce bias, $W_n(\omega)$ should have **smaller sidelobes** than the Fejér kernel.

This typically comes with larger main lobes, which **spread** the main peaks.

- Introduces a new source of bias.
- Increases variance of the estimate due to discarded information.

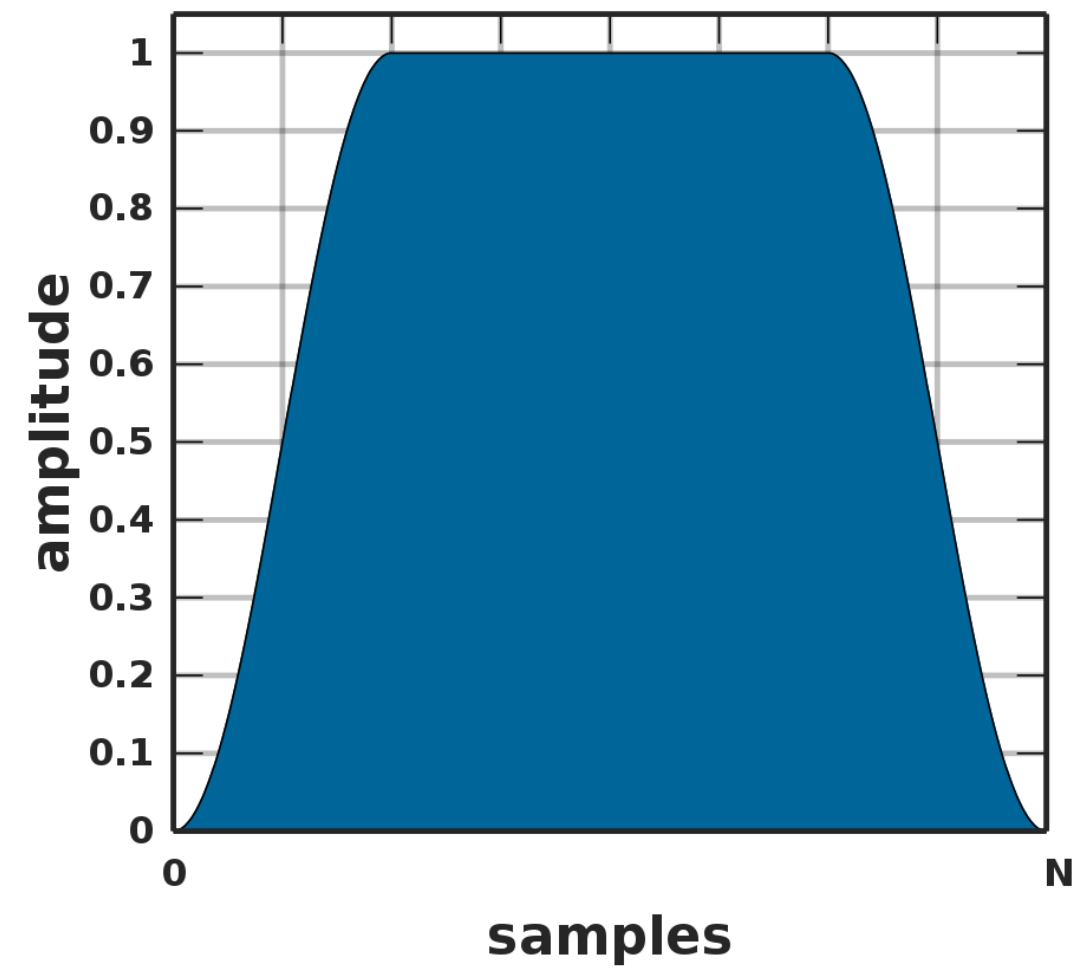
It is preferable to leave most of the data **unchanged** and just **taper the ends**.

- Tapers generally have a symmetric shape that enhances the center of the data relative to the extremities.

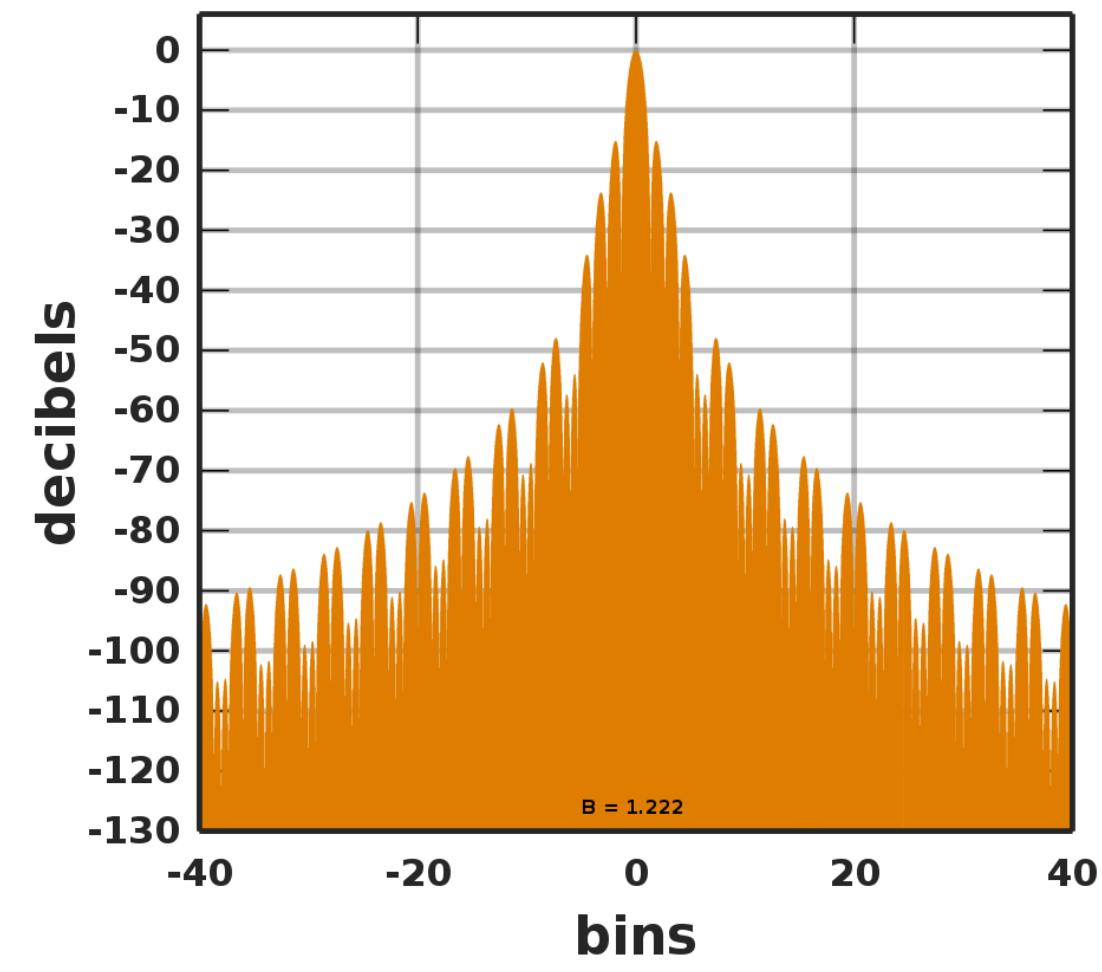
For example, the **Tukey window** tapers a fraction α of the TS start and end values.

$$h_t = \begin{cases} \frac{1}{2} \left(1 - \cos \left(\frac{2\pi t}{\alpha p} \right) \right), & 0 \leq t < \alpha n/2 \\ 1, & \alpha n/2 \leq t \leq n/2 \\ h_{n-t}, & 0 \leq t \leq n/2 \end{cases}$$

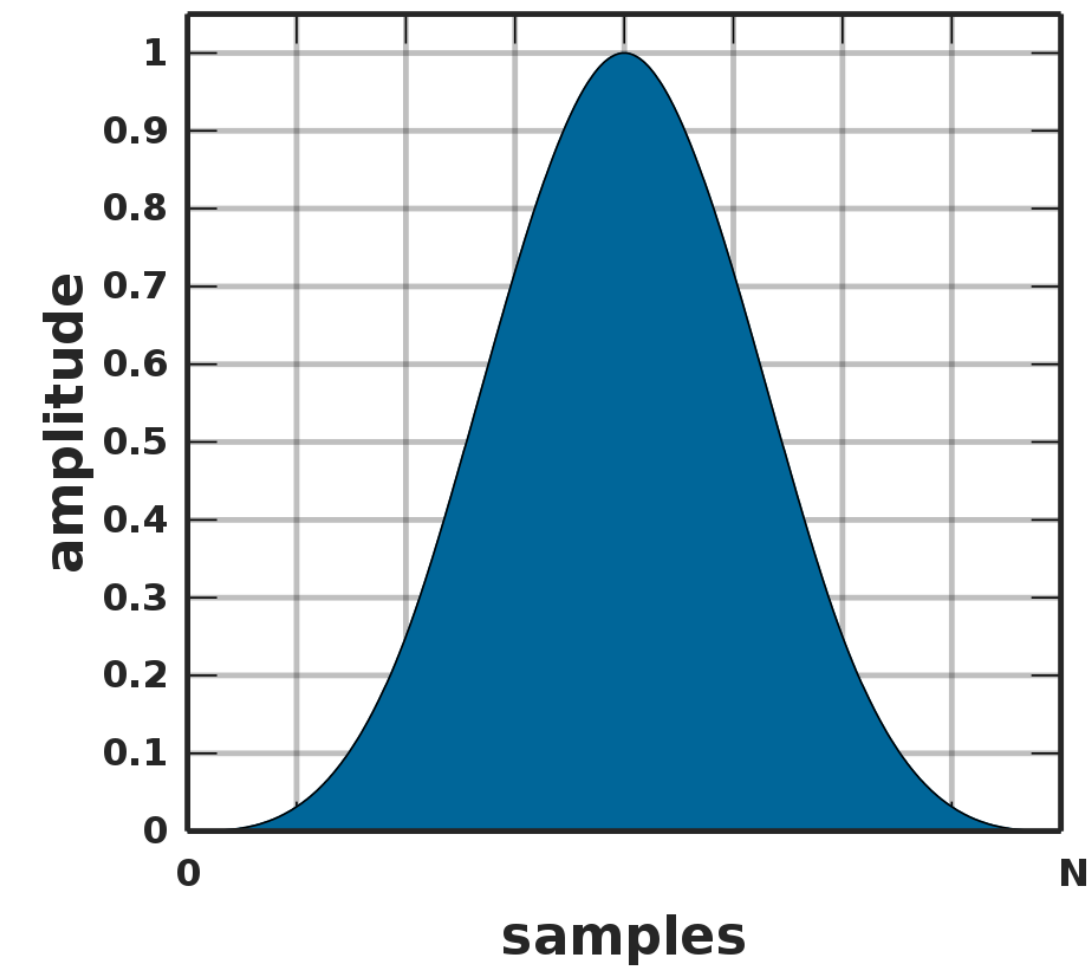
Tukey window ($\alpha = 0.5$)



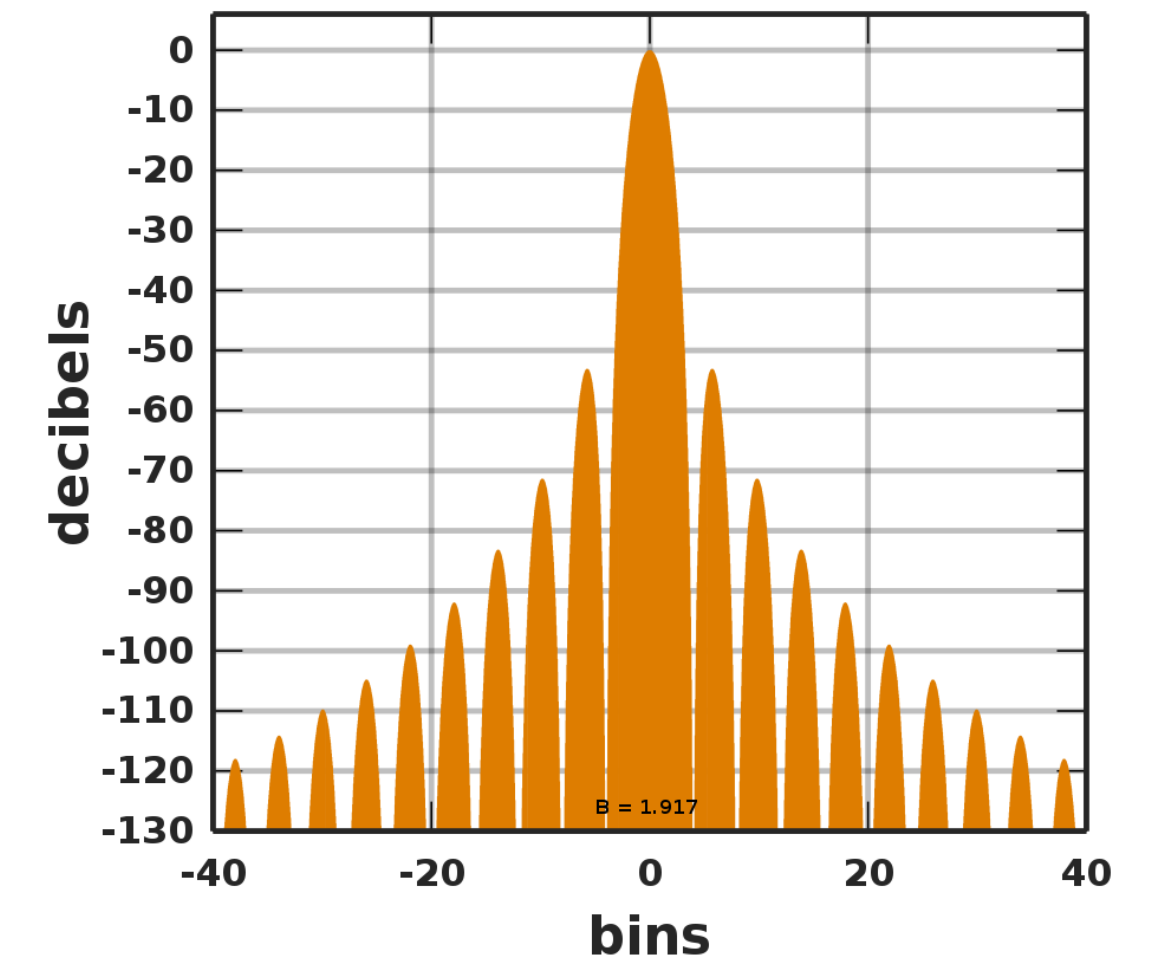
Fourier transform



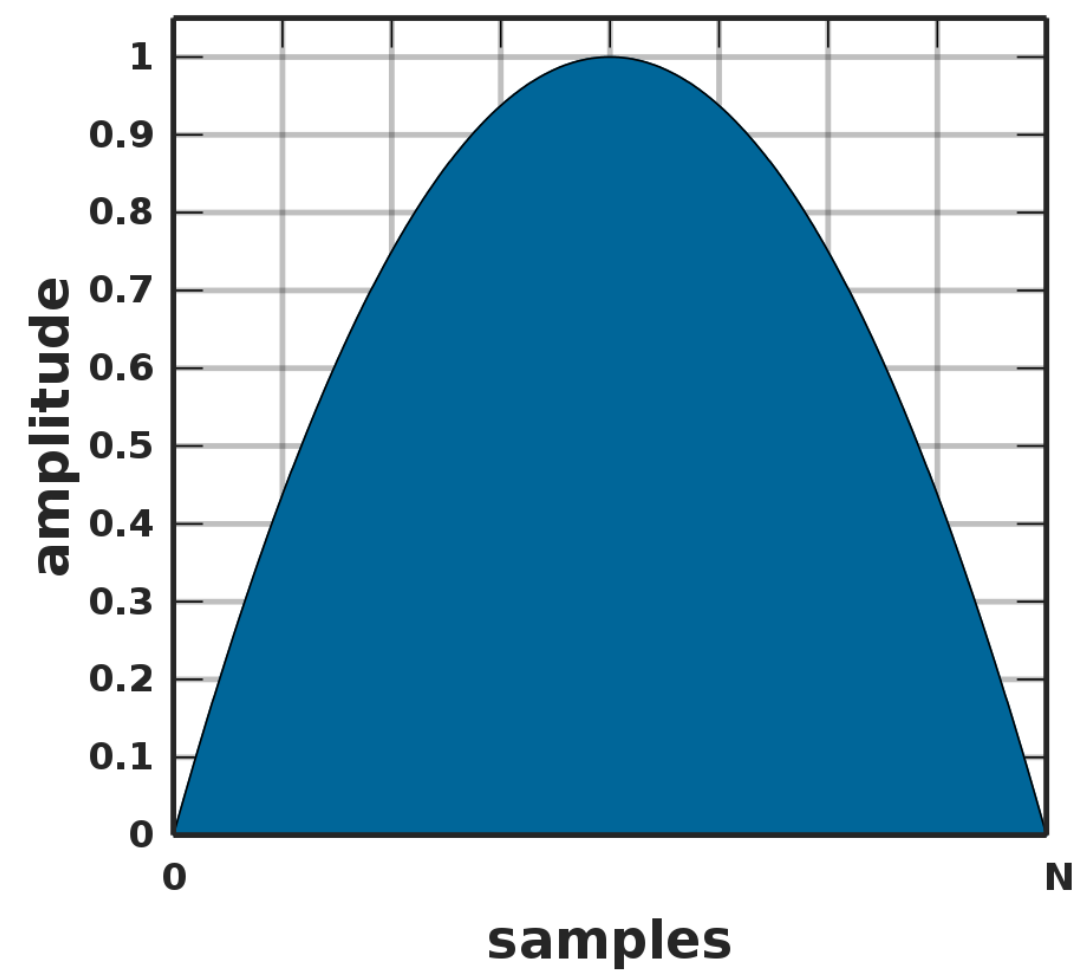
Parzen window



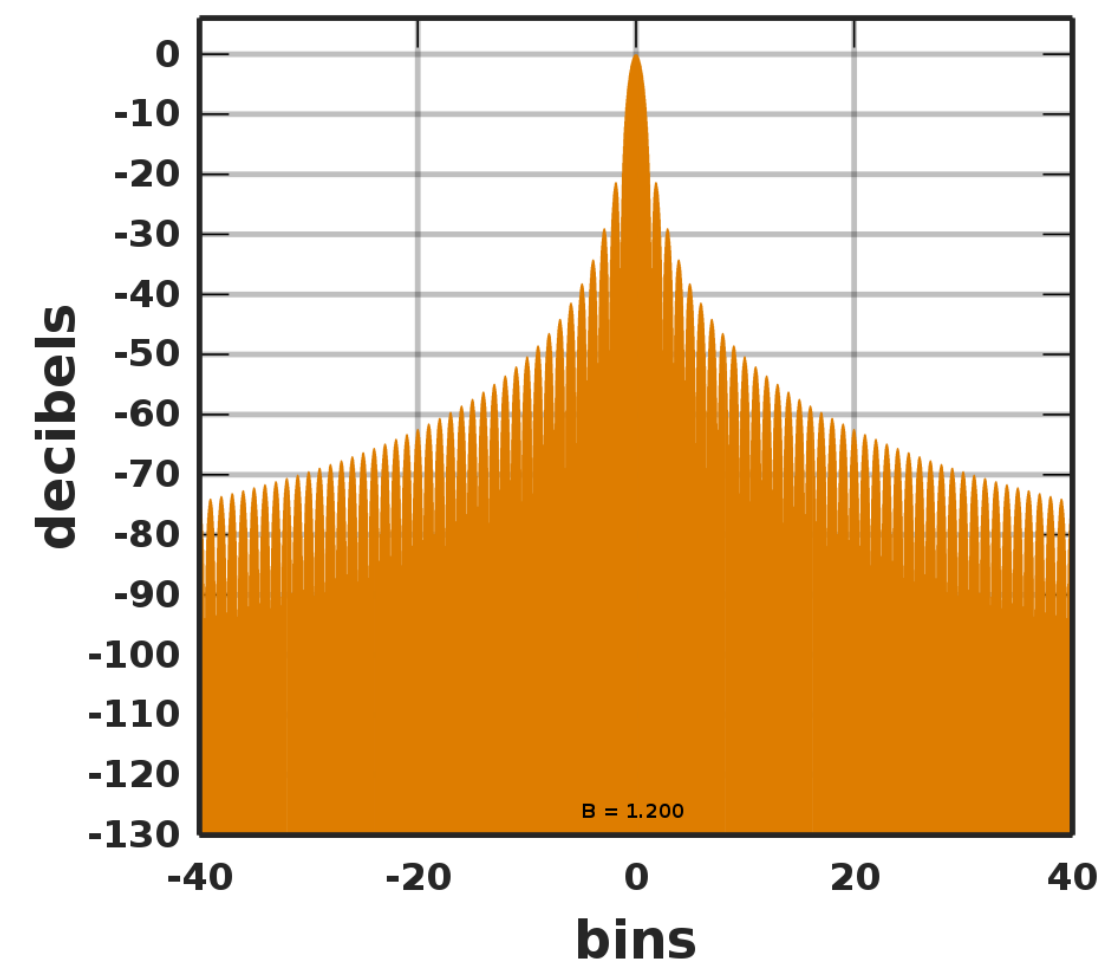
Fourier transform



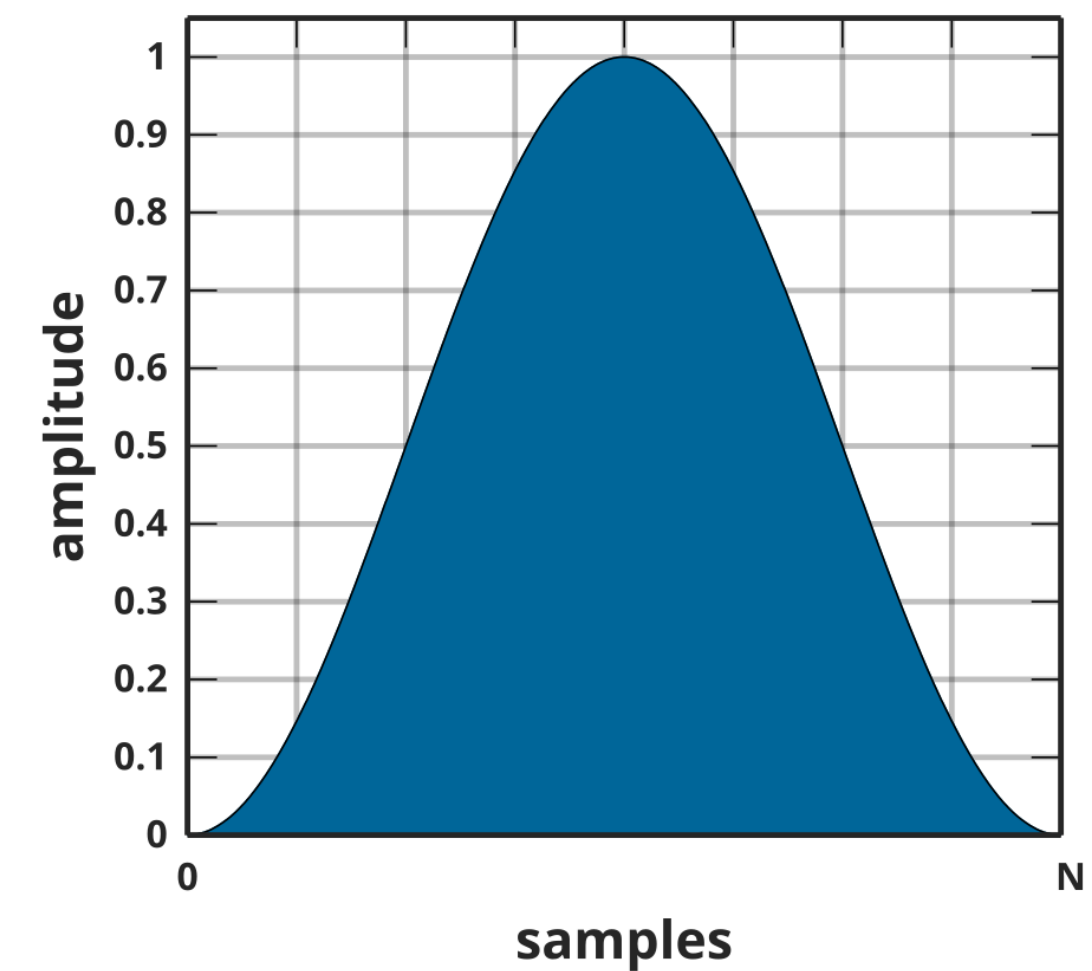
Welch window



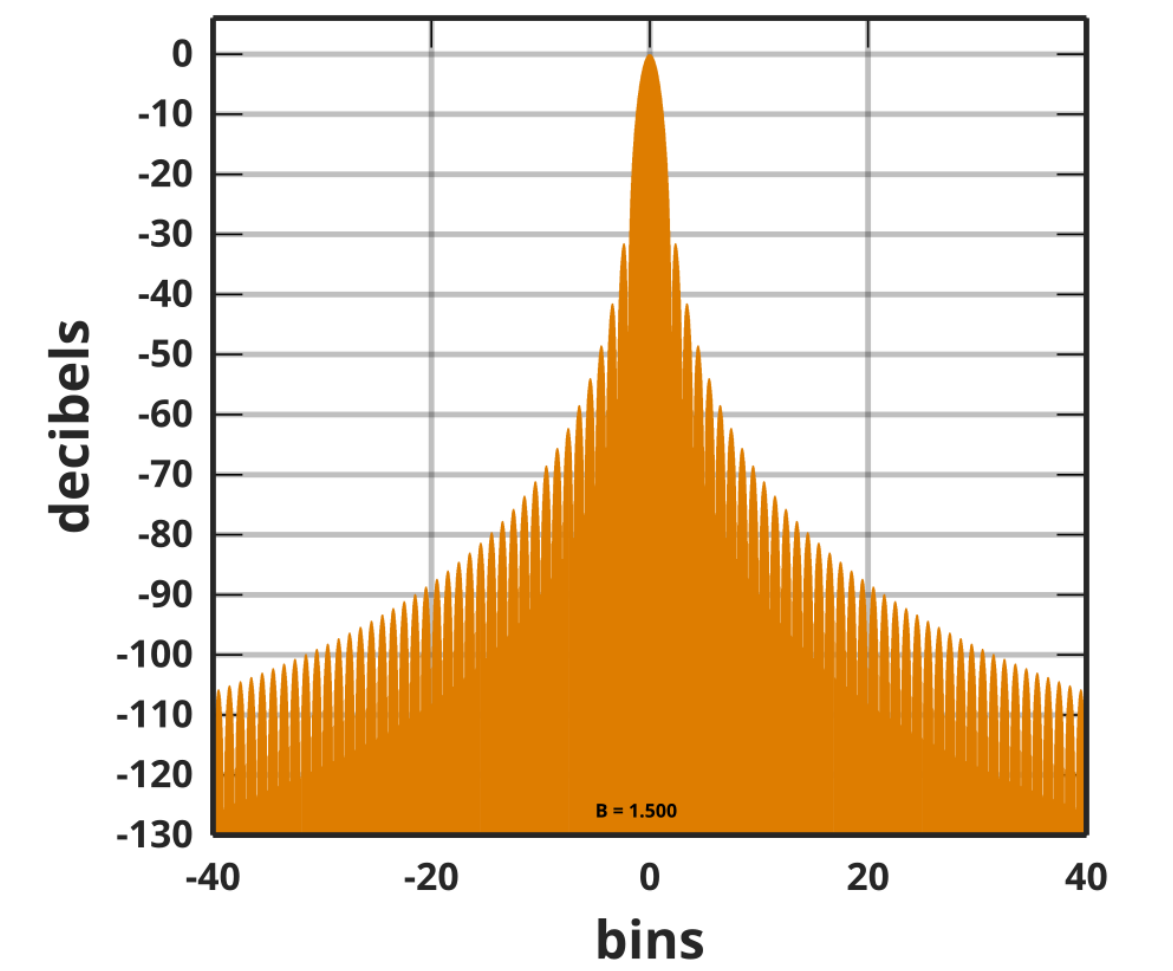
Fourier transform

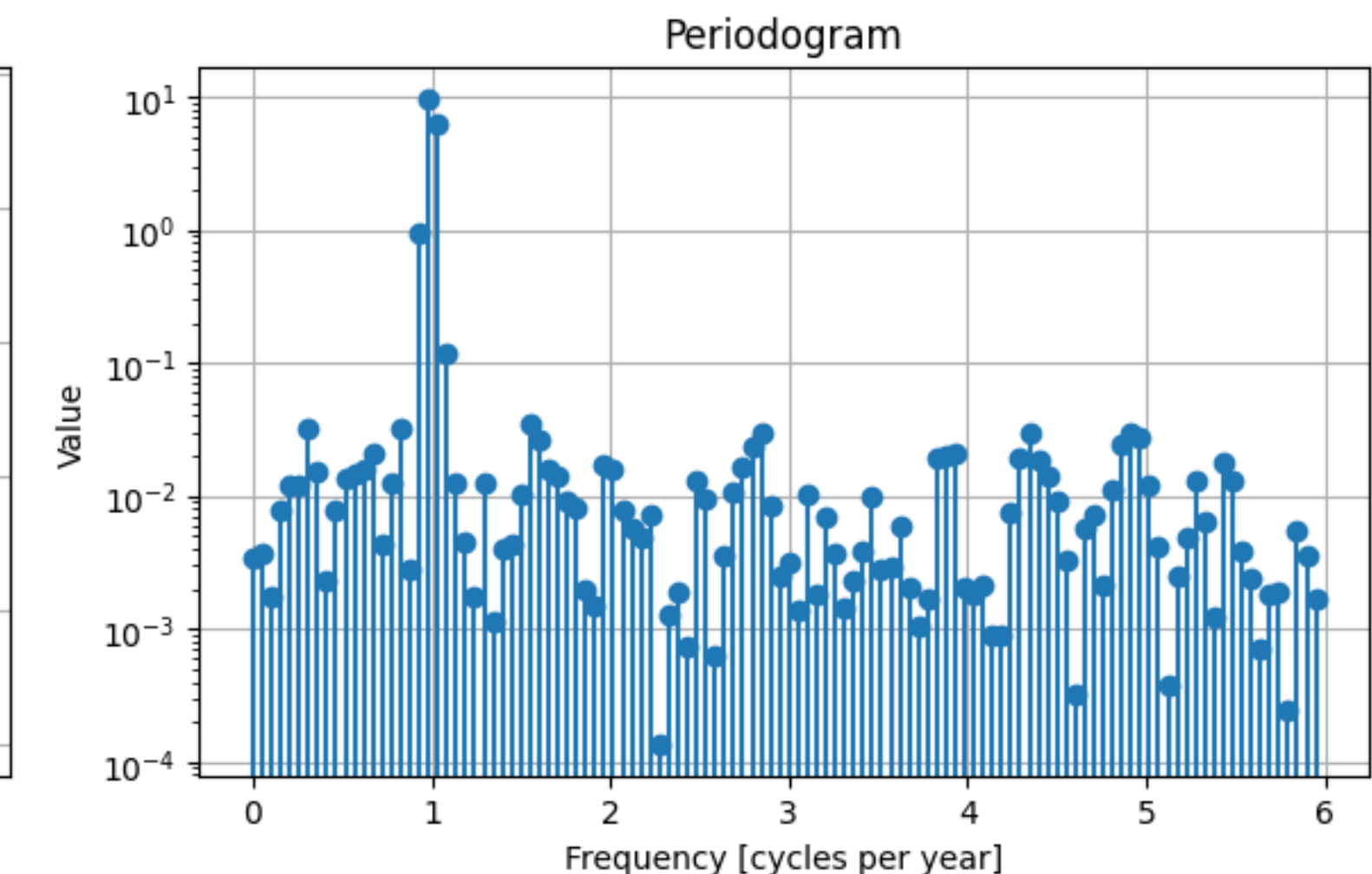
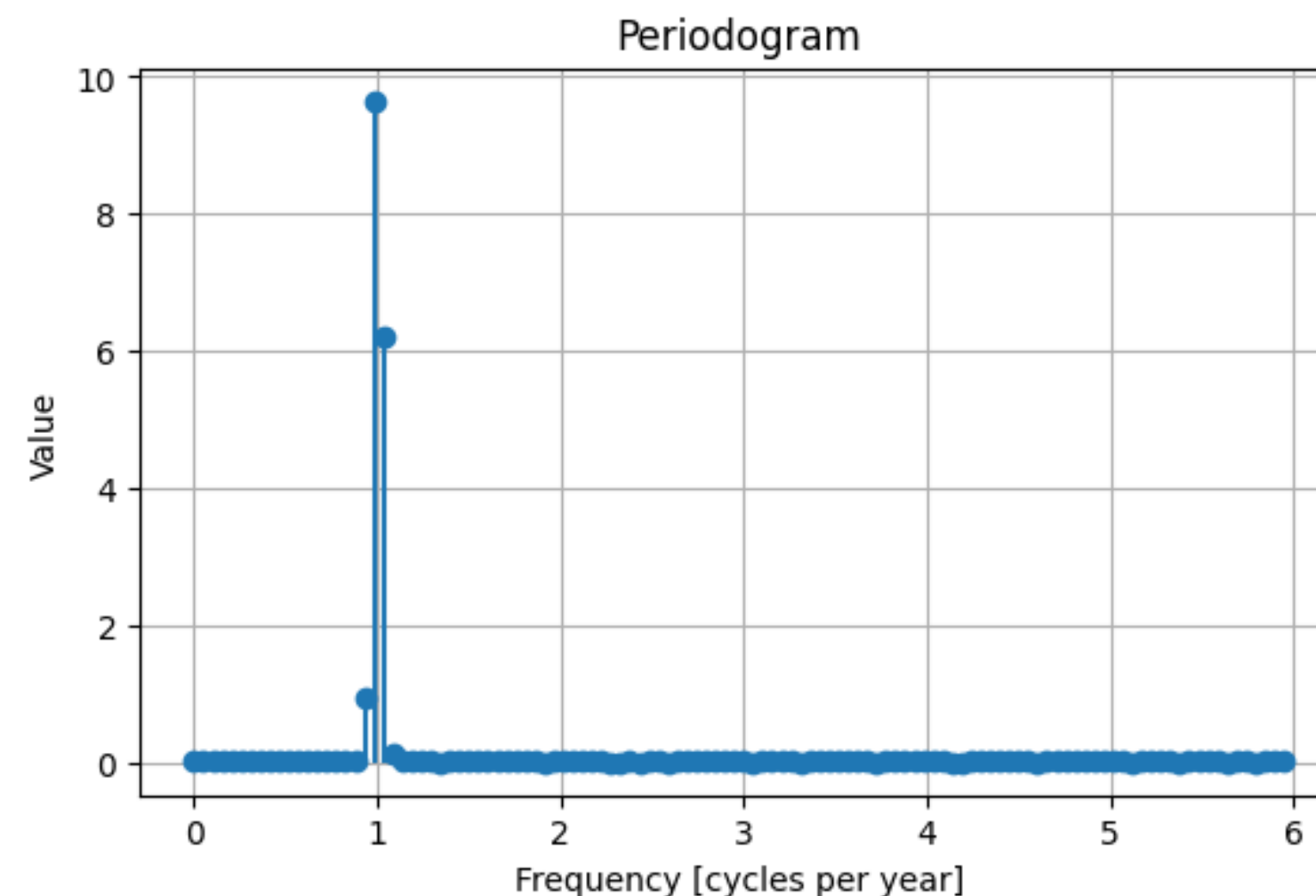
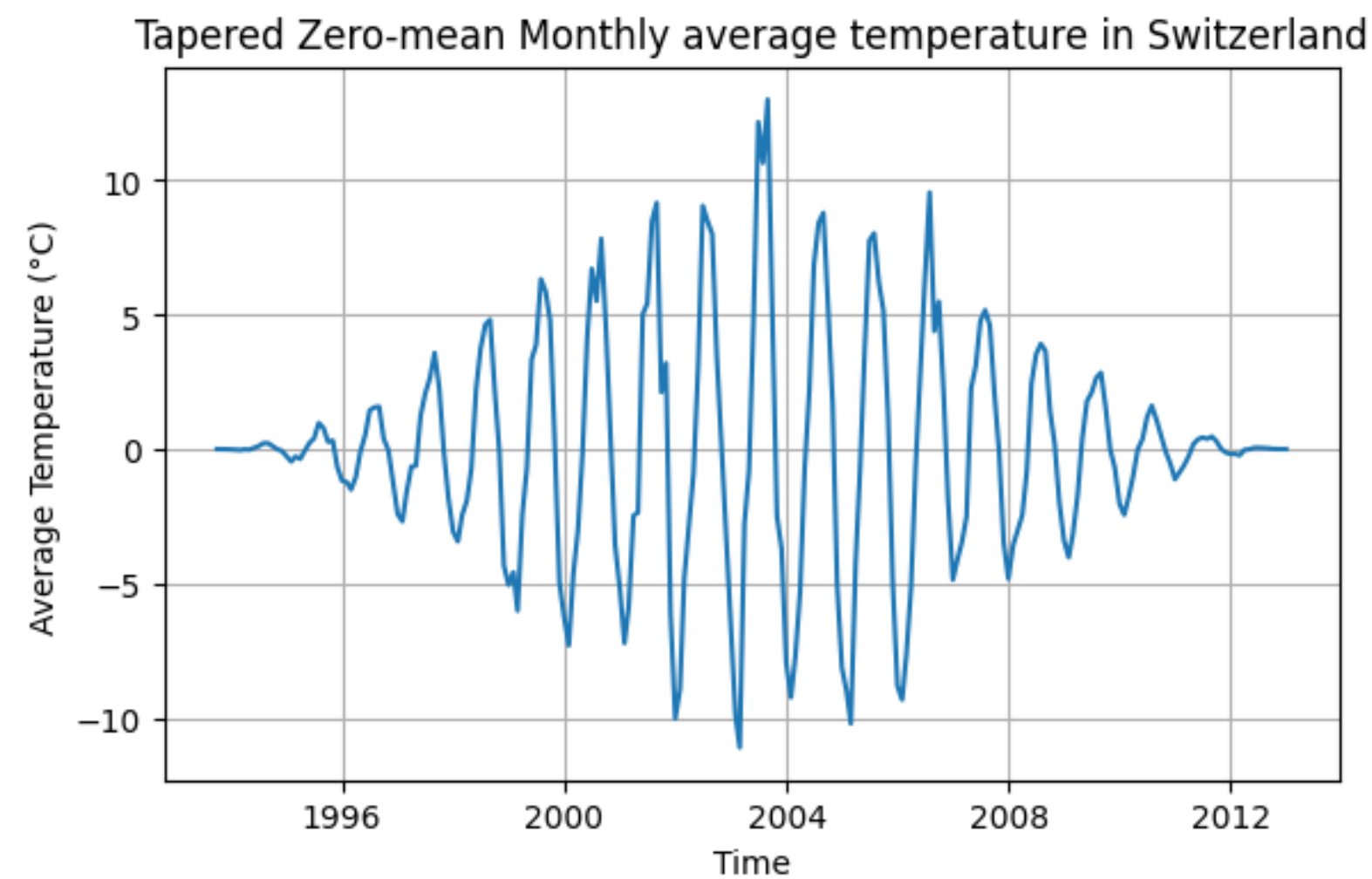
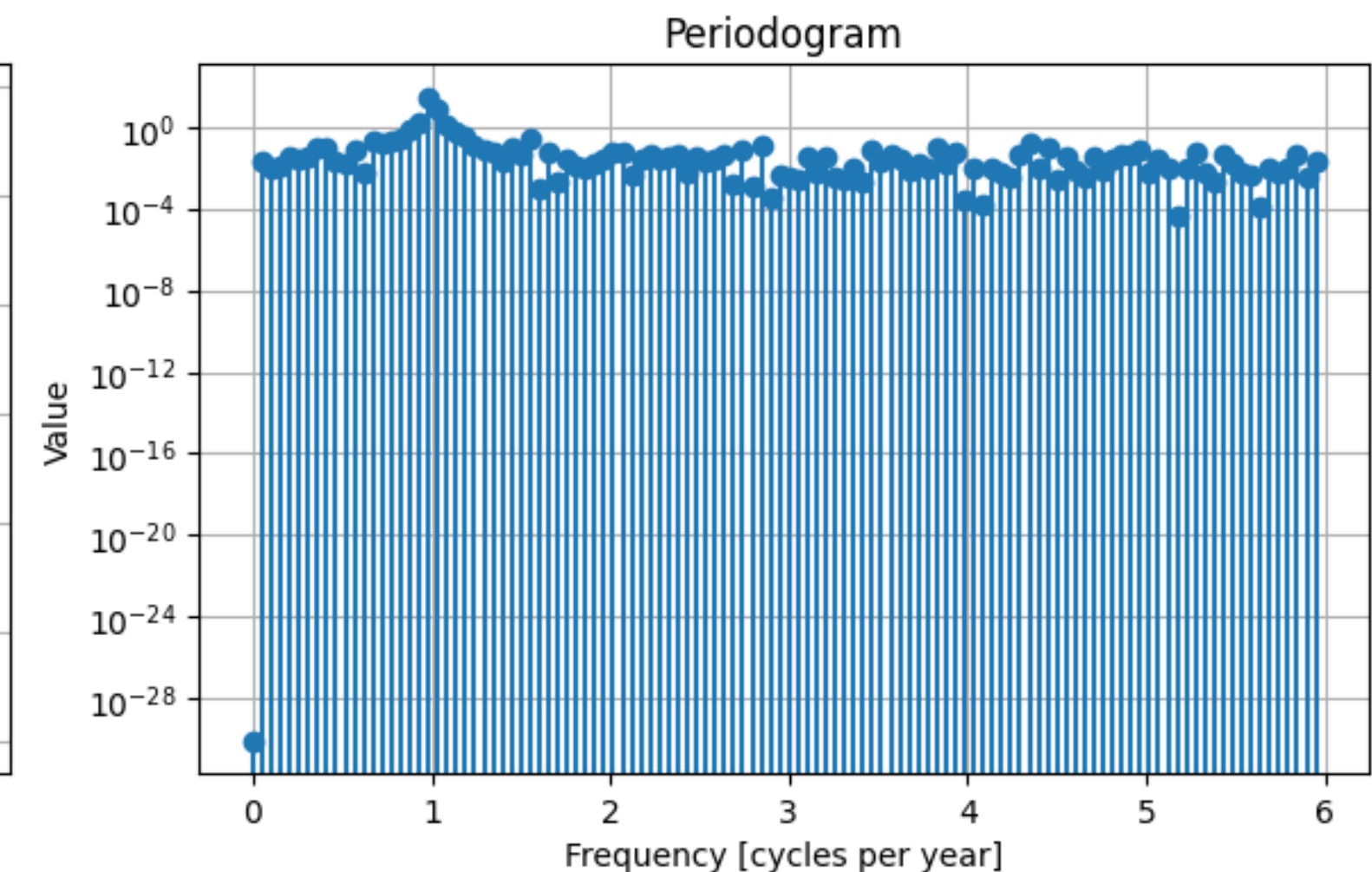
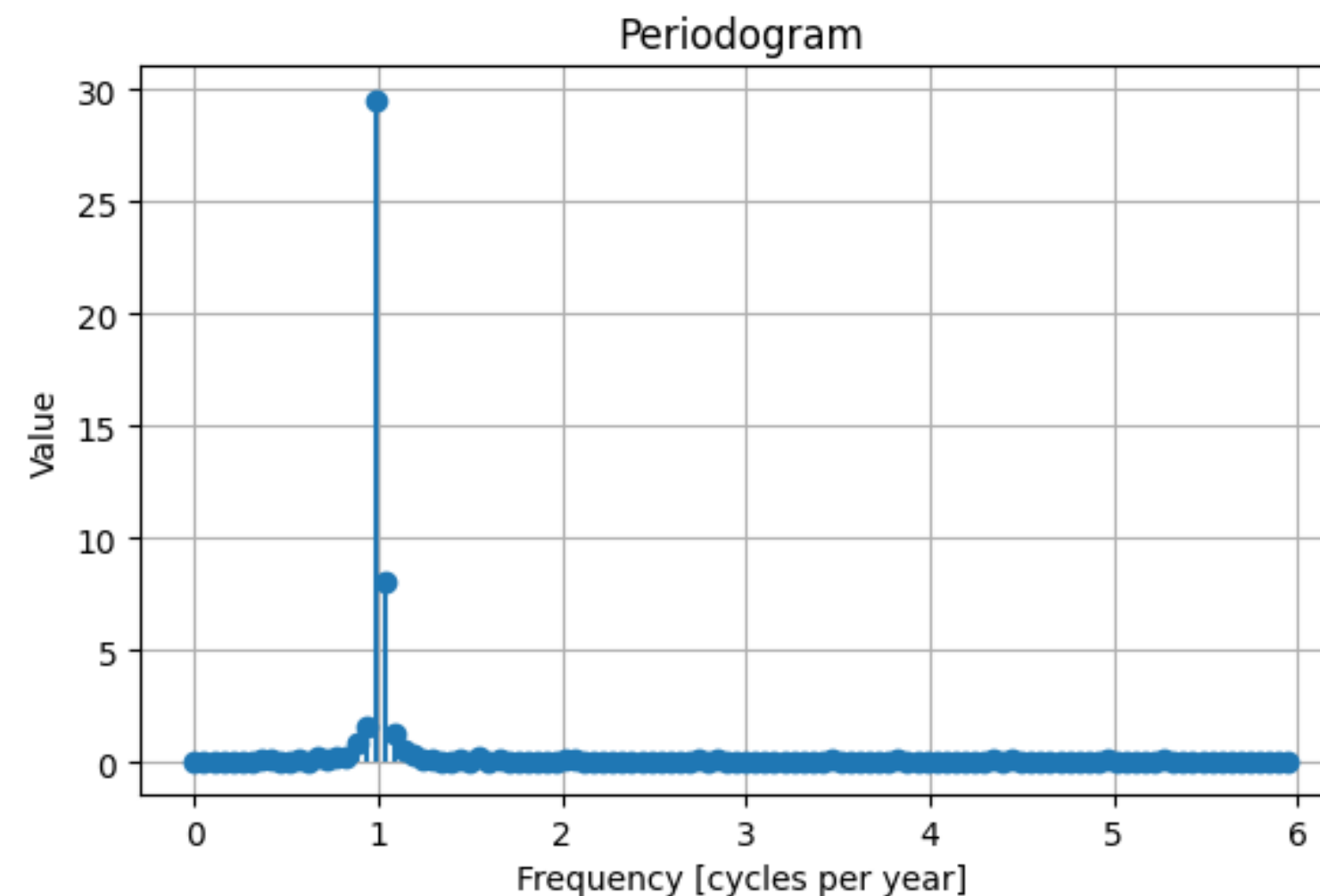
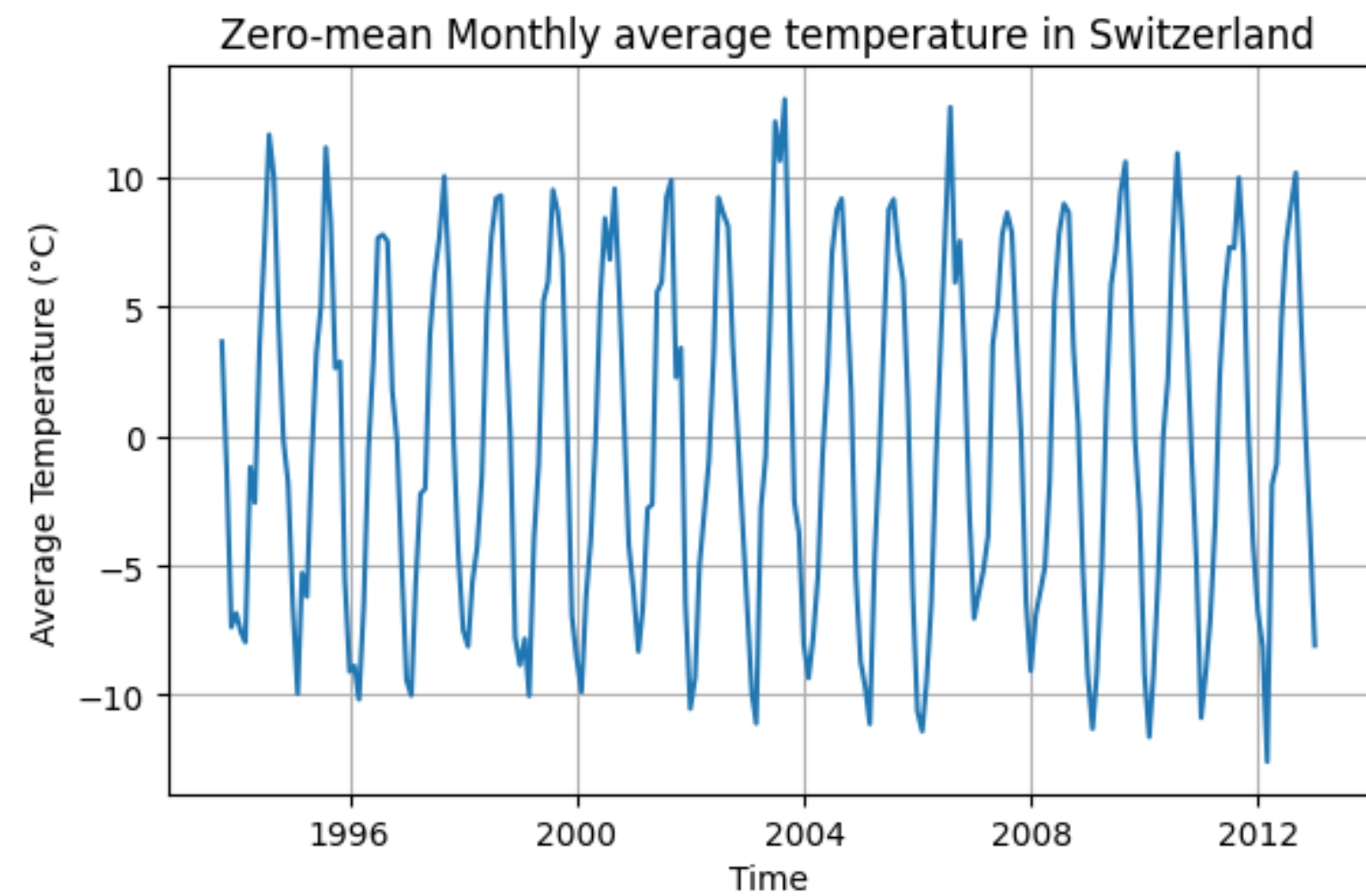


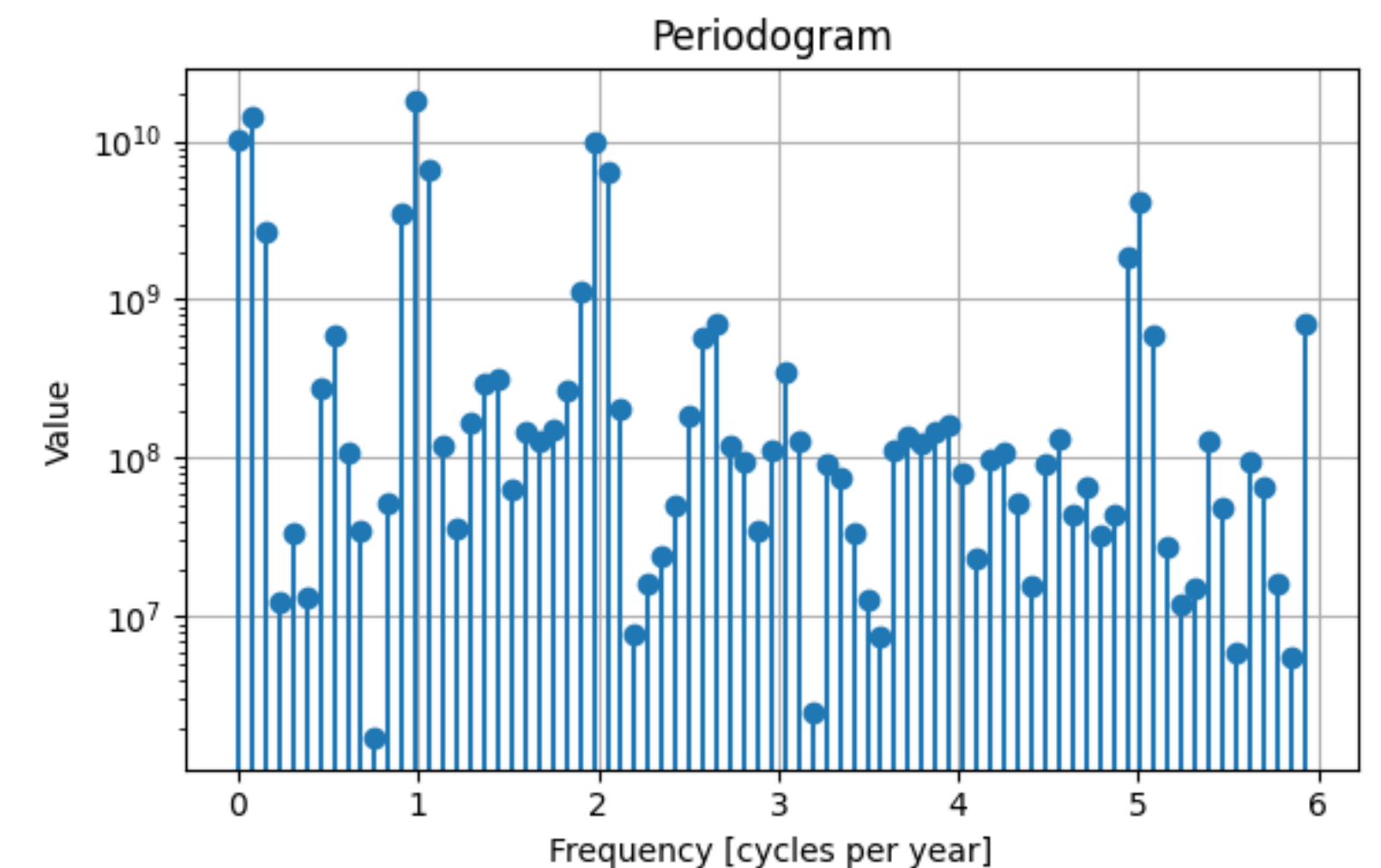
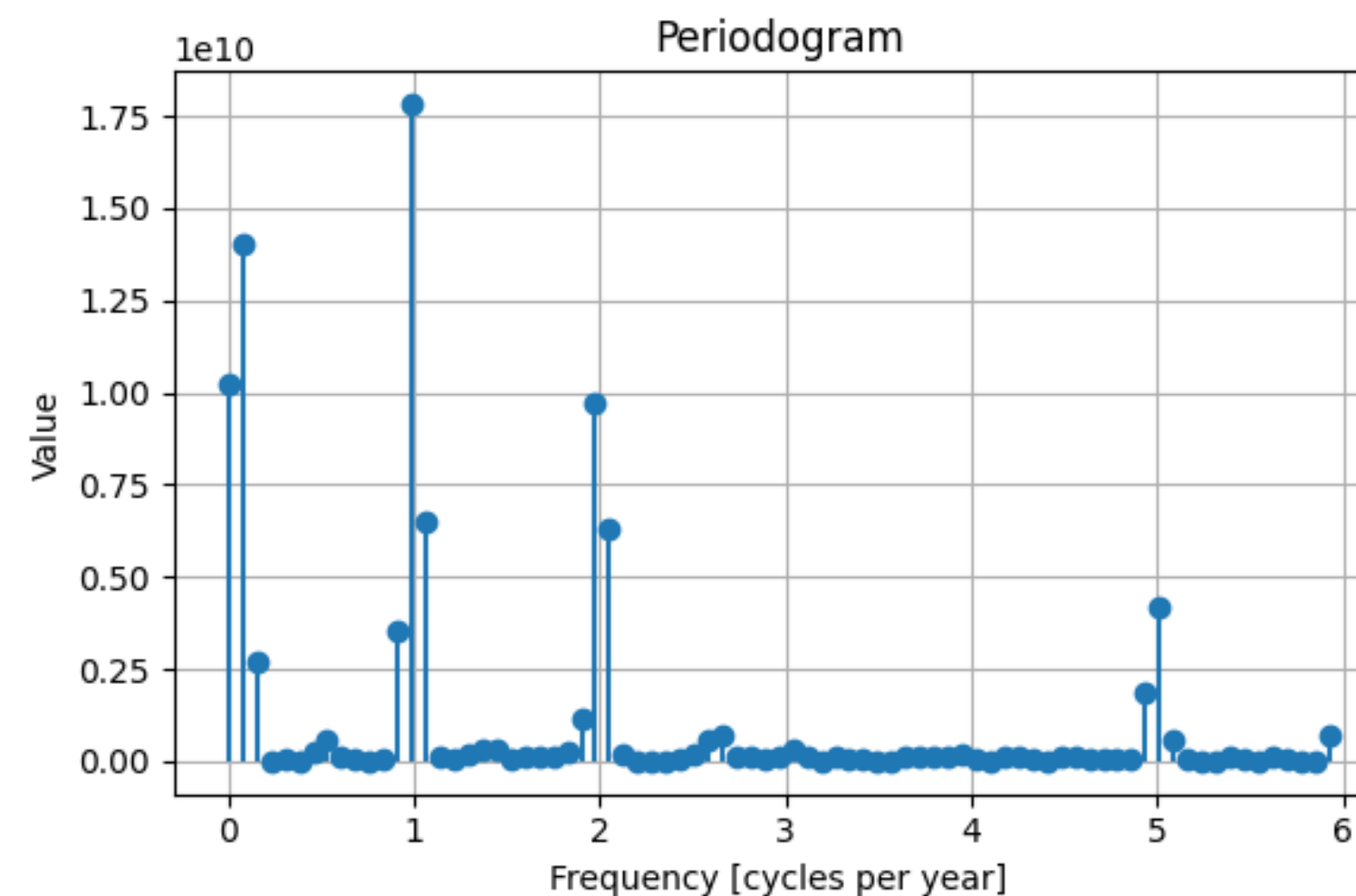
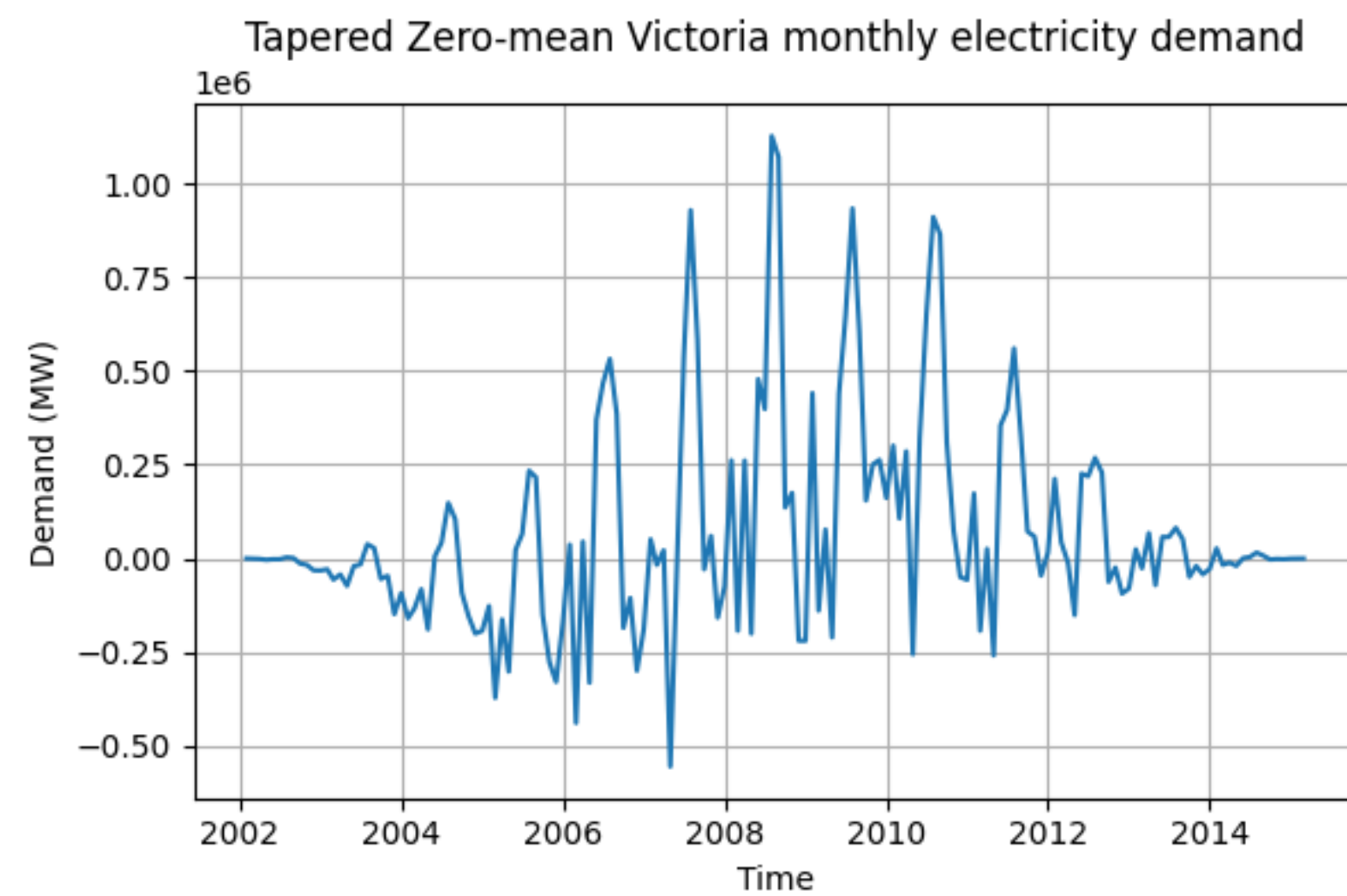
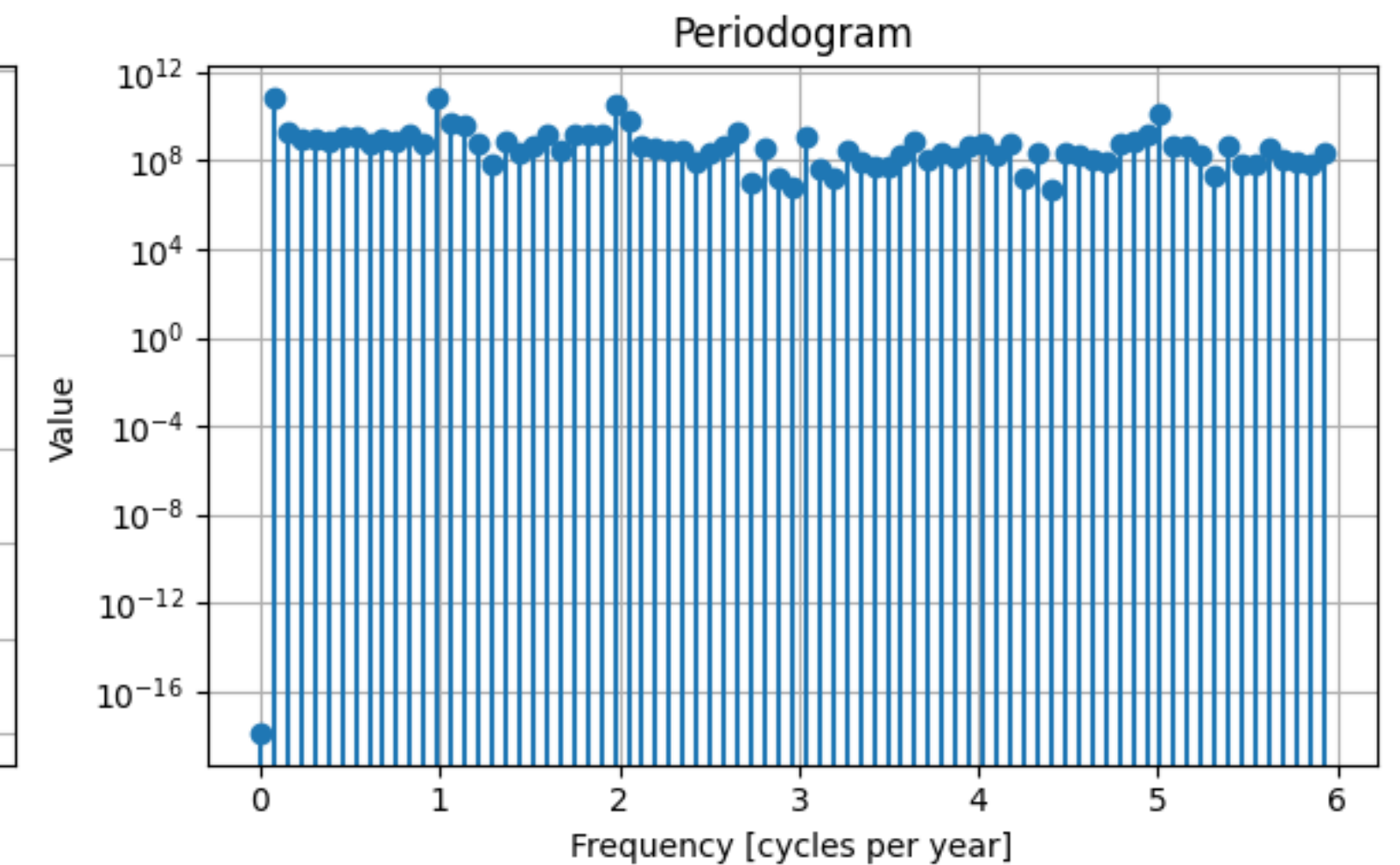
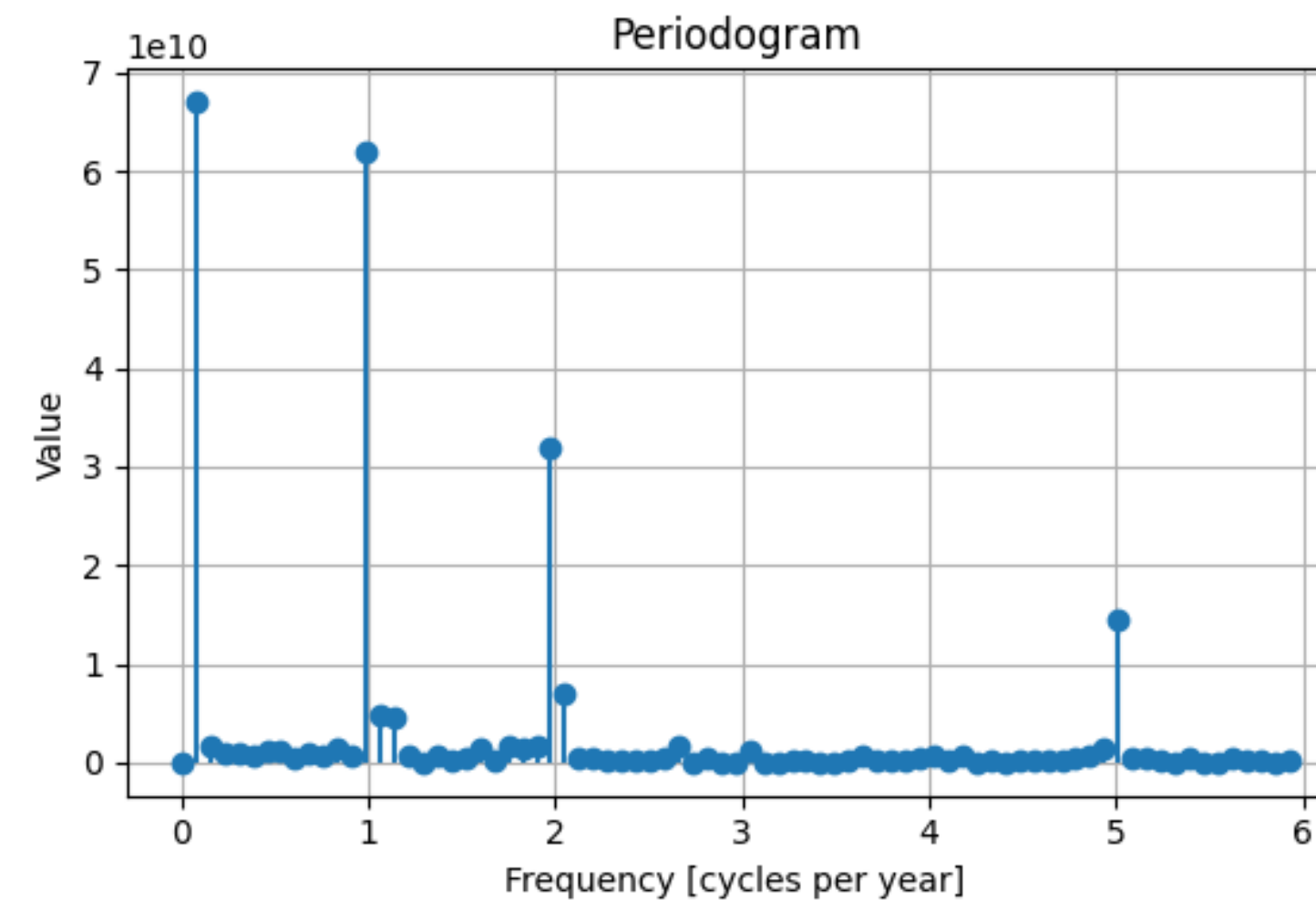
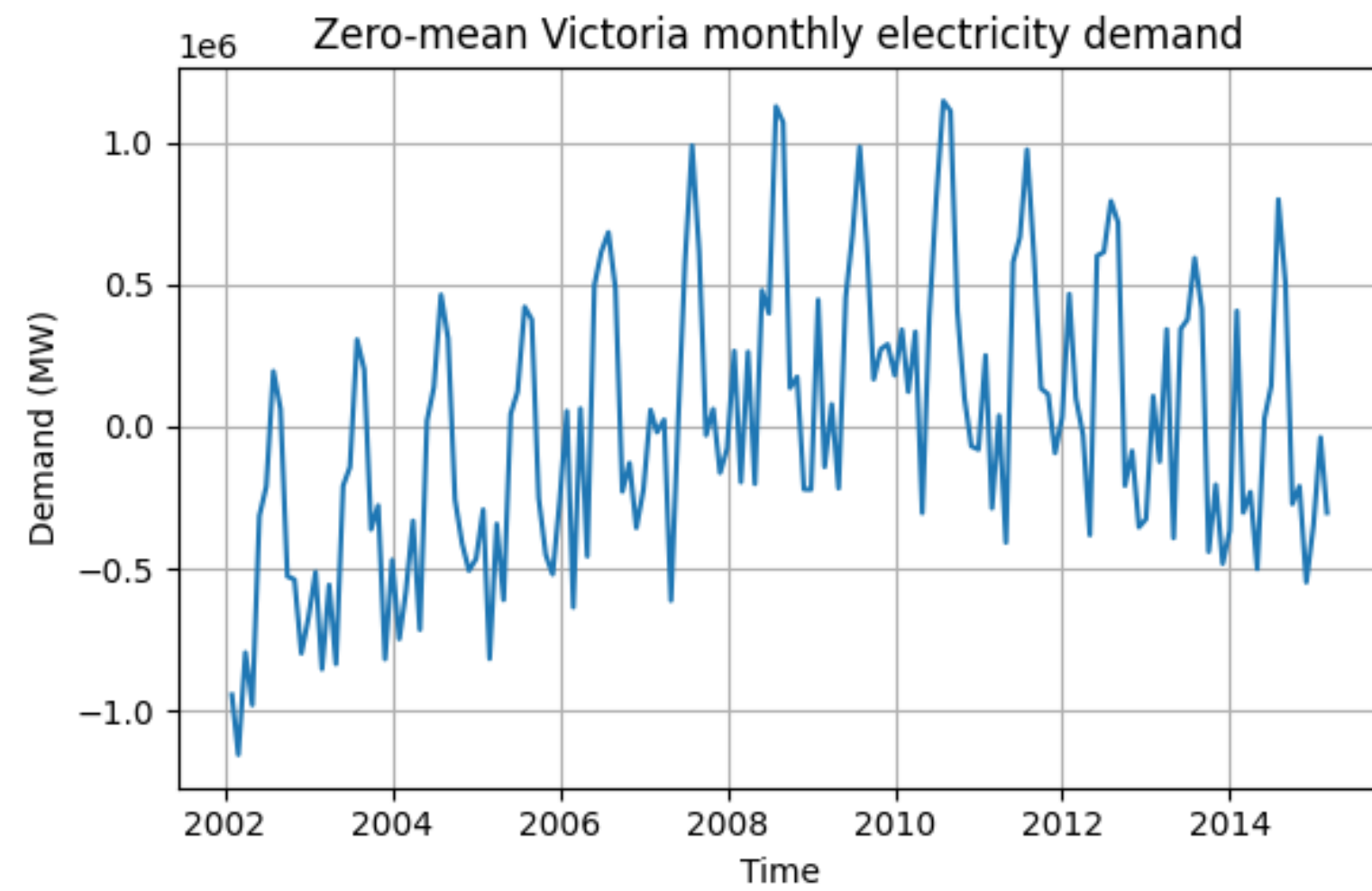
Hann window

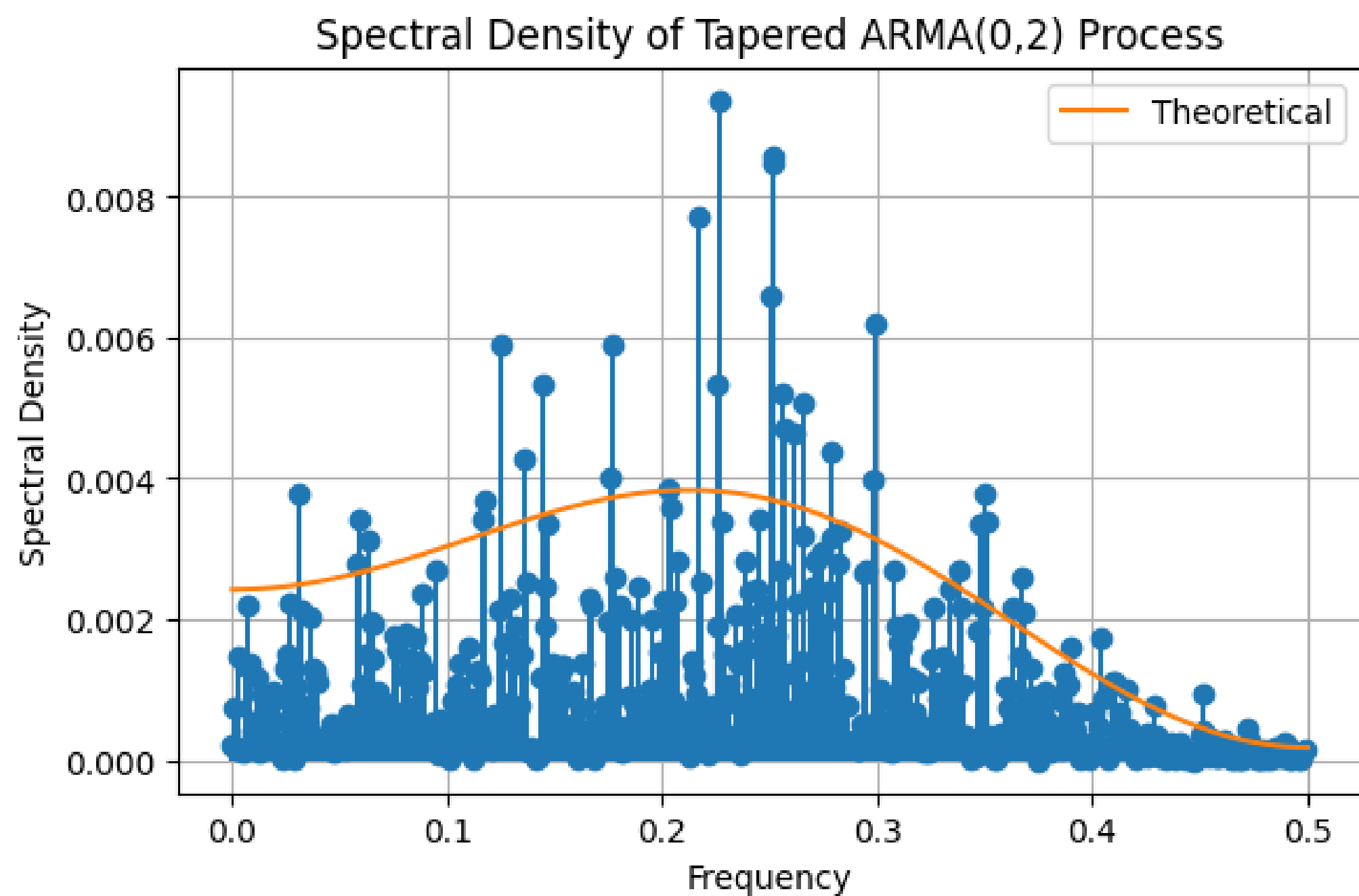
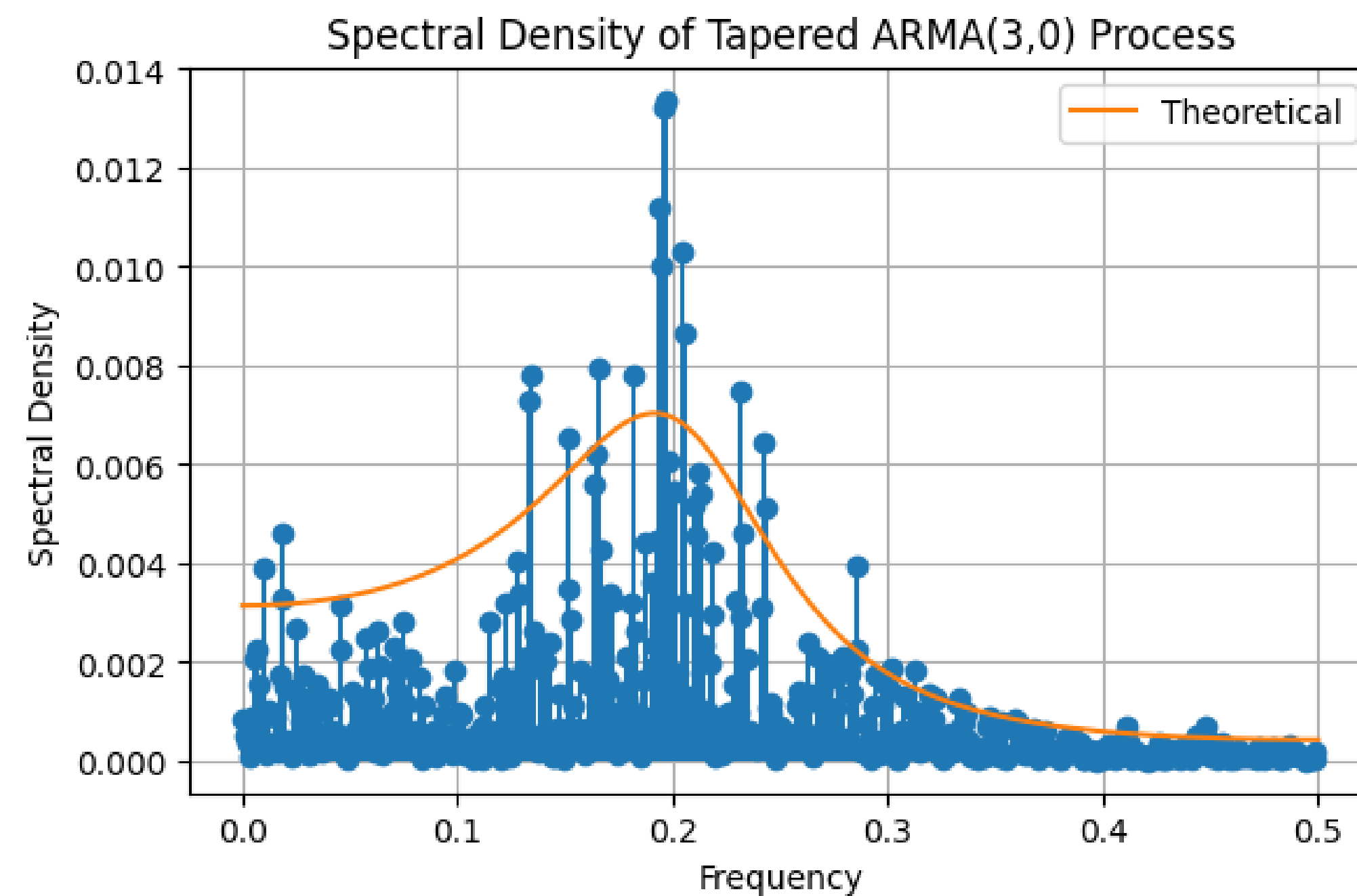
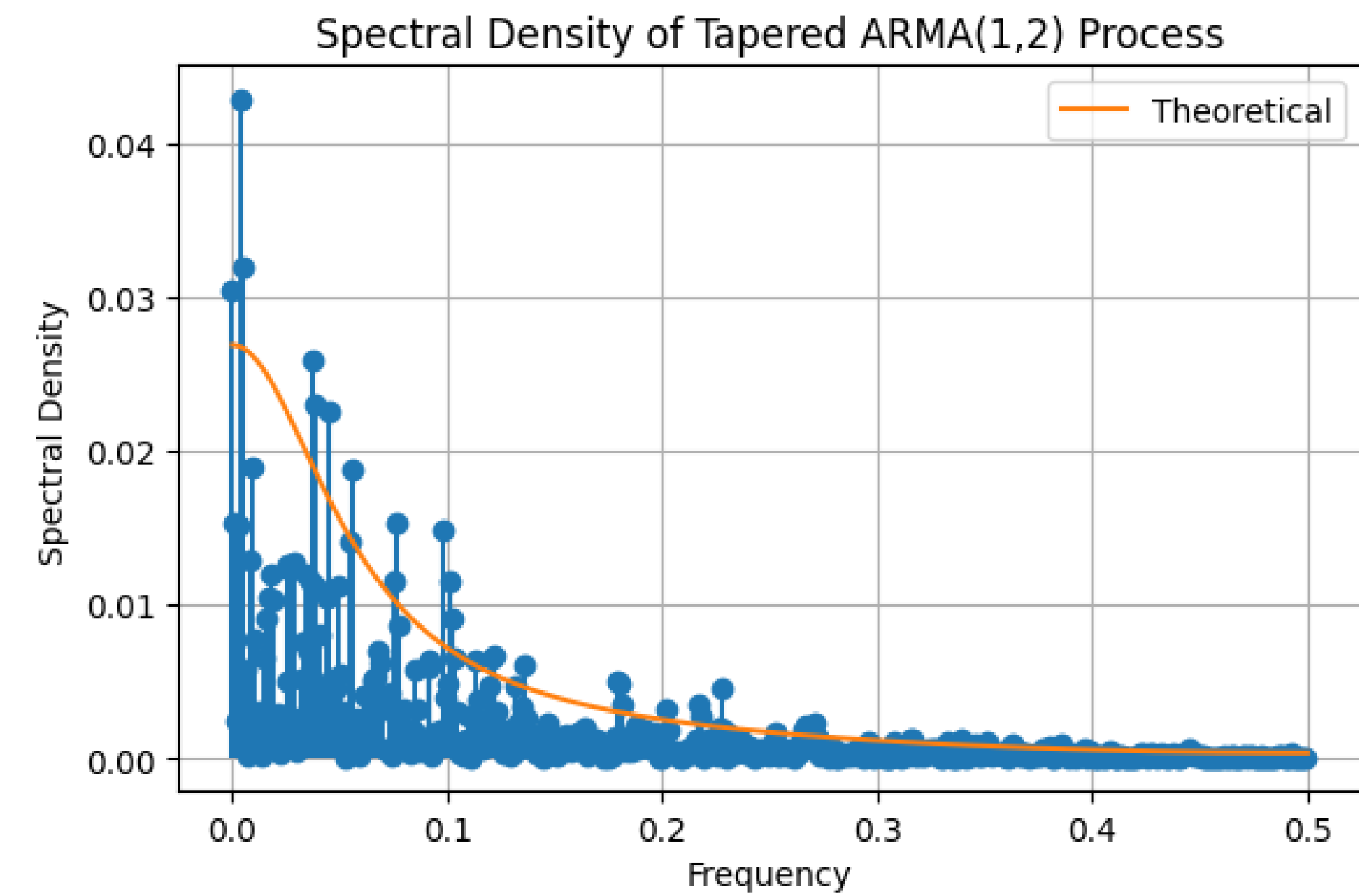
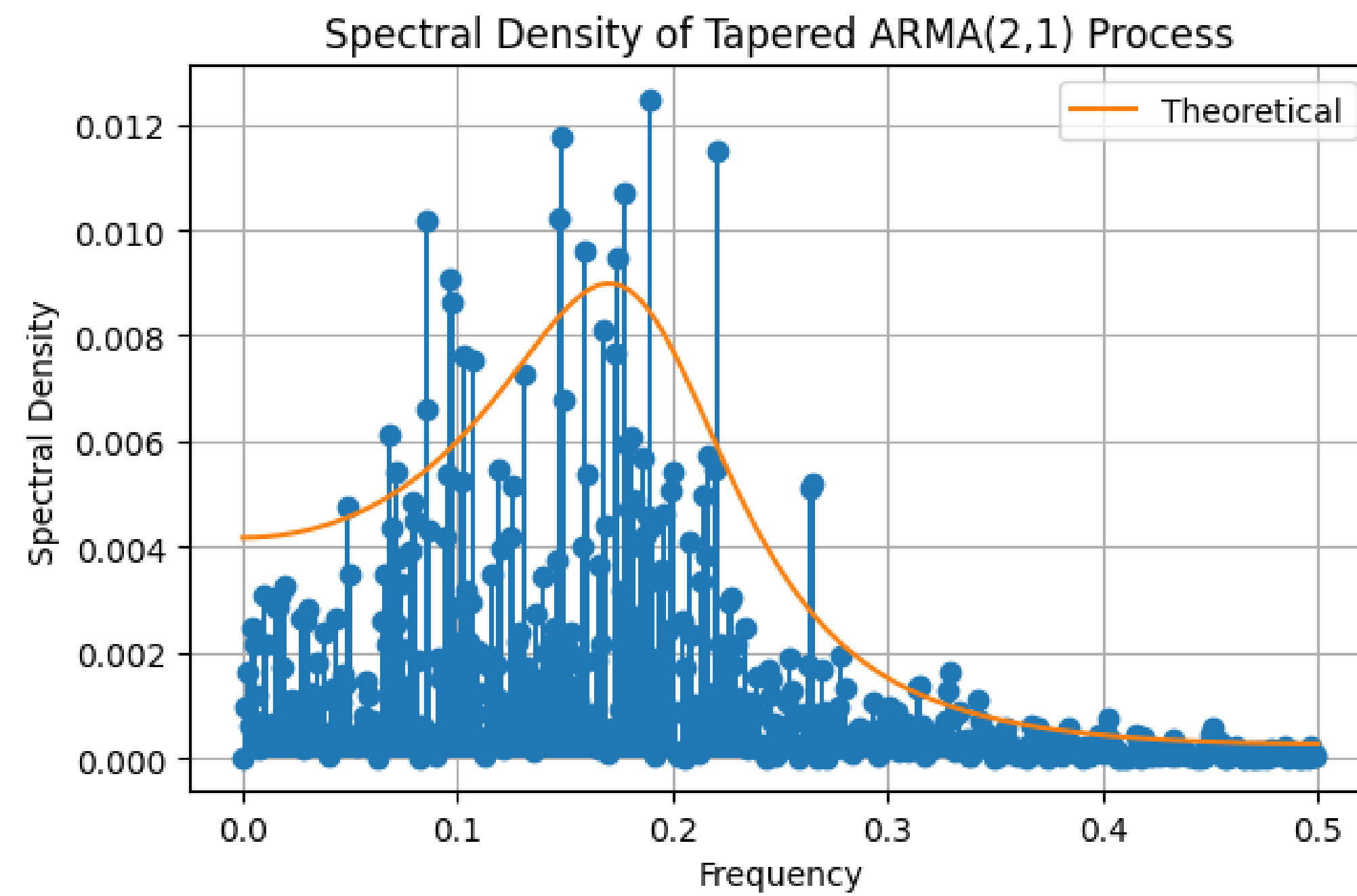


Fourier transform









Periodogram smoothing

Consider the **frequency band** B of $L = 2m + 1$ **contiguous** frequencies centered around $\omega_j = j/n$,

$$B = \left\{ \omega \mid \omega = \omega_j + \frac{k}{n} \right\}, \quad k = -m, \dots, 0, \dots, m$$

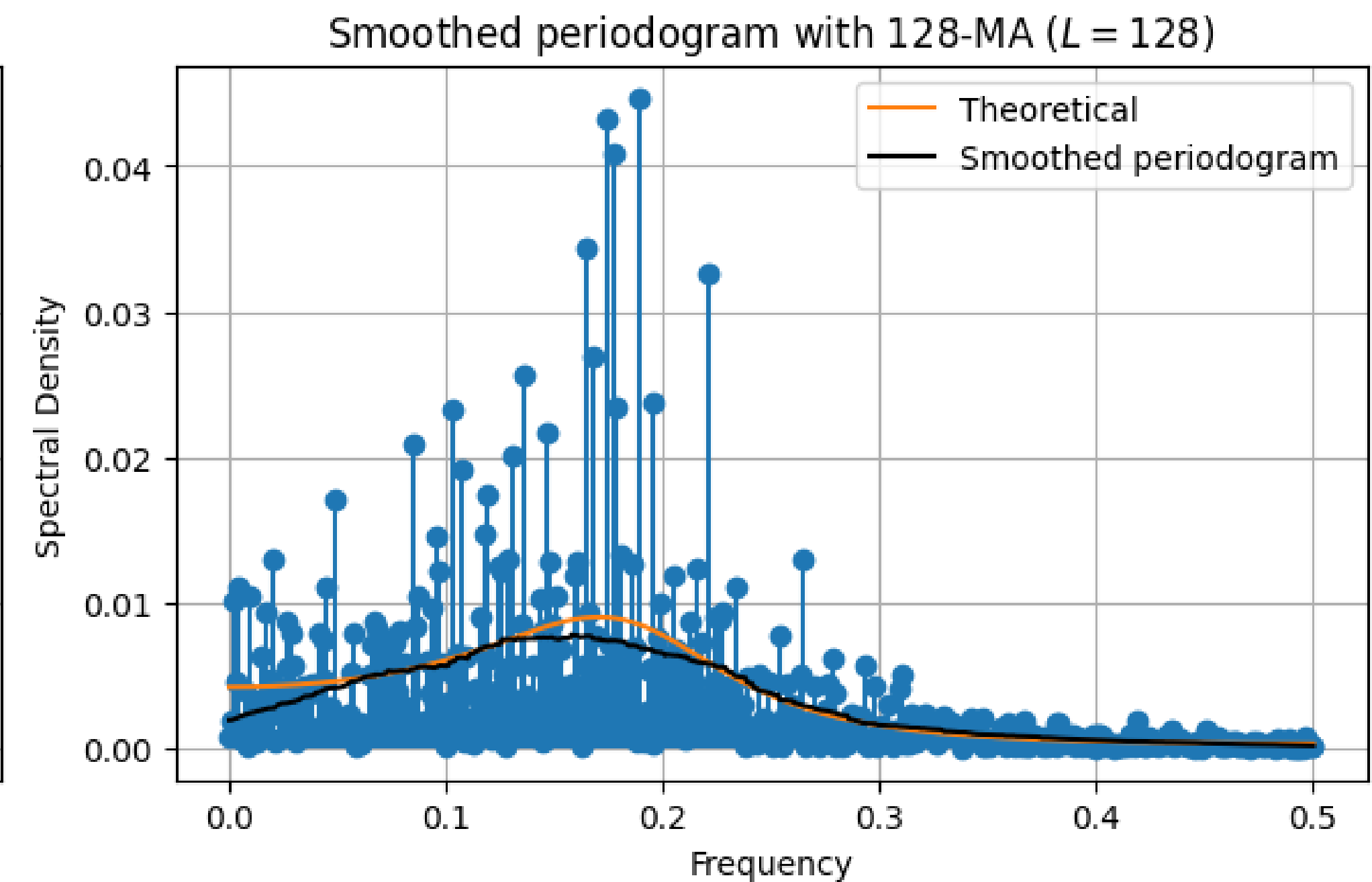
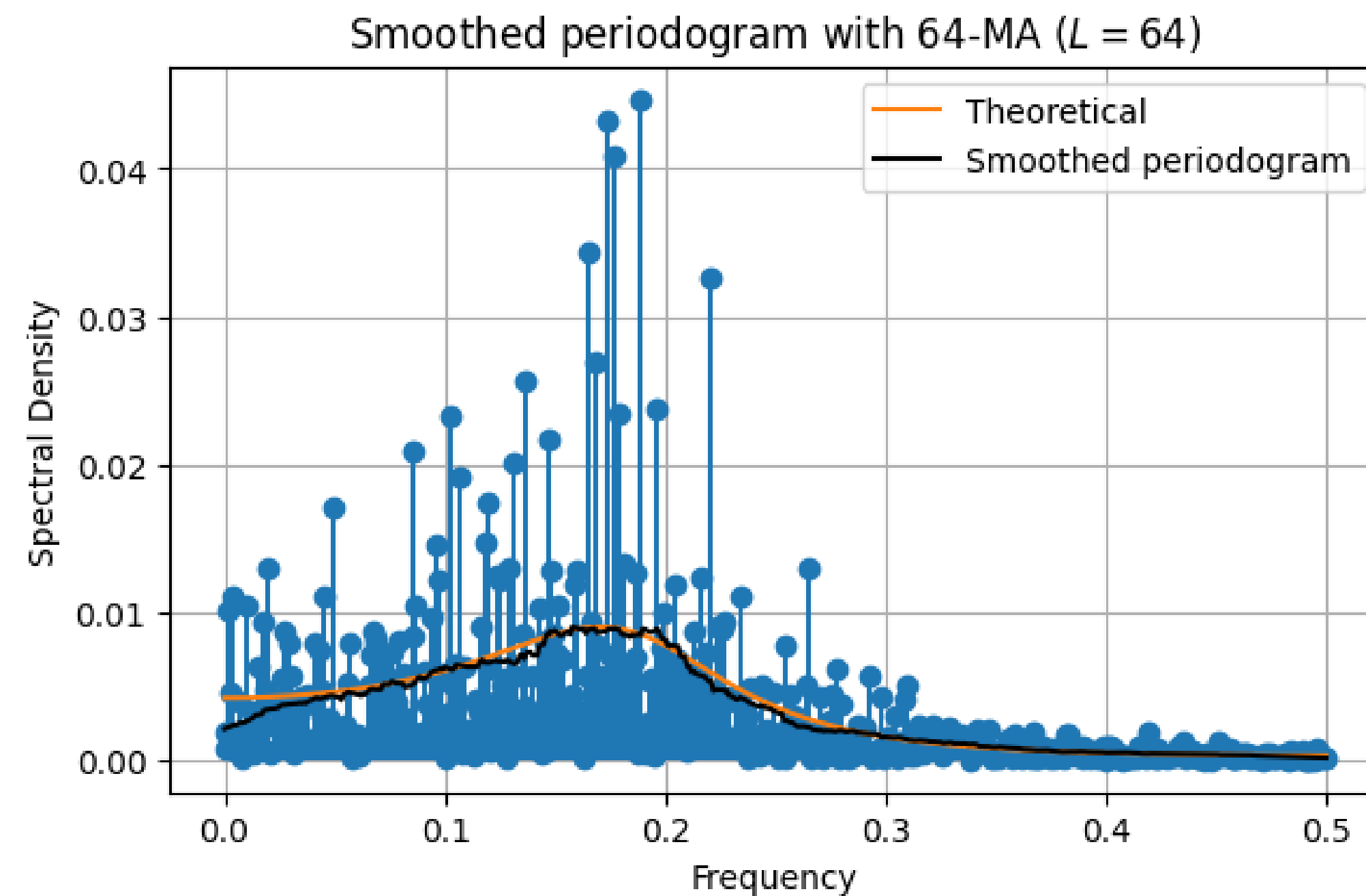
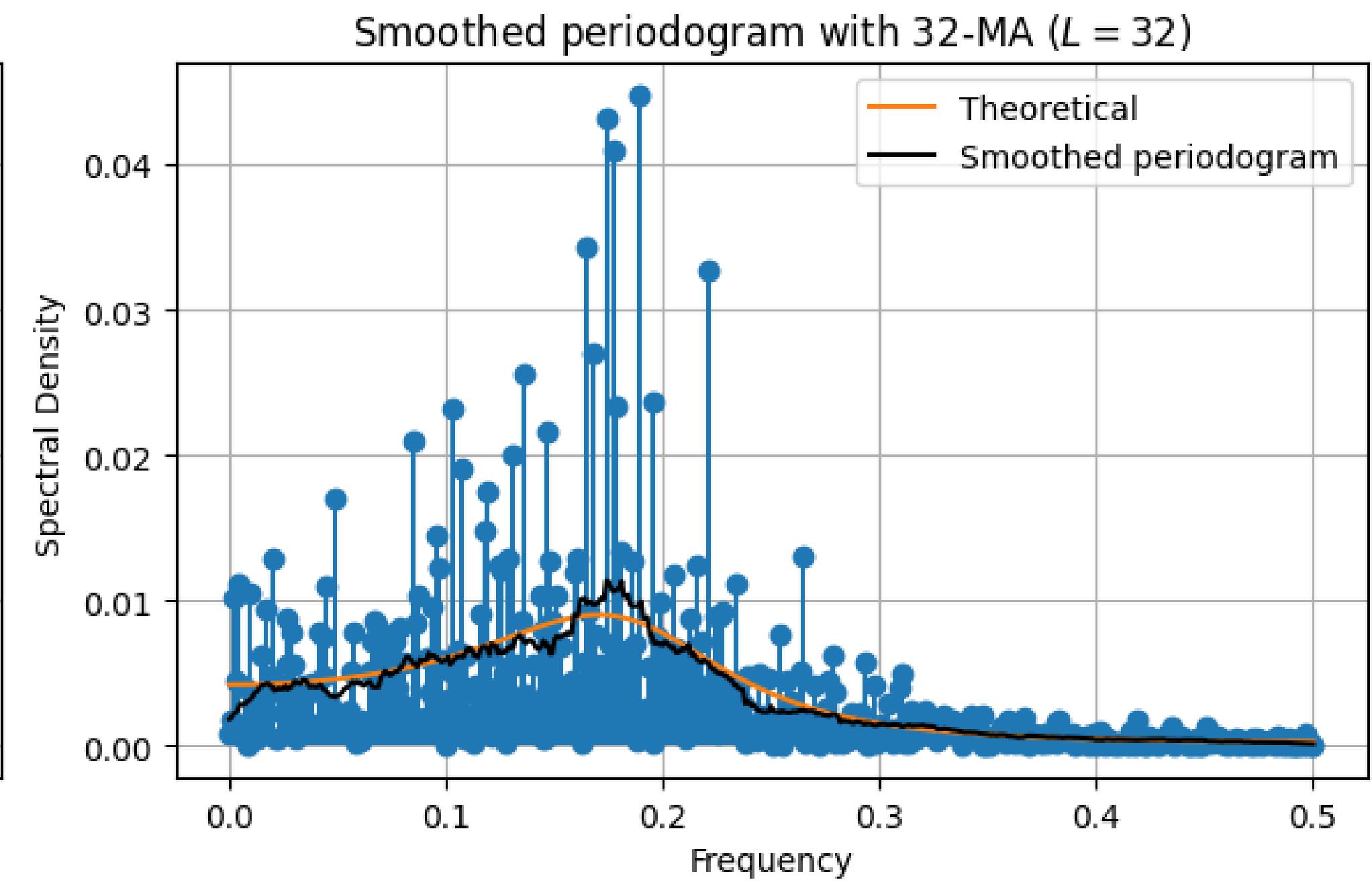
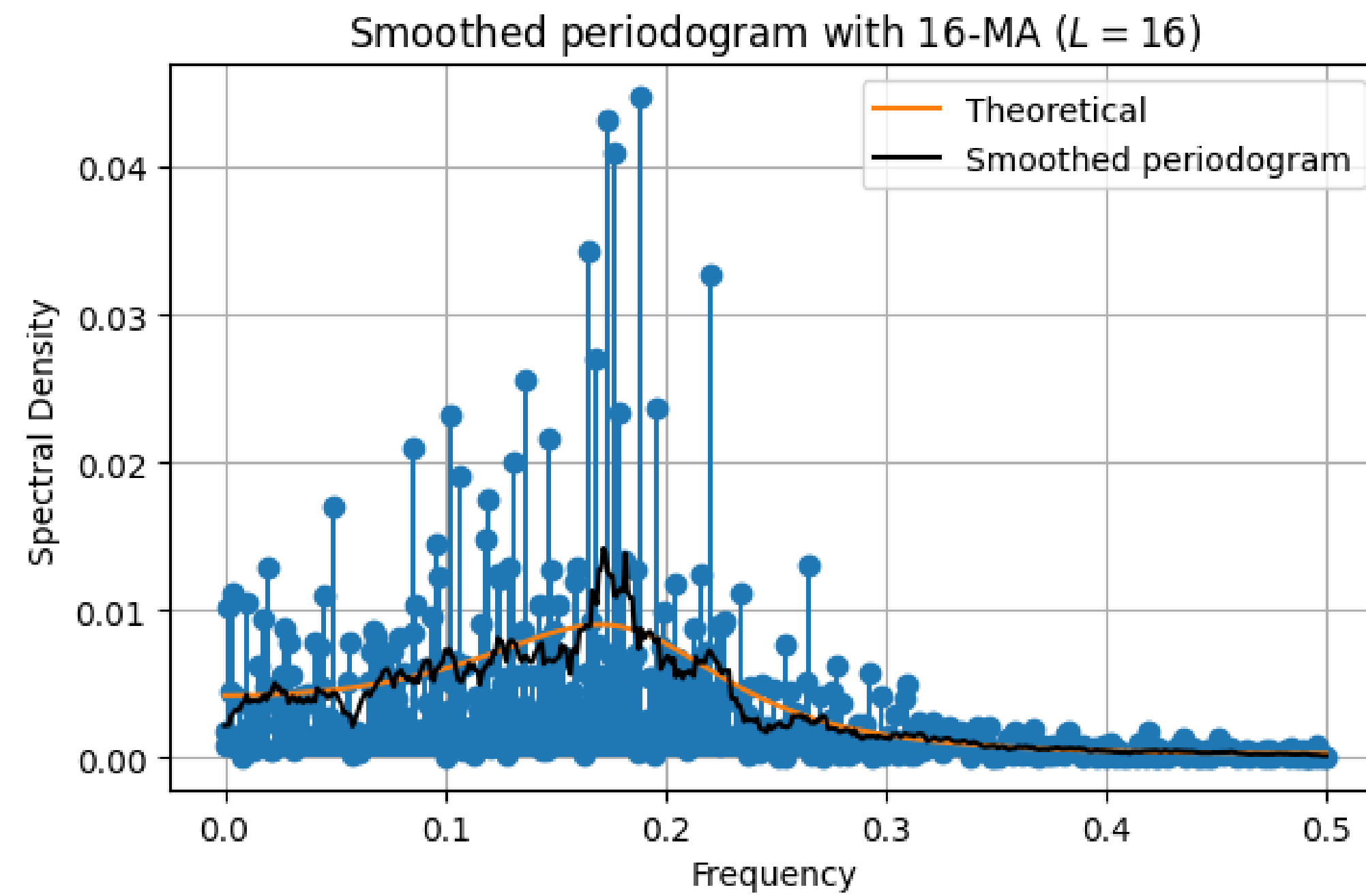
The **smoothed** spectral density (with **equal weights** $1/L$ i.e., moving average smoothing) is defined as

$$\bar{f}(\omega) = \sum_{k=-m}^m \frac{1}{L} P\left(\omega_j + \frac{k}{n}\right)$$

$\bar{f}$ is a good estimator provided $L \ll n$ and f is **relatively constant in the band** B .

Under such conditions and for large n , a **confidence interval** can be computed based on the χ_{2L}^2 distribution.

When the assumptions do not hold, consider **local bootstrap** where the DFT is resampled on a band basis.



Periodogram weighted smoothing

Averaging with equal weights tends to **flatten the peaks**.

This can be mitigated with weighted average:

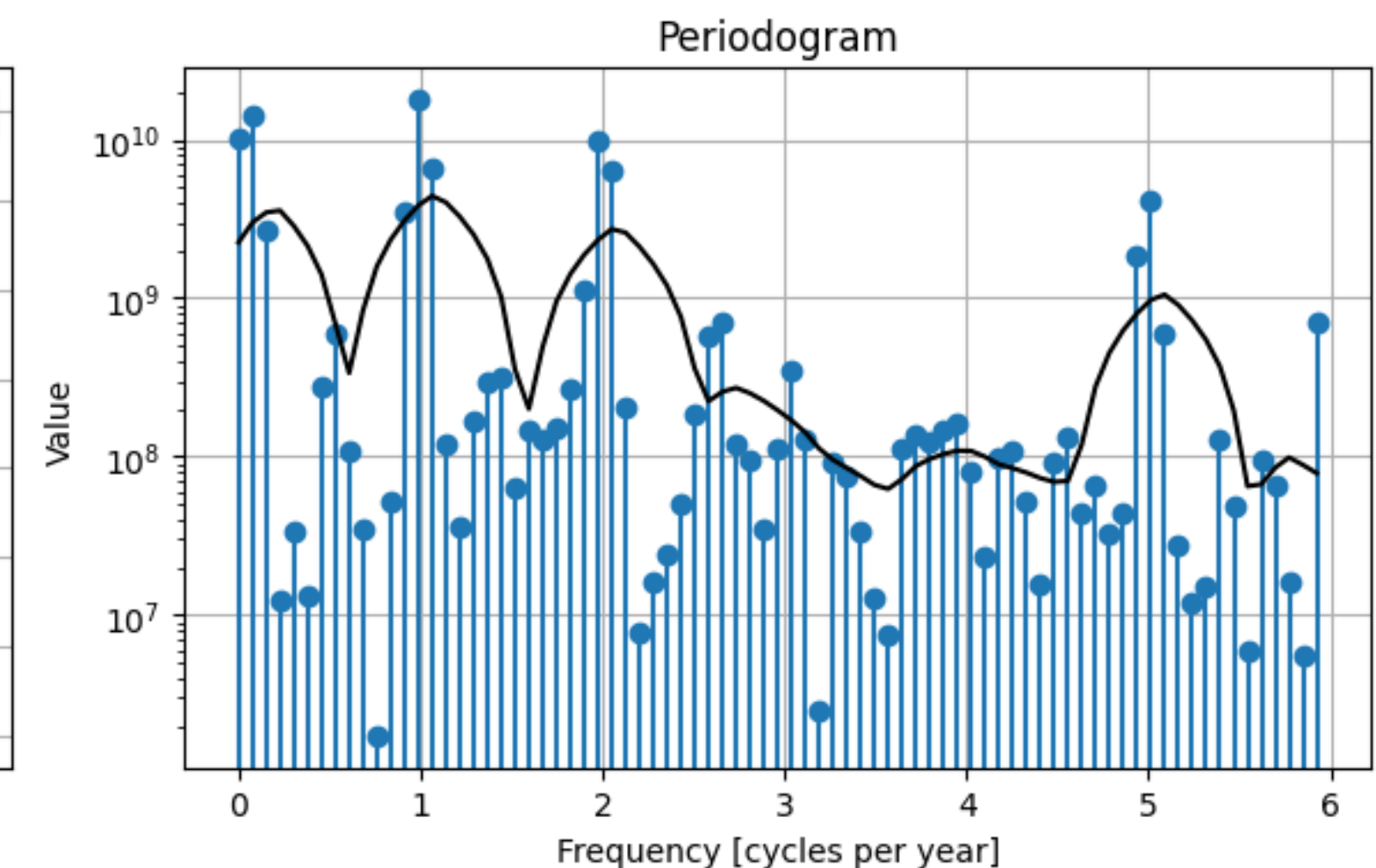
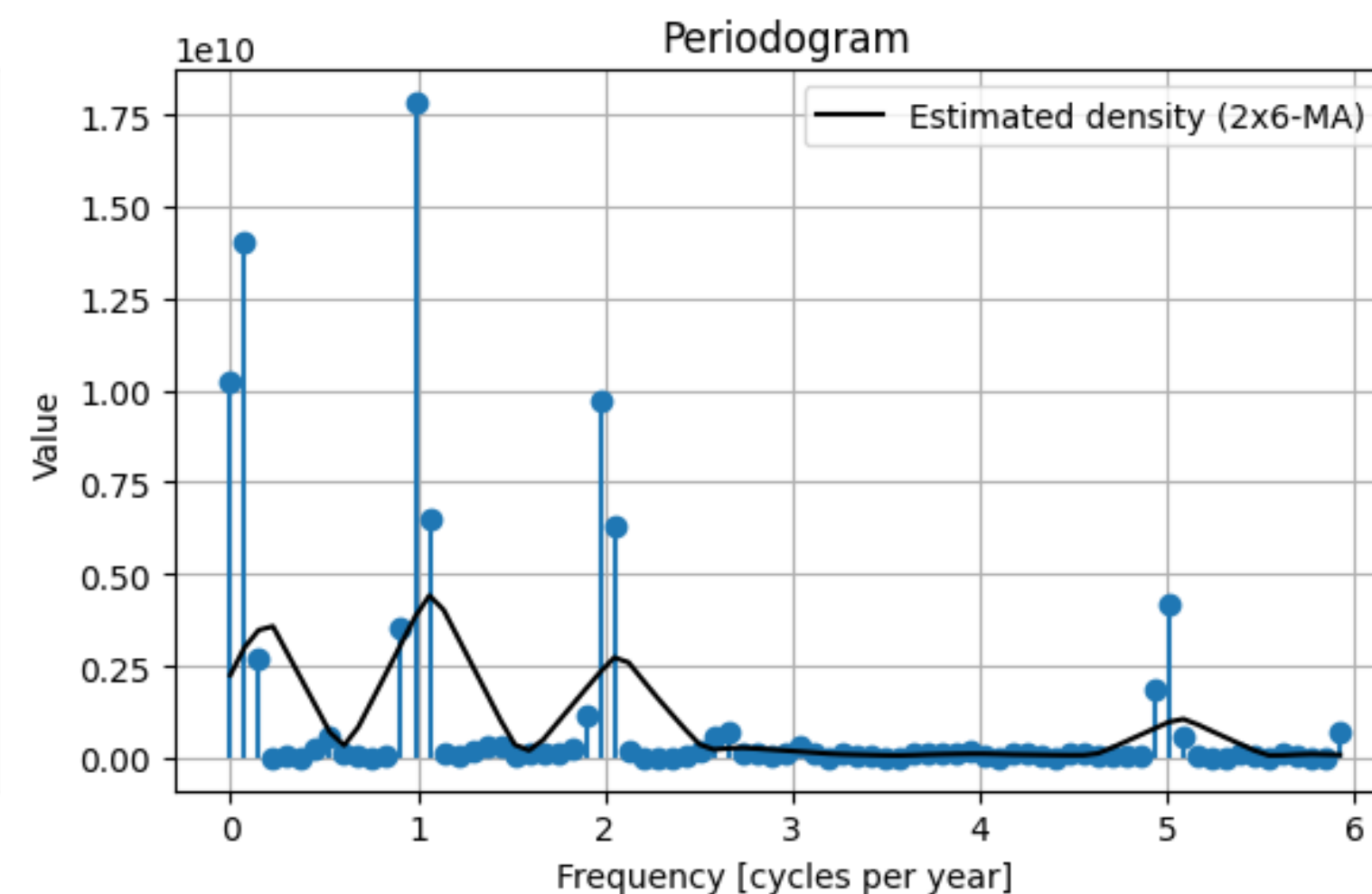
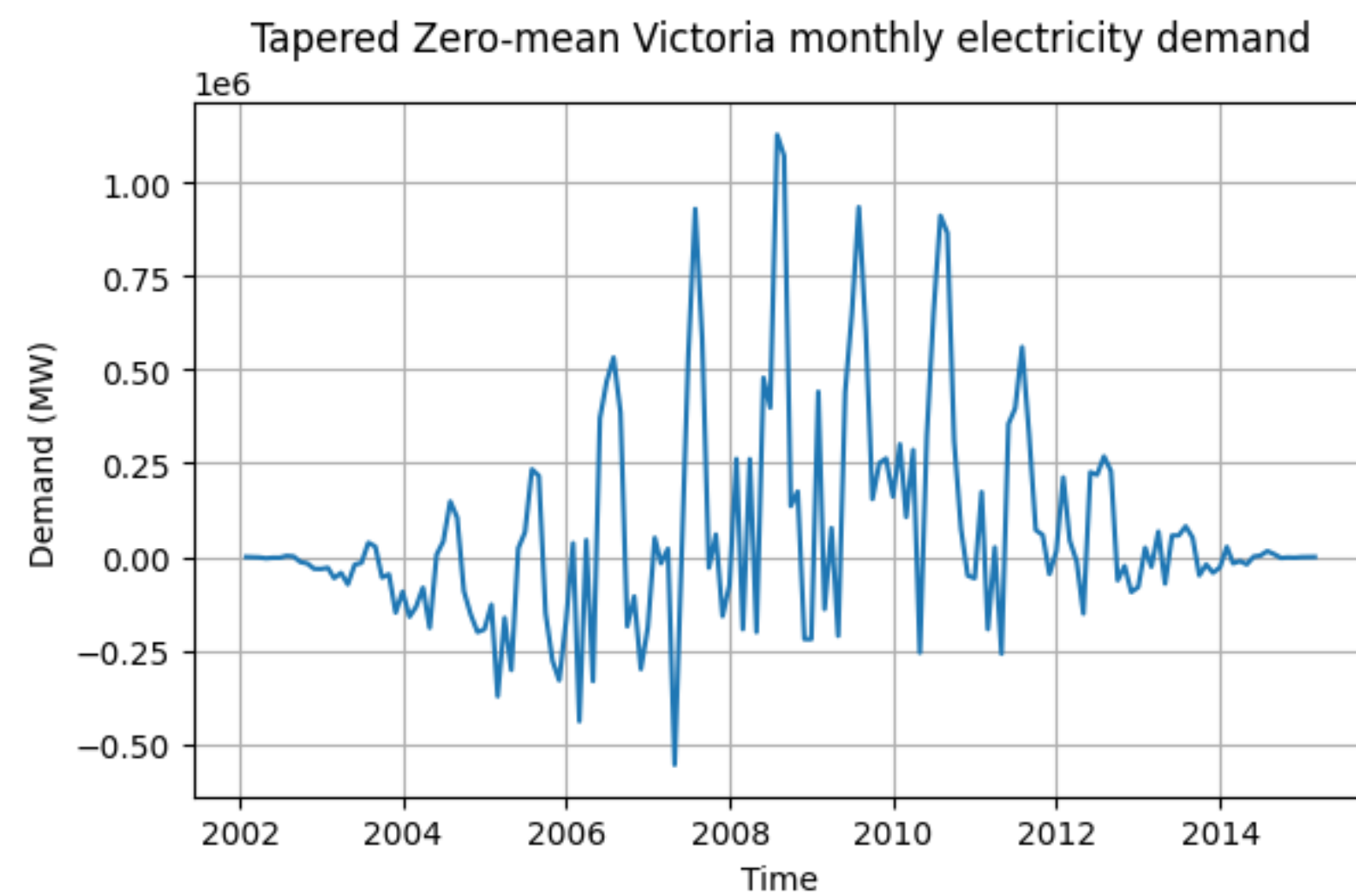
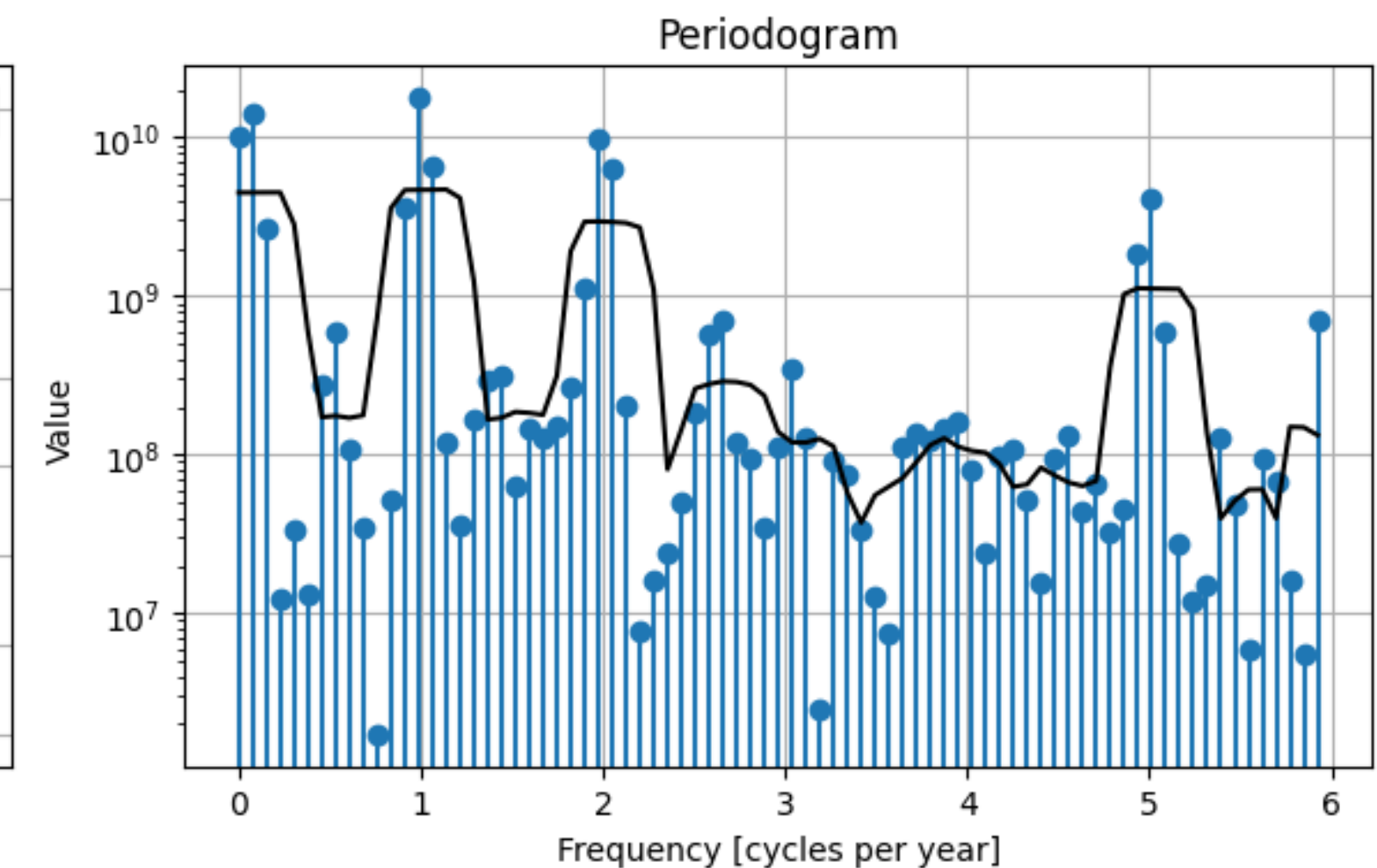
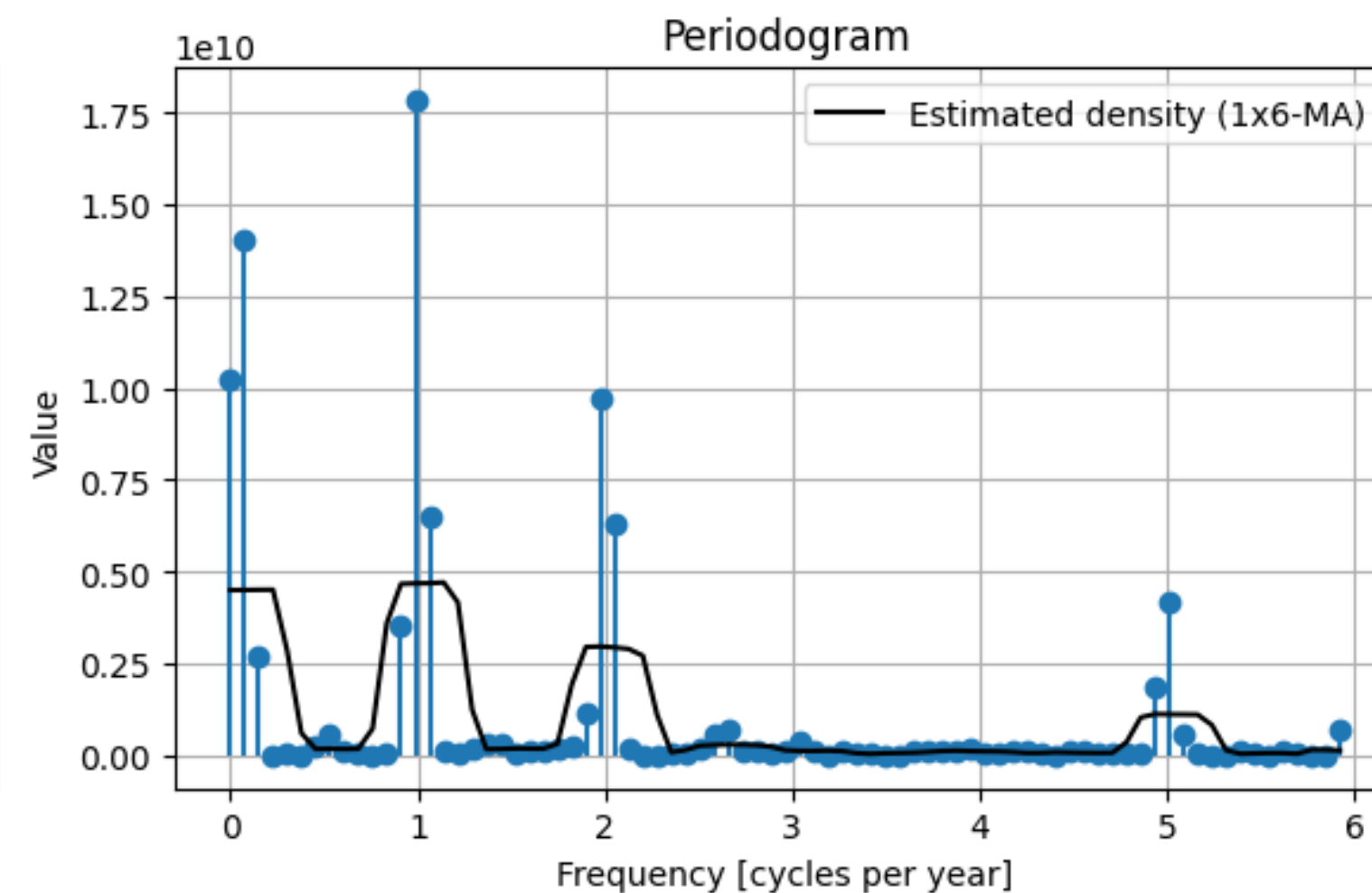
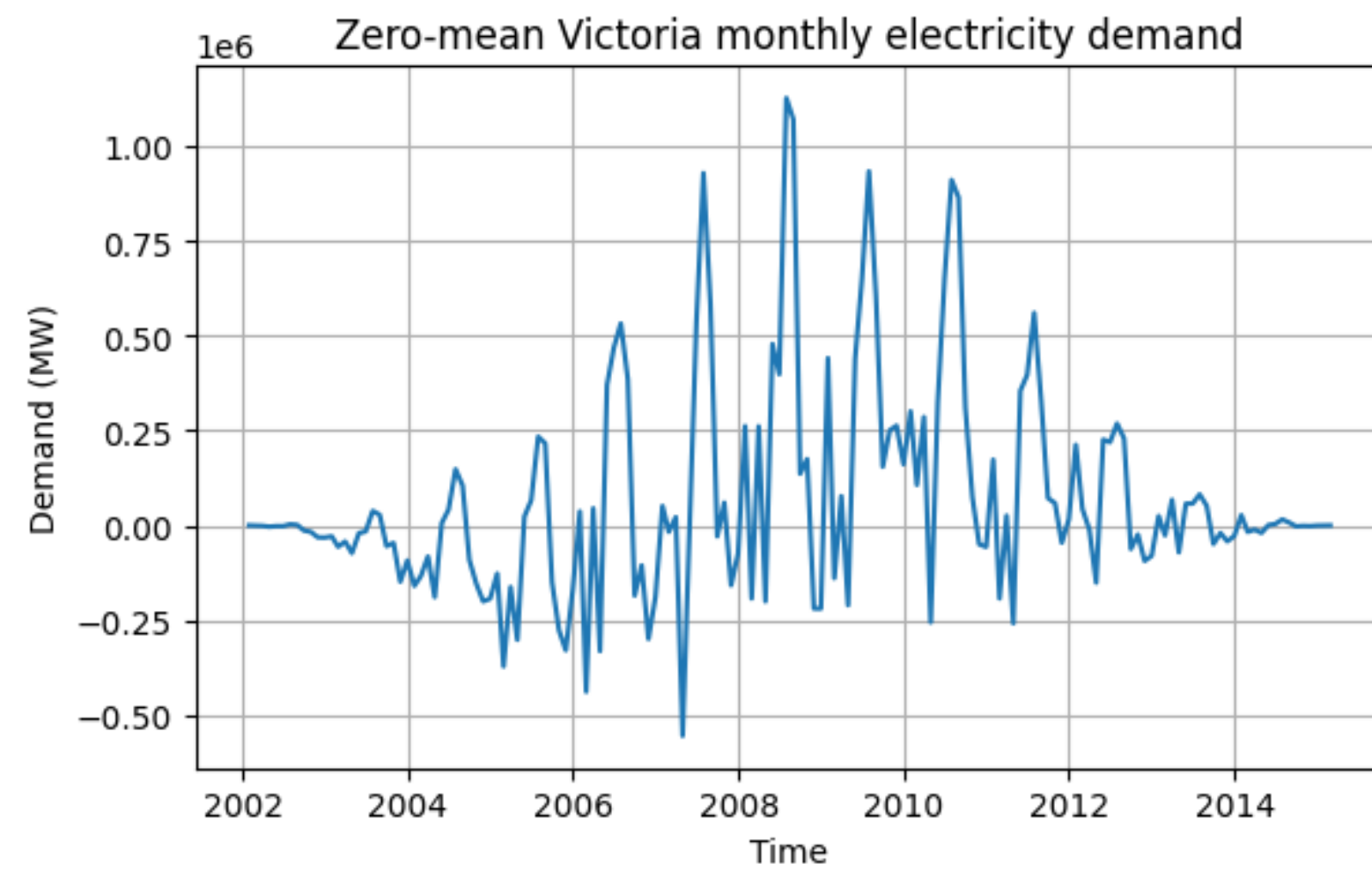
$$\bar{f}(\omega) = \sum_{k=-m}^m h_k P\left(\omega_j + \frac{k}{n}\right), \quad \text{with } h_k \text{ satisfying } \sum_{k=-m}^m h_k = 1$$

Generally, use **weights that decrease** as the **distance from the center increases**.

- Iteratively apply MA smoothing e.g., 2x3-MA yields weights $\left\{\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{2}{9}, \frac{1}{9}\right\}$
- Modified Daniell kernel, which puts half of the weight on the end points e.g., $\left\{\frac{1}{4}, \frac{2}{4}, \frac{1}{4}\right\}$ for $m = 1$.

Confidence intervals can be computed provided that when $m, n \rightarrow \infty$ and $\frac{m}{n} \rightarrow 0$, the weights satisfy $\sum_{k=-m}^m h_k^2 = 0$.

- Otherwise, consider **local bootstrap**.

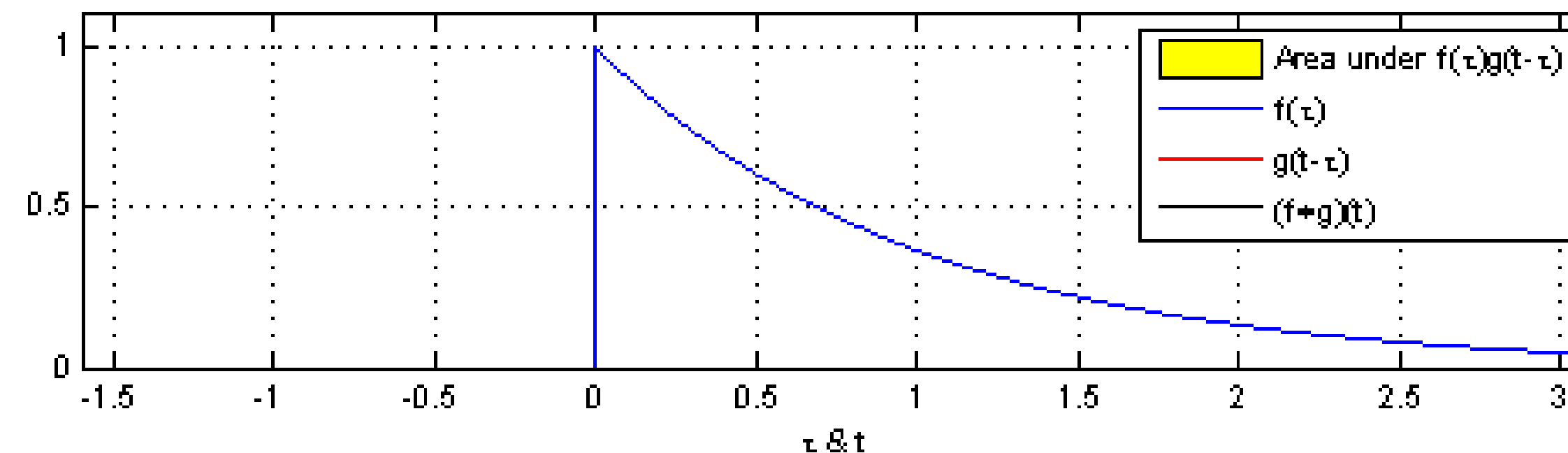


Linear filtering

The spectral density of a TS can be modified in a **determined** manner by applying linear filters to the TS.

A linear filter is defined as a moving linear combination i.e., a **convolution** of a TS with given weights

$$Y_t = \sum_{j=-\infty}^{\infty} a_j X_{t-j} = (\mathbf{a} * \mathbf{X})(t), \quad \sum_{j=-\infty}^{\infty} |a_j| < \infty$$



Tinos, Wikimedia

The coefficients a and their DFT A are resp. called the **impulse response** and the **frequency response** functions.

- For large n , the DFT $d_Y(\omega)$ of Y_t satisfies $d_Y(\omega) = A(\omega)d_X(\omega)$ due to the linearity property.
- Frequency domain multiplication + inverse DFT may be more efficient than the time domain convolution.

Spectral density of filtered series

The spectral density $f_Y(\omega)$ of Y_t satisfies $f_Y(\omega) = |A(\omega)|^2 f_X(\omega)$.

($|A(\omega)|^2$ is called the **power response**)

Proof:

$$\gamma_Y(h) = \text{cov}(X_{t+h}, X_t)$$

(Index difference is $h - r + s$)

$$= \text{cov}\left(\sum_r a_r X_{t+h-r}, \sum_s a_s X_{t-s}\right) = \sum_r \sum_s a_r \gamma_X(h - r + s) a_s$$

$$= \sum_r \sum_s a_r \left[\int_{-1/2}^{1/2} e^{2\pi i \omega (h-r+s)} f_X(\omega) d\omega \right] a_s$$

$$= \int_{-1/2}^{1/2} \left(\underbrace{\sum_r a_r e^{-2\pi i \omega r}}_{A(\omega)} \right) \left(\underbrace{\sum_s a_s e^{2\pi i \omega s}}_{\overline{A(\omega)}} \right) e^{2\pi i \omega h} f_X(\omega) d\omega$$

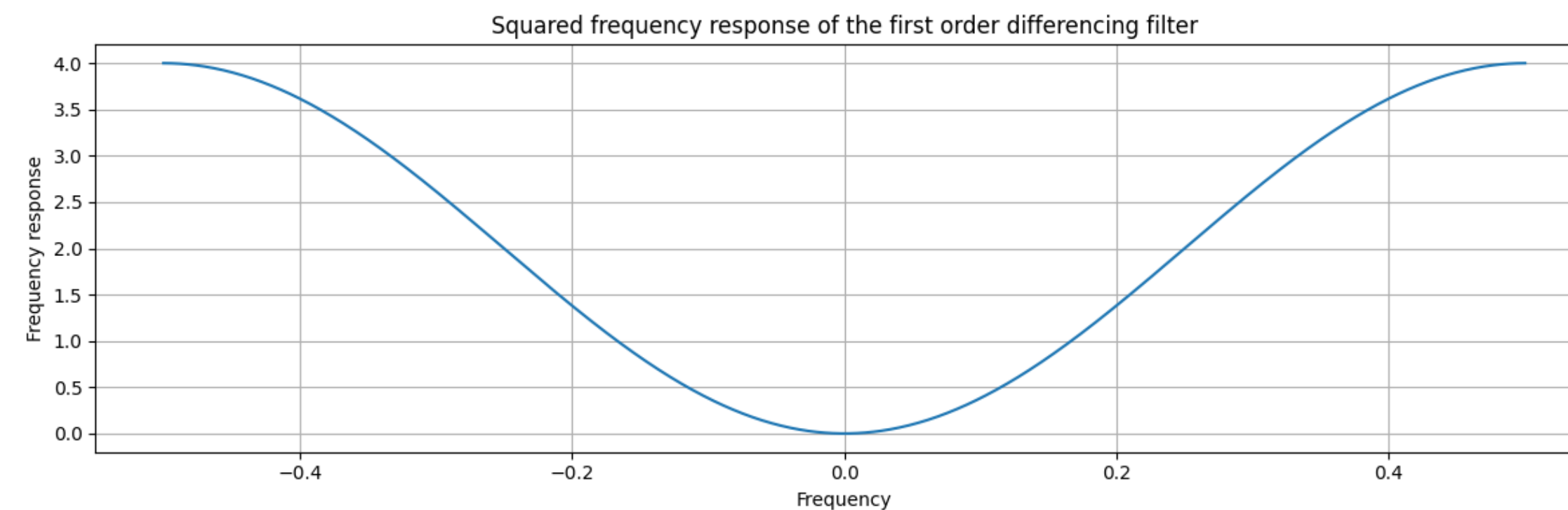
$$= \int_{-1/2}^{1/2} e^{2\pi i \omega h} |A(\omega)|^2 f_X(\omega) d\omega = \int_{-1/2}^{1/2} e^{2\pi i \omega h} f_Y(\omega) d\omega$$

Differencing is a high-pass filter

Consider first order differencing $Y_t = \nabla X_t = X_t - X_{t-1}$.

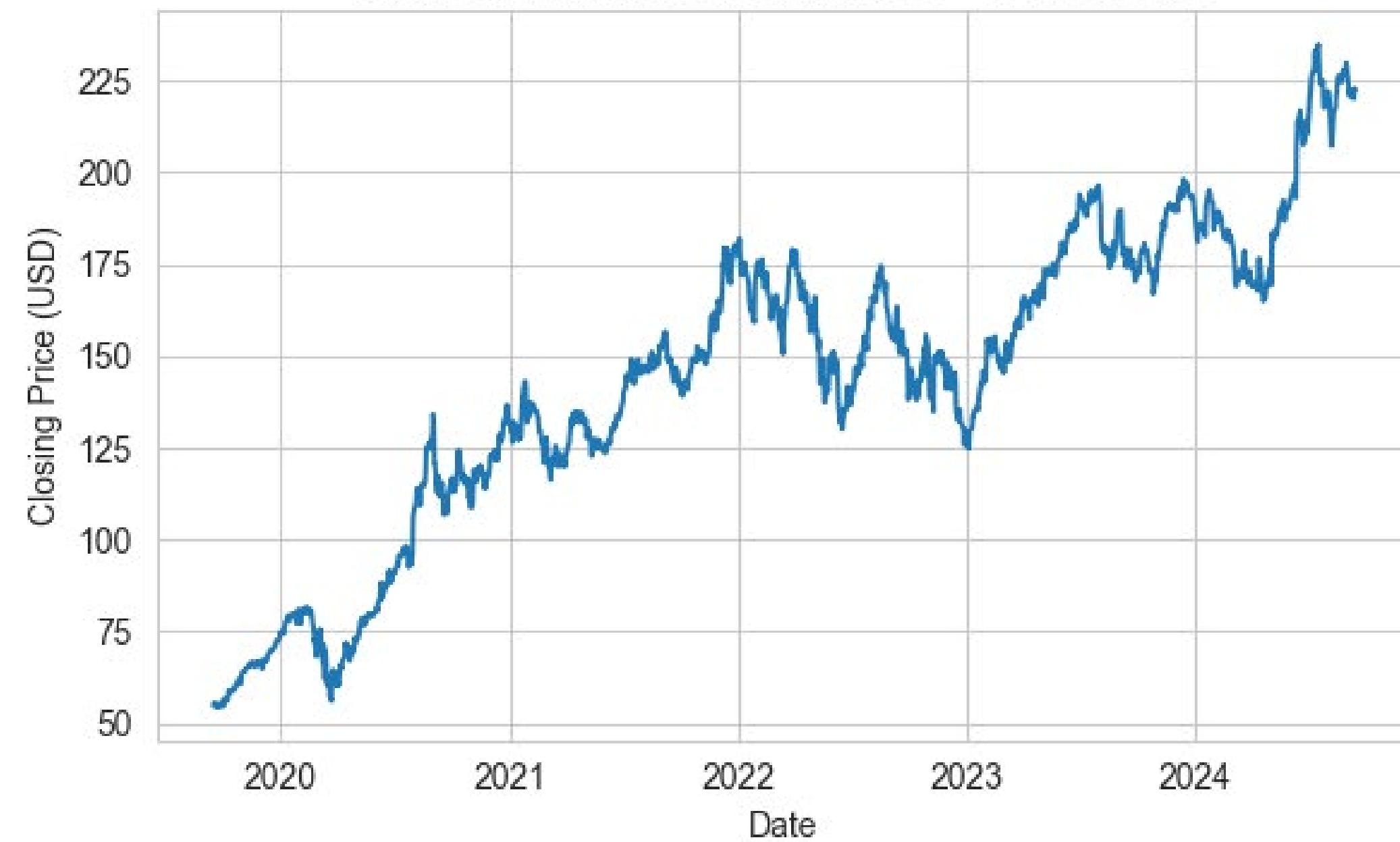
The impulse response function is $a_j = \begin{cases} 1, & j = 0 \\ -1, & j = 1. \\ 0, & \text{otherwise} \end{cases}$

The frequency response function is $A(\omega) = 1 - e^{-2\pi i \omega}$, the power response function is $|A(\omega)|^2 = 2(1 - \cos(2\pi\omega))$

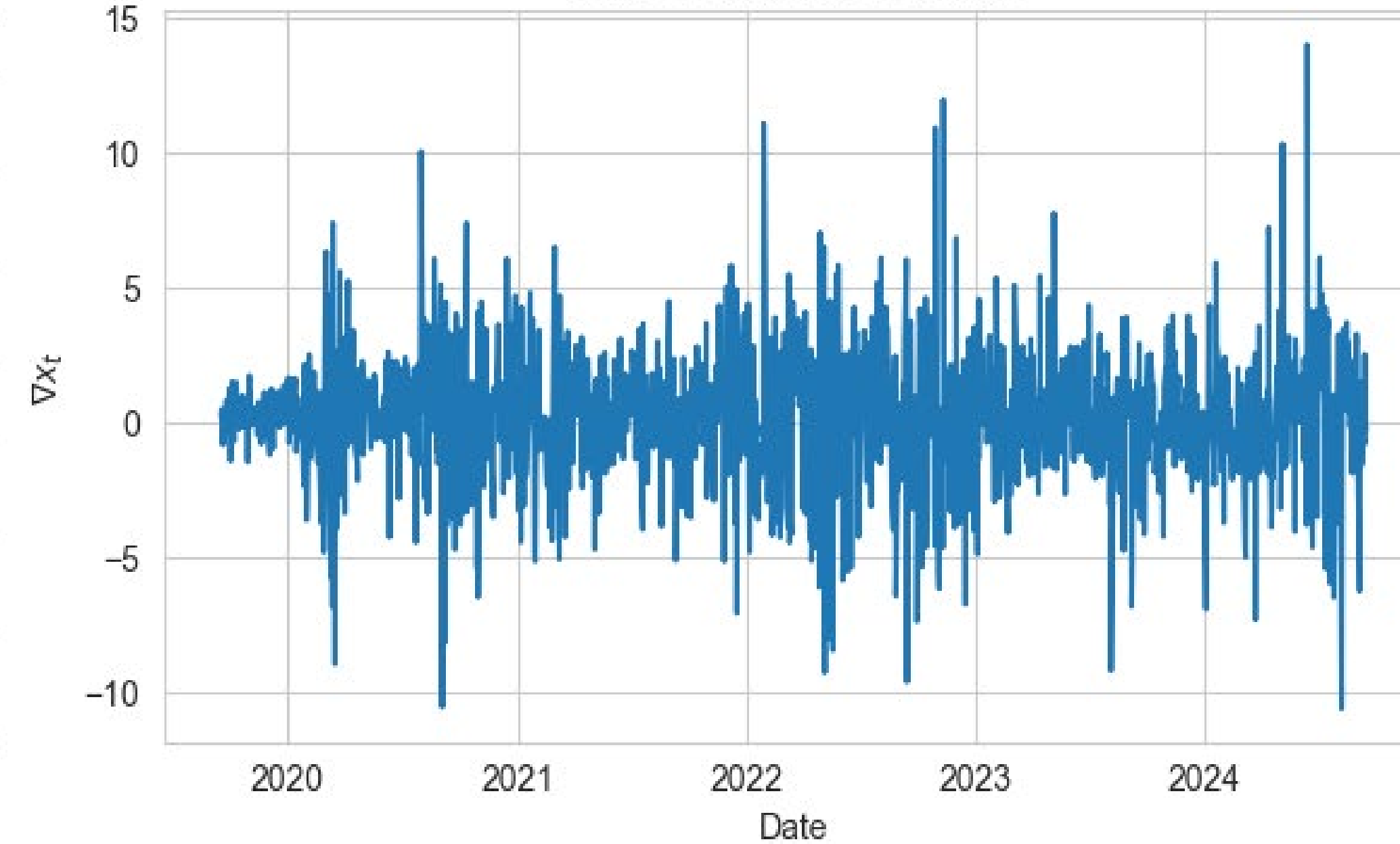


Thus, first order differencing attenuates lower frequencies and enhance higher frequencies. It is a **high-pass filter**.

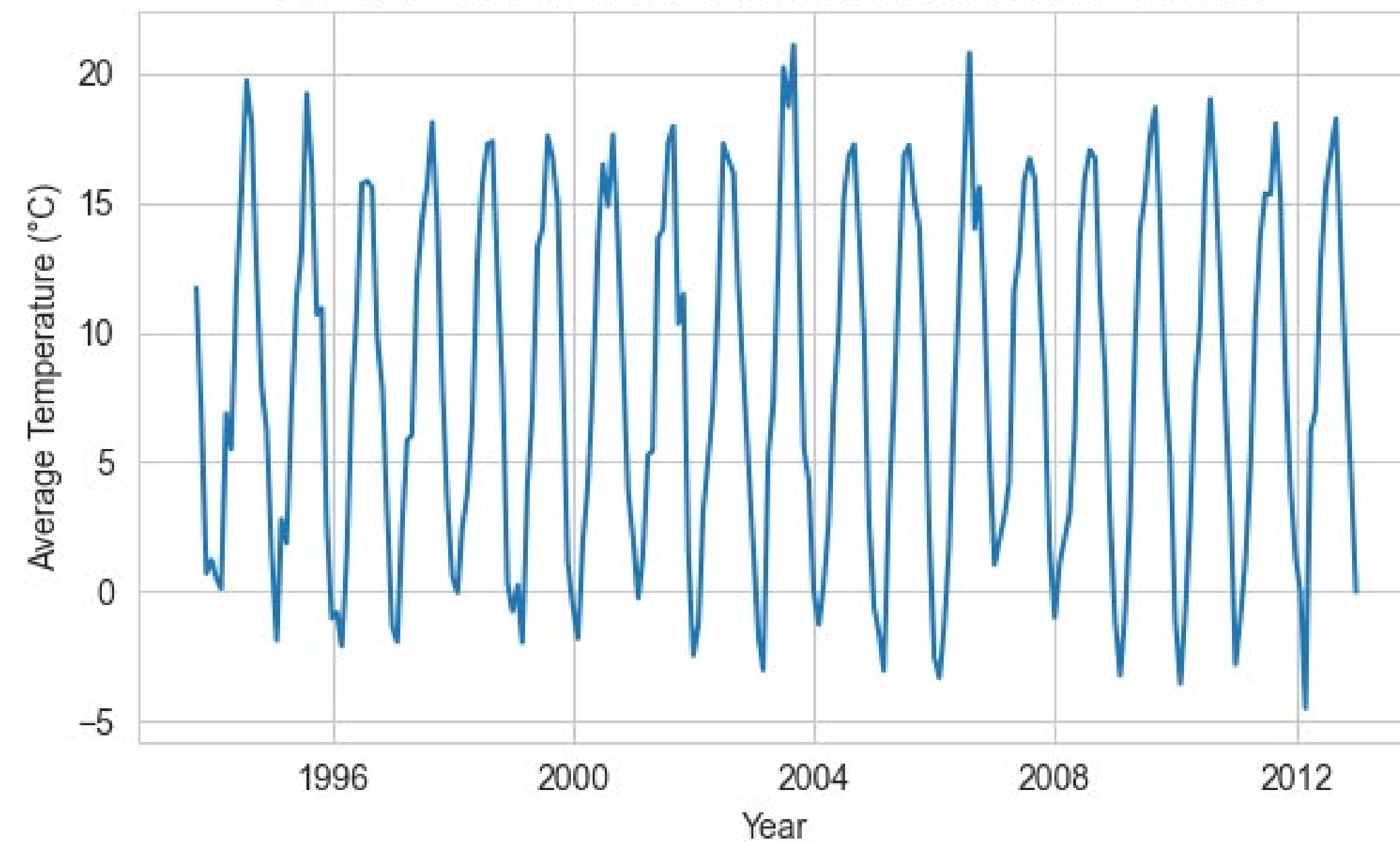
Apple Inc. (AAPL) Closing Prices - Last 5 Years



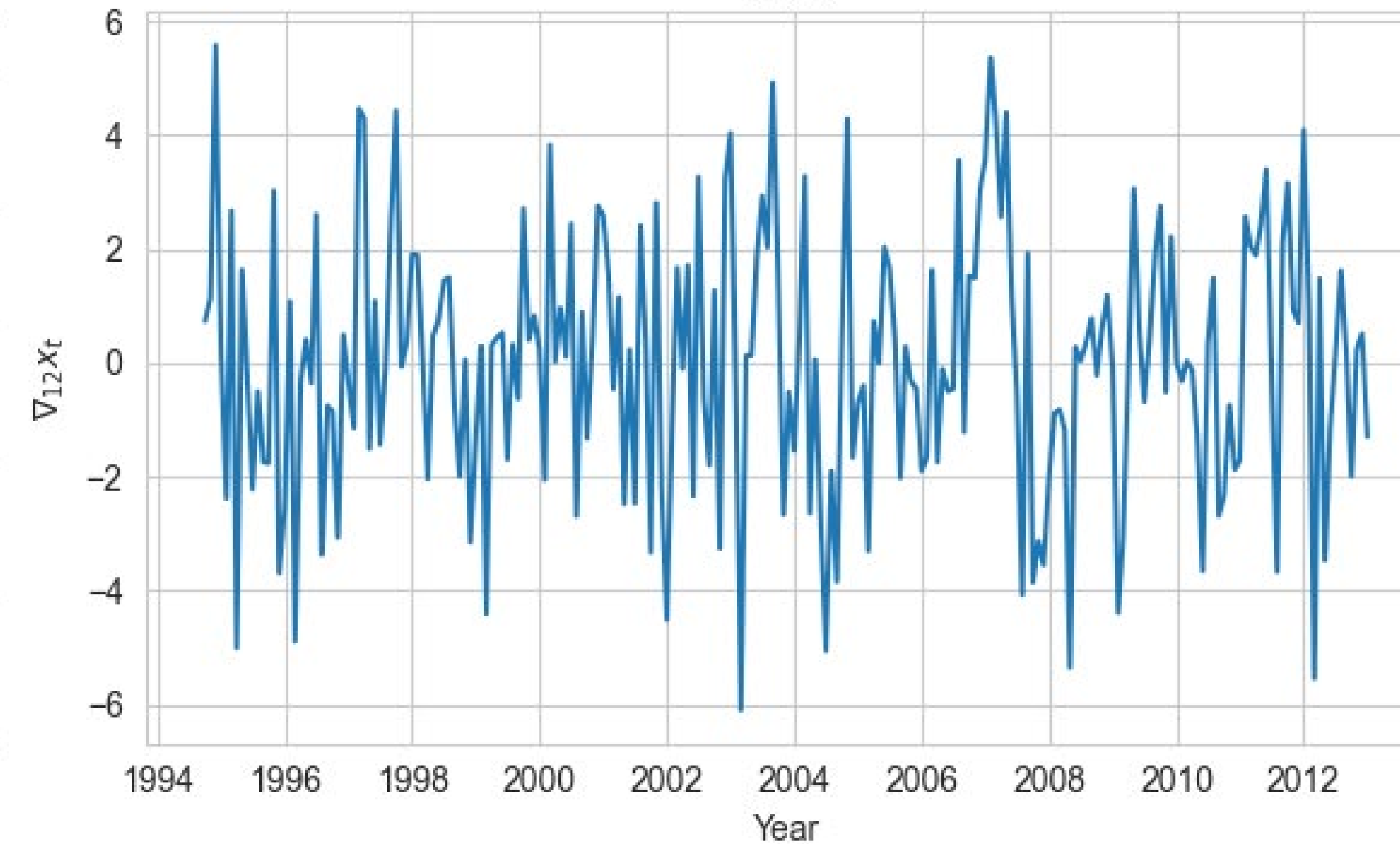
∇X_t : Closing price change



Monthly Average Temperature in Switzerland (20 years)



$\nabla_{12} X_t$

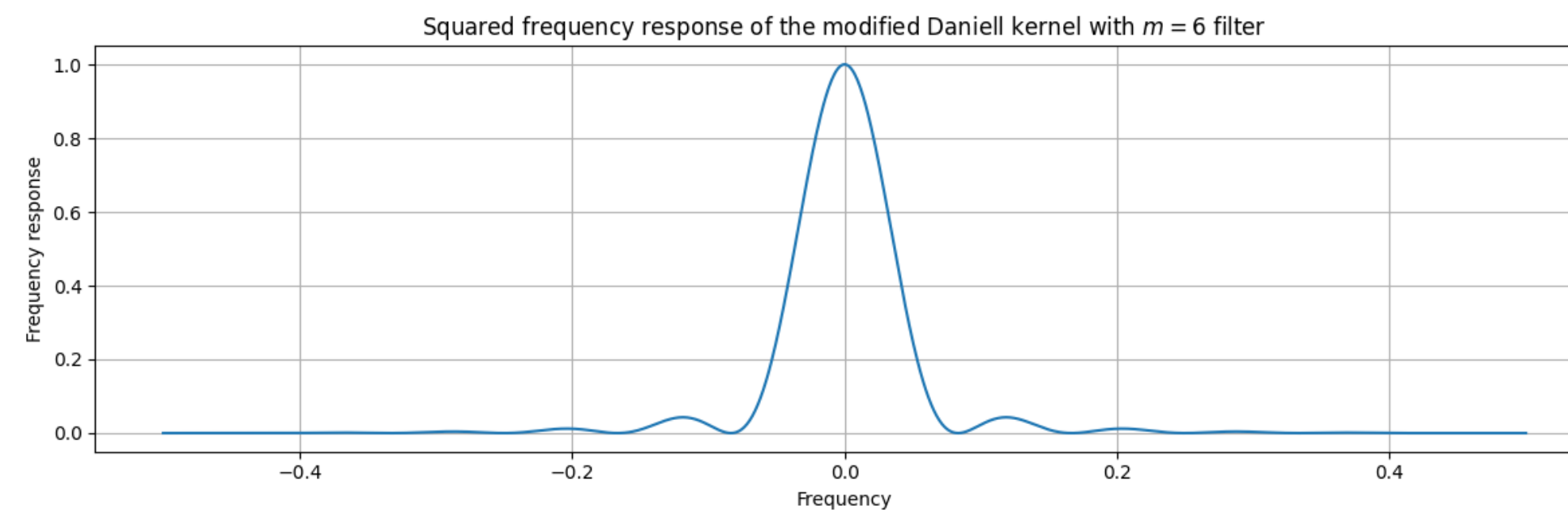


Weighted moving average smoothing is a low-pass filter

Consider the modified Daniell kernel with $m = 6$: $Y_t = \frac{1}{24}(X_{t-6} + X_{t+6}) + \frac{1}{12}\sum_{k=-5}^5 X_{t-k}$.

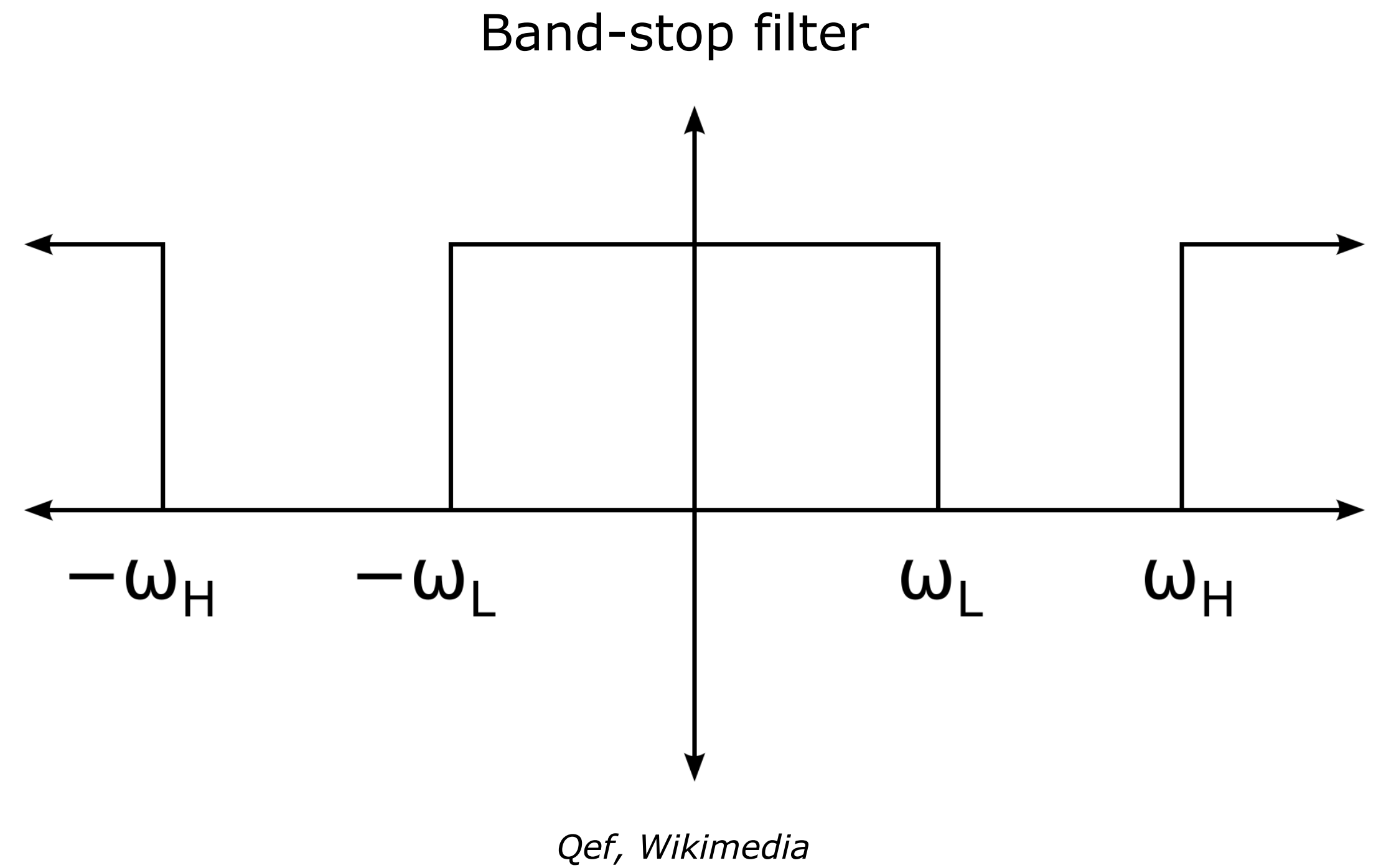
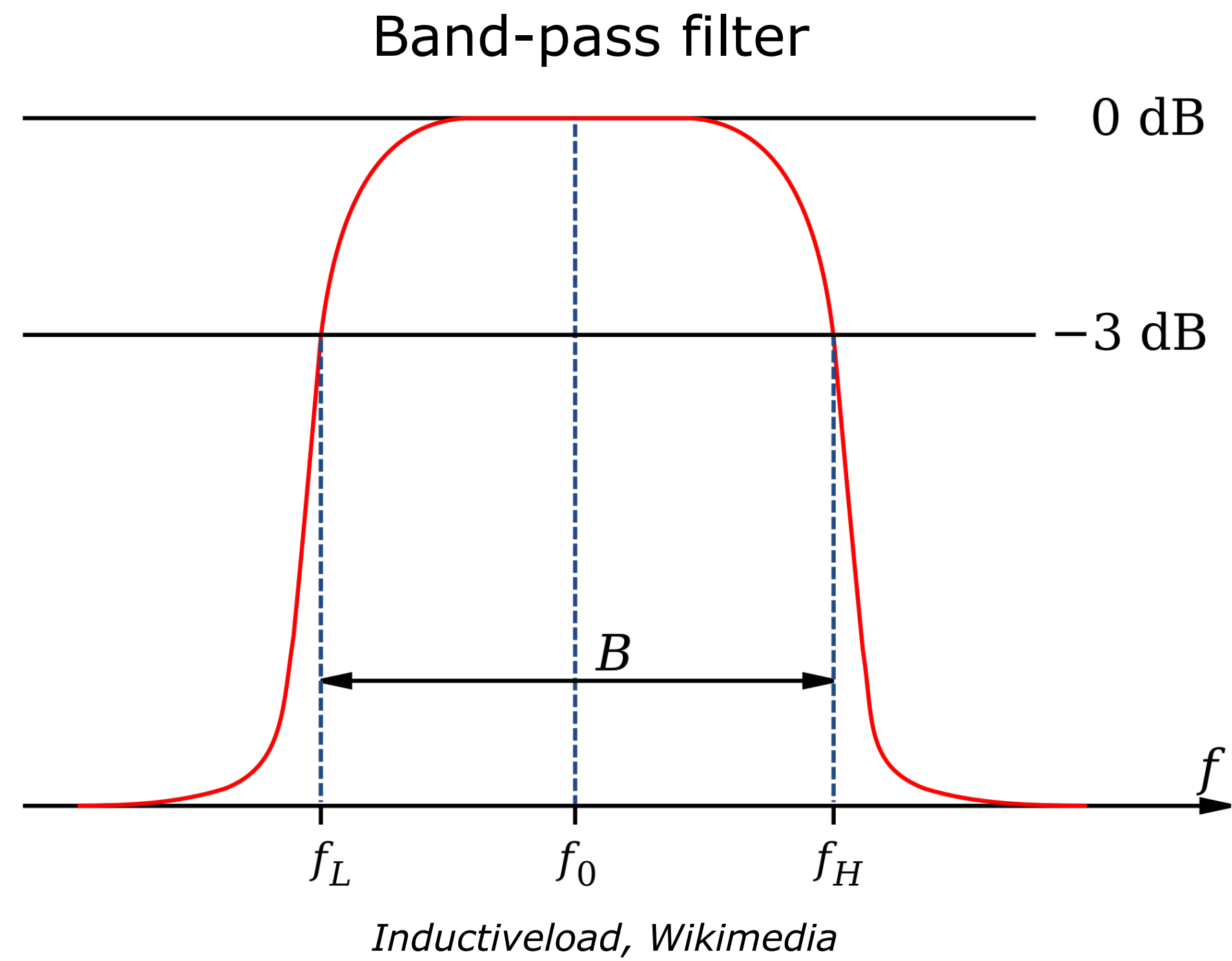
The impulse response function is $a_j = \begin{cases} \frac{1}{12}, & -5 \leq j \leq 5 \\ \frac{1}{24}, & j = \pm 6 \\ 0, & \text{otherwise} \end{cases}$.

The frequency response function is $A(\omega) = \frac{1}{12}(1 + \cos(12\pi\omega) + 2\sum_{k=1}^5 \cos(2\pi\omega k))$.



Thus, the filter removes most frequencies above $\frac{1}{12}$, while keeping other frequencies. It is a **low-pass filter**.

Band-pass and band-stop filters



Exercise

Review Léo Belzile's [lecture notes](#) on periodogram bias and this [article](#) on spectral leakage and windowing

Compare different tapering windows, their properties and use cases.

- How energy spreads around peaks.
- How frequencies are attenuated.

Compare different linear filters, their properties and use cases.

Evaluate the effect of tapering, smoothing, filtering on your real-world data.

- Review the spectrum in log scale.
- Apply different tapering windows.
- Determine appropriate smoothing strength.
- Review the effect of different filters.